Swelio Library

Belgian Electronic ID card access library

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Table of Contents

Symbol Reference 1

Functions 1

ActivateCard Function 10
ActivateCardEx Function 10
AddFileToContainer Function 10
AddRemoveMessageFilter Function 11
AllocateBuffer Function 11
AllocateDefaultHWNDA Function 12
AllocateDefaultHWNDW Function 12
AllocateHWNDA Function 13
AllocateHWNDW Function 13
AllocateLayeredWindowA Function 13
AllocateLayeredWindowW Function 14
AllocateWindowClassA Function 14
AllocateWindowClassW Function 14
AlphaBlendBitmap Function 15
AlphaBlendNative Function 15
BringWindowToFront Function 15
CardDecryptFileA Function 15
CardDecryptFileW Function 16
CardEncryptFileA Function 16
CardEncryptFileW Function 16
CardSignCadesT Function 17
CardSignCadesTEx Function 17
CardSignCMS Function 18
CardSignCMSEx Function 18
CertSignCadesT Function 19
CertSignCadesTData Function 19
CertSignCadesTEx Function 20
CertSignCMS Function 21
CertSignCMSData Function 21
CertSignCMSEx Function 22
CheckMD5 Function 22
CheckSHA1 Function 23
CheckSHA256 Function 23
ClearFileAttributesA Function 23
ClearFileAttributesW Function 24
ClearUnusedMemory Function 24

CloneFont Function 24
ContainerCertificate Function 25
ContainerEidCertificate Function 25
ContainerPickCertificate Function 25
CopyNativeBitmap Function 26
CreateCardBuffer Function 26
CreateNativeBitmap Function 26
CreateUnicodeFileA Function 27
CreateUnicodeFileW Function 27
CreateWindowsFont Function 27
CurrentIPAddressA Function 27
CurrentIPAddressW Function 28
DeactivateCard Function 28
DeactivateCardEx Function 28
DeallocateBuffer Function 29
DeallocateHWNDA Function 29
DeallocateHWNDW Function 29
DecryptFileAESA Function 30
DecryptFileAESW Function 30
DeleteCardBuffer Function 31
DeleteToRecycleBinA Function 31
DeleteToRecycleBinW Function 31
DestroyFont Function 32
DestroyImageBuffer Function 32
DirectoryExistsA Function 32
DirectoryExistsW Function 33
DisplayCertificate Function 33
DpiY Function 33
DrawAlphaText Function 34
DrawAlphaTextRect Function 34
DrawLayeredWindow Function 34
DrawNativeBitmap Function 35
DrawTextDirect Function 35
DrawTextDirectEx Function 35
DrawTextGlow Function 35
DrawTextLine Function 36
DrawTextOutline Function 36
DrawTextRect Function 36
EmptyRecycleBin Function 36
EmToPixels Function 37
EncodeCertificate Function 37
EncodePhoto Function 37

EncryptFileAESA Function 38
EncryptFileAESW Function 38
FileCloseA Function 39
FileCloseW Function 39
FileCopyA Function 39
FileCopyW Function 40
FileCreateRewriteA Function 40
FileCreateRewriteW Function 40
FileDeleteA Function 41
FileDeleteW Function 41
FileExistsA Function 42
FileExistsW Function 42
FileExtensionIsA Function 42
FileExtensionIsW Function 43
FileGetSizeA Function 43
FileGetSizeW Function 43
FileIsExeA Function 44
FileIsExeW Function 44
FileIsIconA Function 45
FileIsIconW Function 45
FileIsImageA Function 45
FileIsImageW Function 46
FileIsLink Function 46
FileOrFolderExistsA Function 46
FileOrFolderExistsW Function 47
FileRenameA Function 47
FileRenameW Function 47
FileWriteA Function 48
FileWriteCharA Function 48
FileWriteCharW Function 48
FileWriteNewLineA Function 49
FileWriteNewLineW Function 49
FileWriteW Function 49
fpreset Function 50
FreeContainer Function 50
FullPathA Function 50
FullPathW Function 51
GenerateAuthenticationSignatureA Function 51
GenerateAuthenticationSignatureExA Function 51
GenerateAuthenticationSignatureExW Function 52
GenerateAuthenticationSignatureW Function 52
GenerateBMPA Function 53

GenerateBMPW Function 53
GenerateNonRepudiationSignatureA Function 54
GenerateNonRepudiationSignatureExA Function 54
GenerateNonRepudiationSignatureExW Function 55
GenerateNonRepudiationSignatureW Function 55
GeneratePNGA Function 56
GeneratePNGW Function 56
GenerateQRCodeA Function 57
GenerateQRCodeExA Function 57
GenerateQRCodeExW Function 57
GenerateQRCodeW Function 58
GetAllFiles Function 58
GetCardBufferA Function 59
GetCardBufferSize Function 59
GetCardBufferW Function 59
GetCardSerialNumber Function 60
GetCardVersion Function 60
GetEncodedCertificateSize Function 61
GetEncodedPhotoSize Function 61
GetFileMD5A Function 62
GetFileMD5W Function 62
GetFilesCountA Function 62
GetFilesCountW Function 63
GetFileSHA1A Function 63
GetFileSHA1W Function 64
GetFileSHA256A Function 64
GetFileSHA256W Function 64
GetHBitmapA Function 65
GetHBitmapW Function 65
GetISOCodeA Function 66
GetISOCodeW Function 66
GetMD5 Function 67
GetPNGA Function 67
GetPNGW Function 68
GetReaderIndexA Function 68
GetReaderIndexW Function 68
GetReaderNameA Function 69
GetReaderNameLenA Function 69
GetReaderNameLenW Function 70
GetReaderNameW Function 70
GetReadersCount Function 70
GetSelectedReaderIndex Function 71

GetSHA1 Function 71
GetSHA256 Function 71
GetStartupA Function 72
GetStartupW Function 72
GetSupportSIS Function 73
GetTextLineSize Function 73
GetTextSize Function 73
GetTextSizeEx Function 73
HibernateWindows Function 74
IgnoreHardwareEvents Function 74
IgnoreServiceEvents Function 74
InitializeContainer Function 75
IsAnimatedGIFA Function 75
IsAnimatedGIFW Function 75
IsCardActivated Function 76
IsCardActivatedEx Function 76
IsCardPresent Function 76
IsCardPresentEx Function 77
IsCardStillInserted Function 77
IsCardStillInsertedEx Function 77
IsCitrixSession Function 78
IsConnectedToInternet Function 78
IsDirectoryA Function 78
IsDirectoryW Function 78
IsEIDCard Function 79
IsEIDCardEx Function 79
IsEngineActive Function 79
IsFemaleA Function 80
IsFemaleW Function 80
IsMaleA Function 80
IsMaleW Function 81
IsMediaCenter Function 81
IsMetroActive Function 81
IsMultiTouchReady Function 82
IsNativeWin64 Function 82
IsRemoteSession Function 82
IsSISCard Function 82
IsSISCardEx Function 83
IsTabletPC Function 83
IsUnicodeFileA Function 83
IsUnicodeFileW Function 84
IsValidFileNameA Function 84

IsValidFileNameW Function 84
IsValidPathNameA Function 85
IsValidPathNameW Function 85
IsWindows10 Function 86
IsWindows7 Function 86
IsWindows8 Function 86
IsWindowsVista Function 86
IsWindowsXP Function 87
IsWindowsXPSP2 Function 87
IsWow64 Function 87
LayeredWndProcA Function 87
LayeredWndProcW Function 87
LoadBitmapJPG Function 88
LoadBitmapPNG Function 88
LoadCertificateA Function 88
LoadCertificateW Function 89
LoadIdentityA Function 89
LoadIdentityW Function 89
LoadPhotoA Function 90
LoadPhotoW Function 90
LoadPNGResource Function 90
MakeCompatibleBitmap Function 91
MakeSoundFromFileA Function 91
MakeSoundFromFileW Function 91
MakeSoundFromResourceA Function 92
MakeSoundFromResourceW Function 92
PointsToPixels Function 92
PortAvailable Function 93
ReadAddressA Function 93
ReadAddressExA Function 93
ReadAddressExW Function 94
ReadAddressW Function 94
ReadAuthenticationCertificate Function 94
ReadAuthenticationCertificateEx Function 95
ReadBufferFromFileA Function 95
ReadBufferFromFileW Function 95
ReadCaCertificate Function 96
ReadCaCertificateEx Function 96
ReadIdentityA Function 97
ReadIdentityExA Function 97
ReadIdentityExW Function 97
ReadIdentityW Function 98

ReadNonRepudiationCertificate Function 98
ReadNonRepudiationCertificateEx Function 99
ReadPhoto Function 99
ReadPhotoAsBitmap Function 99
ReadPhotoAsBitmapEx Function 100
ReadPhotoEx Function 100
ReadRootCaCertificate Function 101
ReadRootCaCertificateEx Function 101
ReadRrnCertificate Function 101
ReadRrnCertificateEx Function 102
ReadSISCardA Function 102
ReadSISCardExA Function 103
ReadSISCardExW Function 103
ReadSISCardW Function 103
RecycleBinEmpty Function 104
ReloadReadersList Function 104
RemoveCallback Function 104
RemoveStartupA Function 105
RemoveStartupW Function 105
RestoreWindowSubclassA Function 105
RestoreWindowSubclassW Function 106
SaveAuthenticationCertificateA Function 106
SaveAuthenticationCertificateExW Function 106
SaveAuthenticationCertificateW Function 107
SaveCaCertificateA Function 107
SaveCaCertificateExW Function 107
SaveCaCertificateW Function 108
SaveCardToToXMLStreamExA Function 108
SaveCardToToXMLStreamExW Function 108
SaveCardToXmlA Function 109
SaveCardToXmlExA Function 109
SaveCardToXmlExW Function 110
SaveCardToXmlW Function 110
SaveContainer Function 111
SaveIdentityA Function 111
SaveIdentityW Function 111
SaveNonRepudiationCertificateA Function 112
SaveNonRepudiationCertificateExW Function 112
SaveNonRepudiationCertificateW Function 112
SavePersonCsvToStreamA Function 113
SavePersonCsvToStreamW Function 113
SavePersonToCsvA Function 114

SavePersonToCsvExA Function 114
SavePersonToCsvExW Function 114
SavePersonToCsvW Function 115
SavePhotoA Function 115
SavePhotoAsBitmapA Function 116
SavePhotoAsBitmapExA Function 116
SavePhotoAsBitmapExW Function 116
SavePhotoAsBitmapW Function 117
SavePhotoAsJpegA Function 117
SavePhotoAsJpegExA Function 117
SavePhotoAsJpegExW Function 118
SavePhotoAsJpegW Function 118
SavePhotoW Function 119
SaveRootCaCertificateA Function 119
SaveRootCaCertificateExW Function 119
SaveRootCaCertificateW Function 120
SaveRrnCertificateA Function 120
SaveRrnCertificateExW Function 120
SaveRrnCertificateW Function 121
SelectReader Function 121
SelectReaderByNameA Function 121
SelectReaderByNameW Function 122
SendAPDU Function 122
SetCallback Function 122
SetMWCompatibility Function 123
SetStartupA Function 123
SetStartupW Function 123
SetSupportSIS Function 124
ShellCopyFileA Function 124
ShellCopyFileW Function 125
ShutdownWindows Function 125
StartEngine Function 126
StopEngine Function 126
StretchNativeBitmap Function 126
StripFileNameA Function 126
StripFileNameW Function 127
SuspendWindows Function 127
TurnMonitorOff Function 127
TurnMonitorOn Function 128
UpdateWindowPosition Function 128
VerifyPinA Function 128
VerifyPinExA Function 128

VerifyPinExW Function 129
VerifyPinW Function 129
VerifySignature Function 130
WriteBufferToFileA Function 130
WriteBufferToFileW Function 130

Structs, Records, Enums 131

structErrorInformation Structure 132
tagCardEventType Enumeration 132
tagEidAddressA Structure 133
tagEidAddressW Structure 133
tagEidCertificate Structure 133
tagEidIdentityA Structure 134
tagEidIdentityExA Structure 135
tagEidIdentityExW Structure 137
tagEidIdentityW Structure 138
tagEidPicture Structure 140
tagSISRecordA Structure 140
tagSISRecordW Structure 141
CardEventType Enumeration 142
EidAddressA Structure 142
EidAddressW Structure 142
EidCertificate Structure 143
EidIdentityA Structure 143
EidIdentityExA Structure 145
EidIdentityExW Structure 146
EidIdentityW Structure 148
EidPicture Structure 149
ErrorInformation Structure 149
PEidAddressA Structure 150
PEidAddressW Structure 150
PEidCertificate Structure 150
PEidIdentityA Structure 151
PEidIdentityExA Structure 152
PEidIdentityExW Structure 154
PEidIdentityW Structure 155
PeidPicture Structure 157
PErrorInformation Structure 157
PSISRecordA Structure 157
PSISRecordW Structure 158
SISRecordA Structure 159
SISRecordW Structure 160

Files **177**

CardEvents.h 177

CardStructures.h 178

Encryption.h 180

FileOperations.h 181

Graphics.h 183

NationalityConverter.h 184

quricol.h 184

Swelio.h 184

System.h 189

SystemInfo.h 190

Index a

1 Symbol Reference

1.1 Functions

The following table lists functions in this documentation.

Functions

	Name	Description
	ActivateCard (see page 10)	Established communication between the card and the reader
	ActivateCardEx (see page 10)	Established communication between the card and the reader
	AddFileToContainer (see page 10)	Add existing file to the container
	AddRemoveMessageFilter (see page 11)	Adds or removes a message from the User Interface Privilege Isolation (UIPI) message filter.
	AllocateBuffer (see page 11)	Allocates the buffer in memory
	AllocateDefaultHWNDA (see page 12)	This function creates the invisible tool window
	AllocateDefaultHWNDW (see page 12)	This function creates the invisible tool window
	AllocateHWNDA (see page 13)	This function creates the invisible tool window using the provided window procedure
	AllocateHWNDW (see page 13)	This function creates the invisible tool window using the provided window procedure
	AllocateLayeredWindowA (see page 13)	This function creates the layered window using the provided window class name
	AllocateLayeredWindowW (see page 14)	This function creates the layered window using the provided window class name
	AllocateWindowClassA (see page 14)	This function creates the standard window using the provided window class name
	AllocateWindowClassW (see page 14)	This function creates the standard window using the provided window class name
	AlphaBlendBitmap (see page 15)	This is function AlphaBlendBitmap.
	AlphaBlendNative (see page 15)	This is function AlphaBlendNative.
	BringWindowToFront (see page 15)	This function brings the specified window to the top of the z-order.
	CardDecryptFileA (see page 15)	Decrypt file using Belgian Id card
	CardDecryptFileW (see page 16)	Decrypt file using Belgian Id card
	CardEncryptFileA (see page 16)	Encrypt file using Belgian Id card
	CardEncryptFileW (see page 16)	Encrypt file using Belgian Id card
	CardSignCadesT (see page 17)	Sign data with eID card according to CADES-T standard
	CardSignCadesTEx (see page 17)	Sign data with eID card according to CADES-T standard
	CardSignCMS (see page 18)	Sign data with eID card according to CMS standard
	CardSignCMSEx (see page 18)	Sign data with eID card according to CMS standard
	CertSignCadesT (see page 19)	Sign data with the certificate file according to CADES-T standard
	CertSignCadesTData (see page 19)	Sign data with the certificate according to CADES-T standard

	CertSignCadesTEx (see page 20)	Sign data with the certificate file according to CADES-T standard
	CertSignCMS (see page 21)	Sign data with the certificate file according to CMS standard
	CertSignCMSData (see page 21)	Sign data with the certificate according to CMS standard
	CertSignCMSEx (see page 22)	Sign data with the certificate file according to CMS standard
	CheckMD5 (see page 22)	Checks the MD5 hash value of the memory buffer
	CheckSHA1 (see page 23)	Checks the SHA1 hash value of the memory buffer
	CheckSHA256 (see page 23)	Checks the SHA256 hash value of the memory buffer
	ClearFileAttributesA (see page 23)	This function sets the file attributes to normal.
	ClearFileAttributesW (see page 24)	This function sets the file attributes to normal.
	ClearUnusedMemory (see page 24)	Clears unused memory and minimized the application memory usage
	CloneFont (see page 24)	This is function CloneFont.
	ContainerCertificate (see page 25)	Assign certificate for signing ASIC container
	ContainerEidCertificate (see page 25)	Select EID card certificate to sign ASIC container
	ContainerPickCertificate (see page 25)	Pick certificate to sign ASIC container
	CopyNativeBitmap (see page 26)	This is function CopyNativeBitmap.
	CreateCardBuffer (see page 26)	Creates XML buffer
	CreateNativeBitmap (see page 26)	This is function CreateNativeBitmap.
	CreateUnicodeFileA (see page 27)	Creates UNICODE file
	CreateUnicodeFileW (see page 27)	Creates UNICODE file
	CreateWindowsFont (see page 27)	This is function CreateWindowsFont.
	CurrentIPAddressA (see page 27)	Returns the IP address
	CurrentIPAddressW (see page 28)	Returns the IP address
	DeactivateCard (see page 28)	Terminates a cennection between a smart card and a reader
	DeactivateCardEx (see page 28)	Terminates a connection between a smart card and a reader
	DeallocateBuffer (see page 29)	Deallocates the memory buffer
	DeallocateHWNDA (see page 29)	This function destroys the specified window.
	DeallocateHWNDW (see page 29)	This function destroys the specified window.
	DecryptFileAESA (see page 30)	Decrypts file using AES algorithm.
	DecryptFileAESW (see page 30)	Decrypts file using AES algorithm.
	DeleteCardBuffer (see page 31)	Deletes XML buffer
	DeleteToRecycleBinA (see page 31)	Deletes file to the Windows Recycle Bin
	DeleteToRecycleBinW (see page 31)	Deletes file to WIndows Recycle Bin
	DestroyFont (see page 32)	This is function DestroyFont.
	DestroyImageBuffer (see page 32)	Destroys the memory buffer
	DirectoryExistsA (see page 32)	Determines whether a specified directory exists.
	DirectoryExistsW (see page 33)	Determines whether a specified directory exists.
	DisplayCertificate (see page 33)	Displays the dialog window with certificate information
	DpiY (see page 33)	This is function DpiY.

	DrawAlphaText (see page 34)	This is function DrawAlphaText.
	DrawAlphaTextRect (see page 34)	This is function DrawAlphaTextRect.
	DrawLayeredWindow (see page 34)	Repaints the surface of the layered window
	DrawNativeBitmap (see page 35)	This is function DrawNativeBitmap.
	DrawTextDirect (see page 35)	This is function DrawTextDirect.
	DrawTextDirectEx (see page 35)	This is function DrawTextDirectEx.
	DrawTextGlow (see page 35)	This is function DrawTextGlow.
	DrawTextLine (see page 36)	This is function DrawTextLine.
	DrawTextOutline (see page 36)	This is function DrawTextOutline.
	DrawTextRect (see page 36)	This is function DrawTextRect.
	EmptyRecycleBin (see page 36)	Empties the recycle bin
	EmToPixels (see page 37)	This is function EmToPixels.
	EncodeCertificate (see page 37)	Performs Base64 encoding of the certificate
	EncodePhoto (see page 37)	Performs Base64 encoding of the photo
	EncryptFileAESA (see page 38)	Encrypts file using AES algorithm.
	EncryptFileAESW (see page 38)	Encrypts file using AES algorithm.
	FileCloseA (see page 39)	Concludes input/output (I/O) to a file opened using the FileCreateRewrite function.
	FileCloseW (see page 39)	Concludes input/output (I/O) to a file opened using the FileCreateRewrite function.
	FileCopyA (see page 39)	The CopyFile function copies an existing file to a new file.
	FileCopyW (see page 40)	The CopyFile function copies an existing file to a new file.
	FileCreateRewriteA (see page 40)	Creates new or overwrites existing file
	FileCreateRewriteW (see page 40)	Creates new or overwrites existing file
	FileDeleteA (see page 41)	Deletes a file from disk.
	FileDeleteW (see page 41)	Deletes a file from disk.
	FileExistsA (see page 42)	Tests whether a specified file exists.
	FileExistsW (see page 42)	Tests whether a specified file exists.
	FileExtensionIsA (see page 42)	Checks the file extension
	FileExtensionIsW (see page 43)	Checks the file extension
	FileGetSizeA (see page 43)	Retrieves the size of a specified file.
	FileGetSizeW (see page 43)	Retrieves the size of a specified file.
	FileIsExeA (see page 44)	Checks if the file is a Windows executable
	FileIsExeW (see page 44)	Checks if the file is a Windows executable
	FileIsIconA (see page 45)	Checks if the file is a Windows icon (.ico) file
	FileIsIconW (see page 45)	Checks if the file is a Windows icon (.ico) file
	FileIsImageA (see page 45)	Checks if the file is an image file
	FileIsImageW (see page 46)	Checks if the file is an image file
	FileIsLink (see page 46)	Checks to see if the file specified by file name is a Microsoft Windows shortcut (.Lnk) file (and is neither a file nor a folder).
	FileOrFolderExistsA (see page 46)	Checks if the file or folder with the given name exists
	FileOrFolderExistsW (see page 47)	Checks if the file or folder with the given name exists
	FileRenameA (see page 47)	Renames the file
	FileRenameW (see page 47)	Renames the file
	FileWriteA (see page 48)	Writes string to the file
	FileWriteCharA (see page 48)	Writes one character to the file

	FileWriteCharW (see page 48)	Writes one character to the file
	FileWriteNewLineA (see page 49)	Writes new line sequence to the file
	FileWriteNewLineW (see page 49)	Writes new line sequence to the file
	FileWriteW (see page 49)	Writes string to the file
	fpreset (see page 50)	This is function fpreset.
	FreeContainer (see page 50)	Deallocates ASIC container
	FullPathA (see page 50)	Gets the full path to the file based on file name
	FullPathW (see page 51)	Gets the full path to the file based on file name
	GenerateAuthenticationSignatureA (see page 51)	Generate authentication signature
	GenerateAuthenticationSignatureExA (see page 51)	Generate authentication signature
	GenerateAuthenticationSignatureExW (see page 52)	Generate authentication signature
	GenerateAuthenticationSignatureW (see page 52)	Generate authentication signature
	GenerateBMPA (see page 53)	Generates Windows Bitmap file with QR Code image
	GenerateBMPW (see page 53)	Generates Windows Bitmap file with QR Code image
	GenerateNonRepudiationSignatureA (see page 54)	Generate non repudiation signature
	GenerateNonRepudiationSignatureExA (see page 54)	Generate non repudiation signature
	GenerateNonRepudiationSignatureExW (see page 55)	Generate non repudiation signature
	GenerateNonRepudiationSignatureW (see page 55)	Generate non repudiation signature
	GeneratePNGA (see page 56)	Generates PNG file with QR Code image
	GeneratePNGW (see page 56)	Generates PNG file with QR Code image
	GenerateQRCodeA (see page 57)	Read eID card and save the identity information and address to PNG QR Code file
	GenerateQRCodeExA (see page 57)	Read eID card and save the identity information and address to PNG QR Code file
	GenerateQRCodeExW (see page 57)	Read eID card and save the identity information and address to PNG QR Code file
	GenerateQRCodeW (see page 58)	Read eID card and save the identity information and address to PNG QR Code file
	GetAllFiles (see page 58)	Returns the names of files in a specified directory.
	GetCardBufferA (see page 59)	Gets XML or CSV information from the memory buffer
	GetCardBufferSize (see page 59)	This function returns the size of the buffer needed to hold the information from the eID card in the XML or CSV format
	GetCardBufferW (see page 59)	Gets XML or CSV information from the memory buffer
	GetCardSerialNumber (see page 60)	Get the serial number of EID card
	GetCardVersion (see page 60)	Get the applet version number for card in the reader with specified number
	GetEncodedCertificateSize (see page 61)	Returns the size of the Base64 encoded certificate
	GetEncodedPhotoSize (see page 61)	Calculates buffer size for Base64 encoded photo
	GetFileMD5A (see page 62)	Gets the MD5 hash value for the file
	GetFileMD5W (see page 62)	Gets the MD5 hash value for the file
	GetFilesCountA (see page 62)	Calculates the number of files in the given folder
	GetFilesCountW (see page 63)	Calculates the number of files in the given folder
	GetFileSHA1A (see page 63)	Gets the SHA1 hash value for the file

	GetFileSHA1W (see page 64)	Gets the SHA1 hash value for the file
	GetFileSHA256A (see page 64)	Gets the SHA256 hash value for the file
	GetFileSHA256W (see page 64)	Gets the SHA256 hash value for the file
	GetHBitmapA (see page 65)	Generates Windows Bitmap in memory with QR Code image
	GetHBitmapW (see page 65)	Generates Windows Bitmap in memory with QR Code image
	GetISOCodeA (see page 66)	Returns the country ISO code based on the nationality string
	GetISOCodeW (see page 66)	Returns the country ISO code based on the nationality string
	GetMD5 (see page 67)	Gets the MD5 hash value for the content of the memory buffer
	GetPNGA (see page 67)	Writes PNG image to the memory buffer.
	GetPNGW (see page 68)	Writes PNG image to the memory buffer.
	GetReaderIndexA (see page 68)	Returns the zero-based reader index with specified name
	GetReaderIndexW (see page 68)	Returns the zero-based reader index with specified name
	GetReaderNameA (see page 69)	Returns the name of the card reader
	GetReaderNameLenA (see page 69)	Returns the length of the reader name
	GetReaderNameLenW (see page 70)	Returns the length of the reader name
	GetReaderNameW (see page 70)	Returns the name of the card reader
	GetReadersCount (see page 70)	Get number of card readers connected to PC
	GetSelectedReaderIndex (see page 71)	Returns the index of the active smart card reader
	GetSHA1 (see page 71)	Gets the SHA1 hash value for the content of the memory buffer
	GetSHA256 (see page 71)	Gets the SHA256 hash value for the content of the memory buffer
	GetStartupA (see page 72)	Checks if the application is registered to run when Windows starts
	GetStartupW (see page 72)	Checks if the application is registered to run when Windows starts
	GetSupportSIS (see page 73)	Checks if the SIS cards are supported by the engine
	GetTextLineSize (see page 73)	This is function GetTextLineSize.
	GetTextSize (see page 73)	This is function GetTextSize.
	GetTextSizeEx (see page 73)	This is function GetTextSizeEx.
	HibernateWindows (see page 74)	Hibernates Windows
	IgnoreHardwareEvents (see page 74)	Ignore USB reader insert / remove events
	IgnoreServiceEvents (see page 74)	Ignore smartcard service stop events when reporting readers list change
	InitializeContainer (see page 75)	Initializes ASIC container
	IsAnimatedGIFA (see page 75)	Checks if the file is an animated GIF image file
	IsAnimatedGIFW (see page 75)	Checks if the file is an animated GIF image file
	IsCardActivated (see page 76)	Checks the connection between a smart card and a reader
	IsCardActivatedEx (see page 76)	Checks the connection between a smart card and a reader
	IsCardPresent (see page 76)	Checks if the card is present in the card reader
	IsCardPresentEx (see page 77)	Checks if the card is present in the card reader
	IsCardStillInserted (see page 77)	Checks if the card is still inserted in the card reader
	IsCardStillInsertedEx (see page 77)	Checks if the card is still inserted in the card reader
	IsCitrixSession (see page 78)	Checks if application is running in Citrix session
	IsConnectedToInternet (see page 78)	Checks if PC is connected to Internet
	IsDirectoryA (see page 78)	Verifies that a path is a valid directory.
	IsDirectoryW (see page 78)	Verifies that a path is a valid directory.
	IsEIDCard (see page 79)	Check if Belgian EID card is inserted into card reader
	IsEIDCardEx (see page 79)	Check if Belgian EID card is inserted into card reader
	IsEngineActive (see page 79)	Checks if the Swelio Engine is activated
	IsFemaleA (see page 80)	Checks if the card owner is female
	IsFemaleW (see page 80)	Checks if the card owner is female

	IsMaleA (see page 80)	Checks if the card owner is male
	IsMaleW (see page 81)	Checks if the card owner is male
	IsMediaCenter (see page 81)	Checks if the Media Center version of Windows is installed
	IsMetroActive (see page 81)	Checks if metro interface is active
	IsMultiTouchReady (see page 82)	Checks if the system is multi touch ready
	IsNativeWin64 (see page 82)	Checks if the application is native 64 bit executable
	IsRemoteSession (see page 82)	Checks if application is running in RDP session
	IsSISCard (see page 82)	Check if Belgian SIS card is inserted into card reader
	IsSISCardEx (see page 83)	Check if Belgian SIS card is inserted into card reader
	IsTabletPC (see page 83)	Checks if the application is running on the Tablet PC
	IsUnicodeFileA (see page 83)	Checks if the file is UNICODE file
	IsUnicodeFileW (see page 84)	Checks if the file is UNICODE file
	IsValidFileNameA (see page 84)	Checks if provided string is a valid file name
	IsValidFileNameW (see page 84)	Checks if provided string is a valid file name
	IsValidPathNameA (see page 85)	Checks if provided string is a valid file path
	IsValidPathNameW (see page 85)	Checks if provided string is a valid file path
	IsWindows10 (see page 86)	Checks if PC is running Windows 10 or better
	IsWindows7 (see page 86)	Checks if PC is running Windows 7 or better
	IsWindows8 (see page 86)	Checks if PC is Running Windows 8 or better
	IsWindowsVista (see page 86)	Checks if PC is running Windows Vista or better
	IsWindowsXP (see page 87)	Checks if PC is running Windows XP
	IsWindowsXPSP2 (see page 87)	Checks if PC is running Windows XP with Service Pack 2 installed
	IsWow64 (see page 87)	Checks if the 32 bit application runs on 64 bit Windows
	LayeredWndProcA (see page 87)	The default window procedure for the layered window
	LayeredWndProcW (see page 87)	The default window procedure for the layered window
	LoadBitmapJPG (see page 88)	This is function LoadBitmapJPG.
	LoadBitmapPNG (see page 88)	This is function LoadBitmapPNG.
	LoadCertificateA (see page 88)	Reads the certificate from a file
	LoadCertificateW (see page 89)	Reads the certificate from a file
	LoadIdentityA (see page 89)	Reads the raw identity information from a file
	LoadIdentityW (see page 89)	Reads the raw identity information from a file
	LoadPhotoA (see page 90)	Loads photo from a file
	LoadPhotoW (see page 90)	Loads photo from a file
	LoadPNGResource (see page 90)	This is function LoadPNGResource.
	MakeCompatibleBitmap (see page 91)	This is function MakeCompatibleBitmap.
	MakeSoundFromFileA (see page 91)	Plays the wave sound from the file
	MakeSoundFromFileW (see page 91)	Plays the wave sound from the file
	MakeSoundFromResourceA (see page 92)	Plays the wave sound from the resource
	MakeSoundFromResourceW (see page 92)	Plays the wave sound from the resource
	PointsToPixels (see page 92)	This is function PointsToPixels.
	PortAvailable (see page 93)	Checks if the port with specified number is available
	ReadAddressA (see page 93)	Read address information from Belgian eID card
	ReadAddressExA (see page 93)	Read address information from Belgian eID card
	ReadAddressExW (see page 94)	Read address information from Belgian eID card
	ReadAddressW (see page 94)	Read address information from Belgian eID card
	ReadAuthenticationCertificate (see page 94)	Read Authentication Certificate to memory

	ReadAuthenticationCertificateEx (see page 95)	Read Authentication Certificate to memory
	ReadBufferFromFileA (see page 95)	Reads the content of the file to the memory buffer
	ReadBufferFromFileW (see page 95)	Reads the content of the file to the memory buffer
	ReadCaCertificate (see page 96)	Read Ca Certificate to memory
	ReadCaCertificateEx (see page 96)	Read Ca Certificate to memory
	ReadIdentityA (see page 97)	Read identity information from Belgian eID card
	ReadIdentityExA (see page 97)	Read identity information from Belgian eID card
	ReadIdentityExW (see page 97)	Read identity information from Belgian eID card
	ReadIdentityW (see page 98)	Read identity information from Belgian eID card
	ReadNonRepudiationCertificate (see page 98)	Read Non Repudiation Certificate to memory
	ReadNonRepudiationCertificateEx (see page 99)	Read Non Repudiation Certificate to memory
	ReadPhoto (see page 99)	Reads a photo from a card
	ReadPhotoAsBitmap (see page 99)	Reads the picture from the card, converts it to bitmap and returns the bitmap handle Decription: Reads the photo from the Belgian eID card and returns the bitmap handle Reading the photo from the card is a time consuming operation. Do it only when needed.
	ReadPhotoAsBitmapEx (see page 100)	Reads the picture from the card, converts it to bitmap and returns the bitmap handle Decription: Reads the photo from the Belgian eID card and returns the Windows bitmap handle Reading the photo from the card is a time consuming operation. Do it only when needed.
	ReadPhotoEx (see page 100)	Reads a photo from a card
	ReadRootCaCertificate (see page 101)	Read Root Ca Certificate to memory
	ReadRootCaCertificateEx (see page 101)	Read Root Ca Certificate to memory
	ReadRrnCertificate (see page 101)	Read Rrn Certificate to memory
	ReadRrnCertificateEx (see page 102)	Read Rrn Certificate to memory
	ReadSISCardA (see page 102)	Read Belgian SIS card.
	ReadSISCardExA (see page 103)	Read Belgian SIS card.
	ReadSISCardExW (see page 103)	Read Belgian SIS card.
	ReadSISCardW (see page 103)	Read Belgian SIS card.
	RecycleBinEmpty (see page 104)	Returns TRUE if Windows Recyce Bin is empty
	ReloadReadersList (see page 104)	Reloads the list of the available card readers
	RemoveCallback (see page 104)	Remove callback procedure for card events
	RemoveStartupA (see page 105)	Removes the application from the list of the automatically started applications
	RemoveStartupW (see page 105)	Removes the application from the list of the automatically started applications
	RestoreWindowSubclassA (see page 105)	Restores window standard procedure
	RestoreWindowSubclassW (see page 106)	Restores window standard procedure
	SaveAuthenticationCertificateA (see page 106)	Save Authentication Certificate to a file
	SaveAuthenticationCertificateExW (see page 106)	Save Authentication Certificate to a file
	SaveAuthenticationCertificateW (see page 107)	Save Authentication Certificate to a file
	SaveCaCertificateA (see page 107)	Save Ca Certificate to a file
	SaveCaCertificateExW (see page 107)	Save Ca Certificate to a file

	SaveCaCertificateW (see page 108)	Save Ca Certificate to a file
	SaveCardToToXMLStreamExA (see page 108)	Read eID card and save the information to XML buffer
	SaveCardToToXMLStreamExW (see page 108)	Read eID card and save the information to XML buffer
	SaveCardToXmlA (see page 109)	Read eID card and save the information to XML file
	SaveCardToXmlExA (see page 109)	Read eID card and save the information to XML file
	SaveCardToXmlExW (see page 110)	Read eID card and save the information to XML file
	SaveCardToXmlW (see page 110)	Read eID card and save the information to XML file
	SaveContainer (see page 111)	Save container to the file
	SaveIdentityA (see page 111)	Saves indentity infornation to a file
	SaveIdentityW (see page 111)	Saves indentity infornation to a file
	SaveNonRepudiationCertificateA (see page 112)	Save Non Repudiation Certificate to a file
	SaveNonRepudiationCertificateExW (see page 112)	Save Non Repudiation Certificate to a file
	SaveNonRepudiationCertificateW (see page 112)	Save Non Repudiation Certificate to a file
	SavePersonCsvToStreamA (see page 113)	Read eID card and save the identity information to CSV memory buffer
	SavePersonCsvToStreamW (see page 113)	Read eID card and save the identity information to CSV memory buffer
	SavePersonToCsvA (see page 114)	Read eID card and save the identity information and address to CSV file
	SavePersonToCsvExA (see page 114)	Read eID card and save the identity information and address to CSV file
	SavePersonToCsvExW (see page 114)	Read eID card and save the identity information and address to CSV file
	SavePersonToCsvW (see page 115)	Read eID card and save the identity information and address to CSV file
	SavePhotoA (see page 115)	Save photo to a file
	SavePhotoAsBitmapA (see page 116)	Save the picture from the card to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsBitmapExA (see page 116)	Reads the picture from the card and saves it to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsBitmapExW (see page 116)	Reads the picture from the card and saves it to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsBitmapW (see page 117)	Save the picture from the card to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegA (see page 117)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegExA (see page 117)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.

	SavePhotoAsJpegExW (see page 118)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegW (see page 118)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoW (see page 119)	Saves photo to a file
	SaveRootCaCertificateA (see page 119)	Save Root Ca Certificate to a file
	SaveRootCaCertificateExW (see page 119)	Save Root Ca Certificate to a file
	SaveRootCaCertificateW (see page 120)	Save Root Ca Certificate to a file
	SaveRrnCertificateA (see page 120)	Save RRN Certificate to a file
	SaveRrnCertificateExW (see page 120)	Save RRN Certificate to a file
	SaveRrnCertificateW (see page 121)	Save RRN Certificate to a file
	SelectReader (see page 121)	When more than 1 reader connected, select the reader with specified number
	SelectReaderByNameA (see page 121)	Select active smart card reader by providing the reader name
	SelectReaderByNameW (see page 122)	Select active smart card reader by providing the reader name
	SendAPDU (see page 122)	This is function SendAPDU.
	SetCallback (see page 122)	Activates callback procedure for card status change event
	SetMWCompatibility (see page 123)	Set the compatibility mode with the old version of the oficial EID MiddleWare
	SetStartupA (see page 123)	Register application to run when Windows starts
	SetStartupW (see page 123)	Register application to run when Windows starts
	SetSupportSIS (see page 124)	Activates or deactivates SIS card support by engine
	ShellCopyFileA (see page 124)	Copies file to the new location
	ShellCopyFileW (see page 125)	Copies file to the new location
	ShutdownWindows (see page 125)	Logs off the interactive user, shuts down the system.
	StartEngine (see page 126)	Activates the Swelio Engine.
	StopEngine (see page 126)	Deactivates the Swelio Engine
	StretchNativeBitmap (see page 126)	This is function StretchNativeBitmap.
	StripFileNameA (see page 126)	Replaces environment variable names with values
	StripFileNameW (see page 127)	Replaces environment variable names with values
	SuspendWindows (see page 127)	Suspends Windows
	TurnMonitorOff (see page 127)	Turns the monitor off
	TurnMonitorOn (see page 128)	Turns the monitor on
	UpdateWindowPosition (see page 128)	Updated the window position
	VerifyPinA (see page 128)	Verify PIN code
	VerifyPinExA (see page 128)	Verify PIN code
	VerifyPinExW (see page 129)	Verify PIN code
	VerifyPinW (see page 129)	Verify PIN code
	VerifySignature (see page 130)	Verifies the signature from the specified hash value.
	WriteBufferToFileA (see page 130)	Writes the memory buffer to file
	WriteBufferToFileW (see page 130)	Writes the memory buffer to file

1.1.1 ActivateCard Function

Established communication between the card and the reader

C++

```
BOOL WINAPI ActivateCard();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE if the card is activated, otherwise returns FALSE

Description

The ActivateCard function establishes a connection between the calling application and a smart card contained by a specific reader. If no card exists in the specified reader, an error is returned.

Example

```
if (IsCardPresent())
{
   ActivateCard();
}
```

1.1.2 ActivateCardEx Function

Established communication between the card and the reader

C++

```
BOOL WINAPI ActivateCardEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader

Returns

Returns TRUE if the card is activated, otherwise returns FALSE

Description

The ActivateCard (see page 10) function establishes a connection between the calling application and a smart card contained by a specific reader. If no card exists in the specified reader, an error is returned.

1.1.3 AddFileToContainer Function

Add existing file to the container

C++

```
BOOL WINAPI AddFileToContainer(LPVOID container, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPVOID container	Container handle, which is allocated by calling InitializeContainer (see page 75) function
LPSTR fileName	The name of the file which will be added to the container

Returns

Returns true if the operation is successful, otherwise returns false

Description

Call this function to include the file to the container. The file must exist.

1.1.4 AddRemoveMessageFilter Function

Adds or removes a message from the User Interface Privilege Isolation (UIPI) message filter.

C++

```
void WINAPI AddRemoveMessageFilter(UINT message, DWORD dwFlags);
```

File

System.h (see page 189)

Parameters

Parameters	Description
UINT message	Specifies the message to add to or remove from the filter.
DWORD dwFlags	Specifies the action to be performed. One of the following values. • MSGFLT_ADD - Adds the message to the filter. This has the effect of allowing the message to be received. • MSGFLT_REMOVE - Removes the message from the filter. This has the effect of blocking the message.

Description

This function changes the message filter for Windows Vista or better. UIPI is a security feature that prevents messages from being received from a lower integrity level sender. All such messages with a value above WM_USER are blocked by default. The filter, somewhat contrary to intuition, is a list of messages that are allowed through. Therefore, adding a message to the filter allows that message to be received from a lower integrity sender, while removing a message blocks that message from being received.

Certain messages with a value less than WM_USER are required to pass through the filter regardless of the filter setting. You can call this function to remove one of those messages from the filter and it will return TRUE. However, the message will still be received by the calling process.

1.1.5 AllocateBuffer Function

Allocates the buffer in memory

C++

```
void* WINAPI AllocateBuffer(int bufferSize);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
int bufferSize	The size of the buffer

Returns

The pointer to the allocated memory block

Description

AllocateBuffer allocates a block of the given size on the heap, and returns the address of this memory.The bytes of the allocated buffer are not set to zero. To dispose of the buffer, use DeallocateBuffer (see page 29) function.

1.1.6 AllocateDefaultHWNDA Function

This function creates the invisible tool window

C++

```
HWND WINAPI AllocateDefaultHWNDA();
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the invisible zero-size tool window that can be used for internal purposes, like processing the special Windows messages for synchronization, etc...

1.1.7 AllocateDefaultHWNDW Function

This function creates the invisible tool window

C++

```
HWND WINAPI AllocateDefaultHWNDW();
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the invisible zero-size tool window that can be used for internal purposes, like processing the special Windows messages for synchronization, etc...

1.1.8 AllocateHWNDA Function

This function creates the invisible tool window using the provided window procedure

C++

```
HWND WINAPI AllocateHWNDA(LONG_PTR method);
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the invisible zero-size tool window that can be used for internal purposes, like processing the special Windows messages for synchronization, etc...

1.1.9 AllocateHWNDW Function

This function creates the invisible tool window using the provided window procedure

C++

```
HWND WINAPI AllocateHWNDW(LONG_PTR method);
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the invisible zero-size tool window that can be used for internal purposes, like processing the special Windows messages for synchronization, etc...

1.1.10 AllocateLayeredWindowA Function

This function creates the layered window using the provided window class name

C++

```
HWND WINAPI AllocateLayeredWindowA(LPCSTR className);
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the layered window using the provided window class name

1.1.11 AllocateLayeredWindowW Function

This function creates the layered window using the provided window class name

C++

```
HWND WINAPI AllocateLayeredWindowW(LPCWSTR className);
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the layered window using the provided window class name

1.1.12 AllocateWindowClassA Function

This function creates the standard window using the provided window class name

C++

```
HWND WINAPI AllocateWindowClassA(LPCSTR className);
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the standard window using the provided window class name

1.1.13 AllocateWindowClassW Function

This function creates the standard window using the provided window class name

C++

```
HWND WINAPI AllocateWindowClassW(LPCWSTR className);
```

File

System.h (see page 189)

Returns

If the function succeeds, the return value is a handle to the new window. If the function fails, the return value is NULL.

Description

This function creates the standard window using the provided window class name

1.1.14 AlphaBlendBitmap Function

This is function AlphaBlendBitmap.

C++

```
void WINAPI AlphaBlendBitmap(HBITMAP src, HDC hdc, int left, int top, int width, int height, int alpha);
```

File

Graphics.h (see page 183)

1.1.15 AlphaBlendNative Function

This is function AlphaBlendNative.

C++

```
void WINAPI AlphaBlendNative(HDC hdcDest, int xoriginDest, int yoriginDest, int wDest, int hDest, HDC hdcSrc, int xoriginSrc, int yoriginSrc, int wSrc, int hSrc, BYTE alpha);
```

File

Graphics.h (see page 183)

1.1.16 BringWindowToFront Function

This function brings the specified window to the top of the z-order.

C++

```
void WINAPI BringWindowToFront(HWND window);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND window	Handle to the window to bring to the top of the z-order.

1.1.17 CardDecryptFileA Function

Decrypt file using Belgian Id card

C++

```
BOOL WINAPI CardDecryptFileA(LPSTR szSource, LPSTR szDestination);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR szSource	The name of the encrypted file
LPSTR szDestination	The name of the decrypted file

Description

Decrypt file which was encrypted using CardEncryptFile function

1.1.18 CardDecryptFileW Function

Decrypt file using Belgian Id card

C++

```
BOOL WINAPI CardDecryptFileW(LPWSTR szSource, LPWSTR szDestination);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR szSource	The name of the encrypted file
LPWSTR szDestination	The name of the decrypted file

Description

Decrypt file which was encrypted using CardEncryptFile function

1.1.19 CardEncryptFileA Function

Encrypt file using Belgian Id card

C++

```
BOOL WINAPI CardEncryptFileA(LPSTR szSource, LPSTR szDestination);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR szSource	The name of the source file
LPSTR szDestination	The name of the encrypted file

Description

Encrypt file using Belgian Id card. The card must be inserted in the reader

1.1.20 CardEncryptFileW Function

Encrypt file using Belgian Id card

C++

```
BOOL WINAPI CardEncryptFileW(LPWSTR szSource, LPWSTR szDestination);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR szSource	The name of the source file
LPWSTR szDestination	The name of the encrypted file

Description

Encrypt file using Belgian Id card. The card must be inserted in the reader

1.1.21 CardSignCadesT Function

Sign data with eID card according to CADES-T standard

C++

```
BOOL WINAPI CardSignCadesT(int readerNumber, BYTE * data, UINT dataLen, BYTE * signature,
UINT * signatureLen);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
BYTE * data	the data to sign
UINT dataLen	the size of the data buffer
BYTE * signature	the signature buffer
UINT * signatureLen	the size of the signature buffer

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CADES-T signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.22 CardSignCadesTEx Function

Sign data with eID card according to CADES-T standard

C++

```
BOOL WINAPI CardSignCadesTEx(int readerNumber, BYTE* data, UINT dataLen, BYTE* signature,
UINT* signatureLen, ErrorInformation* error);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
BYTE* data	the data to sign
UINT dataLen	the size of the data buffer
BYTE* signature	the signature buffer
UINT* signatureLen	the size of the signature buffer
ErrorInformation* error	error information

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CADES-T signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.23 CardSignCMS Function

Sign data with eID card according to CMS standard

C++

```
BOOL WINAPI CardSignCMS(int readerNumber, BYTE * data, UINT dataLen, BYTE * signature, UINT * signatureLen);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
BYTE * data	the data to sign
UINT dataLen	the size of the data buffer
BYTE * signature	the signature buffer
UINT * signatureLen	the size of the signature buffer

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CMS signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.24 CardSignCMSEx Function

Sign data with eID card according to CMS standard

C++

```
BOOL WINAPI CardSignCMSEx(int readerNumber, BYTE* data, UINT dataLen, BYTE* signature, UINT* signatureLen, ErrorInformation* Error);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
BYTE* data	the data to sign
UINT dataLen	the size of the data buffer
BYTE* signature	the signature buffer
UINT* signatureLen	the size of the signature buffer
error	error information

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CMS signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.25 CertSignCadesT Function

Sign data with the certificate file according to CADES-T standard

C++

```
BOOL WINAPI CertSignCadesT(LPWSTR certificate, LPWSTR password, BYTE * data, UINT dataLen,
BYTE * signature, UINT * signatureLen);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR certificate	The name of the certificate file
LPWSTR password	The private key password
BYTE * data	the data to sign
UINT dataLen	the size of the data buffer
BYTE * signature	the signature buffer
UINT * signatureLen	the size of the signature buffer

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CADES-T signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.26 CertSignCadesTData Function

Sign data with the certificate according to CADES-T standard

C++

```
BOOL WINAPI CertSignCadesTData(BYTE* certificate, DWORD certLen, LPWSTR password, BYTE*
data, UINT dataLen, BYTE* signature, UINT* signatureLen, ErrorInformation* error);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
BYTE* certificate	The byte array with the signing certificate content
DWORD certLen	The size of th signing certificate
LPWSTR password	The password of the signing certificate
BYTE* data	The data to be signed
UINT dataLen	The size of the data to be signed
BYTE* signature	The value of the signature
UINT* signatureLen	The size of the signature
ErrorInformation* error	The error information

Returns

Returns true if the operation is successful, otherwise returns false

1.1.27 CertSignCadesTEx Function

Sign data with the certificate file according to CADES-T standard

C++

```
BOOL WINAPI CertSignCadesTEx(LPWSTR certificate, LPWSTR password, BYTE* data, UINT dataLen,
BYTE* signature, UINT* signatureLen, ErrorInformation* error);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR certificate	The name of the certificate file
LPWSTR password	The private key password
BYTE* data	the data to sign
UINT dataLen	the size of the data buffer
BYTE* signature	the signature buffer
UINT* signatureLen	the size of the signature buffer
ErrorInformation* error	error information

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CADES-T signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.28 CertSignCMS Function

Sign data with the certificate file according to CMS standard

C++

```
BOOL WINAPI CertSignCMS(LPWSTR certificate, LPWSTR password, BYTE * data, UINT dataLen,
BYTE * signature, UINT * signatureLen);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR certificate	The name of the certificate file
LPWSTR password	The private key password
BYTE * data	the data to sign
UINT dataLen	the size of the data buffer
BYTE * signature	the signature buffer
UINT * signatureLen	the size of the signature buffer

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CMS signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.29 CertSignCMSData Function

Sign data with the certificate according to CMS standard

C++

```
BOOL WINAPI CertSignCMSData(BYTE* certificate, DWORD certLen, LPWSTR password, BYTE* data,
UINT dataLen, BYTE* signature, UINT* signatureLen, ErrorInformation* error);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
BYTE* certificate	The byte array with the signing certificate content
DWORD certLen	The size of th signing certificate
LPWSTR password	The password of the signing certificate
BYTE* data	The data to be signed
UINT dataLen	The size of the data to be signed
BYTE* signature	The value of the signature
UINT* signatureLen	The size of the signature
ErrorInformation* error	The error information

Returns

Returns true if the operation is successful, otherwise returns false

1.1.30 CertSignCMSEx Function

Sign data with the certificate file according to CMS standard

C++

```
BOOL WINAPI CertSignCMSEx(LPWSTR certificate, LPWSTR password, BYTE* data, UINT dataLen,
BYTE* signature, UINT* signatureLen, ErrorInformation* error);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR certificate	The name of the certificate file
LPWSTR password	The private key password
BYTE* data	the data to sign
UINT dataLen	the size of the data buffer
BYTE* signature	the signature buffer
UINT* signatureLen	the size of the signature buffer
ErrorInformation* error	error information

Returns

Returns true if the operation is successful, otherwise returns false

Description

Create CMS signature for data buffer. Can be used for digital signature of PDF documents in combination with external PDF library

1.1.31 CheckMD5 Function

Checks the MD5 hash value of the memory buffer

C++

```
BOOL WINAPI CheckMD5(BYTE* source, int sourceSize, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
BYTE* source	The source bytes
int sourceSize	The size of the source buffer
BYTE* buffer	The hash value buffer
int bufferSize	The size of the hash value buffer

Returns

Returns TRUE if the hash value is correct, otherwise returns false

Description

This function checks if the provided value of the hash is valid

1.1.32 CheckSHA1 Function

Checks the SHA1 hash value of the memory buffer

C++

```
BOOL WINAPI CheckSHA1(BYTE* source, int sourceSize, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
BYTE* source	The source bytes
int sourceSize	The size of the source buffer
BYTE* buffer	The hash value buffer
int bufferSize	The size of the hash value buffer

Returns

Returns TRUE if the hash value is correct, otherwise returns false

Description

This function checks if the provided value of the hash is valid

1.1.33 CheckSHA256 Function

Checks the SHA256 hash value of the memory buffer

C++

```
BOOL WINAPI CheckSHA256(BYTE* source, int sourceSize, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
BYTE* source	The source bytes
int sourceSize	The size of the source buffer
BYTE* buffer	The hash value buffer
int bufferSize	The size of the hash value buffer

Returns

Returns TRUE if the hash value is correct, otherwise returns false

Description

This function checks if the provided value of the hash is valid

1.1.34 ClearFileAttributesA Function

This function sets the file attributes to normal.

C++

```
void WINAPI ClearFileAttributesA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file

Description

This function removed additional file attributes, like system, read-only and hidden.

1.1.35 ClearFileAttributesW Function

This function sets the file attributes to normal.

C++

```
void WINAPI ClearFileAttributesW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file

Description

This function removed additional file attributes, like system, read-only and hidden.

1.1.36 ClearUnusedMemory Function

Clears unused memory and minimized the application memory usage

C++

```
void WINAPI ClearUnusedMemory();
```

File

System.h (see page 189)

1.1.37 CloneFont Function

This is function CloneFont.

C++

```
HFONT WINAPI CloneFont(HFONT hFont);
```

File

Graphics.h (see page 183)

1.1.38 ContainerCertificate Function

Assign certificate for signing ASIC container

C++

```
BOOL WINAPI ContainerCertificate(LPVOID container, LPWSTR fileName, LPWSTR password);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPVOID container	Container handle, which is allocated by calling InitializeContainer (see page 75) function

Returns

Returns true if the operation is successful, otherwise returns false

Description

Use this function to assign certificate which will be used to sign container at the moment when container will be saved to the file

1.1.39 ContainerEidCertificate Function

Select EID card certificate to sign ASIC container

C++

```
BOOL WINAPI ContainerEidCertificate(LPVOID container, int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPVOID container	Container handle, which is allocated by calling InitializeContainer (see page 75) function

Returns

Returns true if the operation is successful, otherwise returns false

Description

Call this function to select EID certificate to sign ASIC container

1.1.40 ContainerPickCertificate Function

Pick certificate to sign ASIC container

C++

```
BOOL WINAPI ContainerPickCertificate(LPVOID container);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPVOID container	Container handle, which is allocated by calling InitializeContainer (see page 75) function

Returns

Returns true if the operation is successful, otherwise returns false

Description

Use this function to let user pick the certificate to sign ASIC contaner The dialog to choose the certificate will be shown to the user

1.1.41 CopyNativeBitmap Function

This is function CopyNativeBitmap.

C++

```
void WINAPI CopyNativeBitmap(HBITMAP src, HDC dstDC, int width, int height, int left, int top);
```

File

Graphics.h (see page 183)

1.1.42 CreateCardBuffer Function

Creates XML buffer

C++

```
void* WINAPI CreateCardBuffer();
```

File

Swelio.h (see page 184)

Returns

The memory buffer to store information

Description

Use this function to create XML buffer

1.1.43 CreateNativeBitmap Function

This is function CreateNativeBitmap.

C++

```
HBITMAP WINAPI CreateNativeBitmap(INT width, INT height, __deref_opt_out void ** ppvBits);
```

File

Graphics.h (see page 183)

1.1.44 CreateUnicodeFileA Function

Creates UNICODE file

C++

```
void WINAPI CreateUnicodeFileA(LPCSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCSTR fileName	The name of the file

1.1.45 CreateUnicodeFileW Function

Creates UNICODE file

C++

```
void WINAPI CreateUnicodeFileW(LPCWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The name of the file

1.1.46 CreateWindowsFont Function

This is function CreateWindowsFont.

C++

```
HFONT WINAPI CreateWindowsFont(LPCWSTR fontFamily, INT size, INT fontStyle, INT
fontQuality);
```

File

Graphics.h (see page 183)

1.1.47 CurrentIPAddressA Function

Returns the IP address

C++

```
void WINAPI CurrentIPAddressA(LPSTR address, UINT len);
```

File

SystemInfo.h (see page 190)

1.1.48 CurrentIPAddressW Function

Returns the IP address

C++

```
void WINAPI CurrentIPAddressW(LPWSTR address, UINT len);
```

File

SystemInfo.h (see page 190)

1.1.49 DeactivateCard Function

Terminates a cennection between a smart card and a reader

C++

```
void WINAPI DeactivateCard();
```

File

Swelio.h (see page 184)

Description

The DeactivateCard function terminates a connection previously opened between the calling application and a smart card in the target reader.

1.1.50 DeactivateCardEx Function

Terminates a connection between a smart card and a reader

C++

```
void WINAPI DeactivateCardEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader

Description

The DeactivateCard (see page 28) function terminates a connection previously opened between the calling application and a smart card in the target reader.

1.1.51 DeallocateBuffer Function

Deallocates the memory buffer

C++

```
void WINAPI DeallocateBuffer(void* buffer);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
void* buffer	The pointer to the memory buffer

Description

DeallocateBuffer frees a memory block previously allocated with AllocateBuffer (see page 11). Use this procedure to dispose of a memory block obtained with AllocateBuffer (see page 11).

1.1.52 DeallocateHWNDA Function

This function destroys the specified window.

C++

```
BOOL WINAPI DeallocateHWNDA(HWND hwnd);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND hwnd	Handle to the window to be destroyed

Description

This function restores the window default procedure and destroys the window

1.1.53 DeallocateHWNDW Function

This function destroys the specified window.

C++

```
BOOL WINAPI DeallocateHWNDW(HWND hwnd);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND hwnd	Handle to the window to be destroyed

Description

This function restores the window default procedure and destroys the window

1.1.54 DecryptFileAESA Function

Decrypts file using AES algorithm.

C++

```
BOOL WINAPI DecryptFileAESA(LPSTR szSource, LPSTR szDestination, LPSTR szPassword);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR szSource	The source file name
LPSTR szDestination	The decrypted file name
LPSTR szPassword	The password

Returns

Returns TRUE if the file is successfully decrypted, otherwise returns FALSE.

Description

Use this function to decrypt the file using AES algorithm

1.1.55 DecryptFileAESW Function

Decrypts file using AES algorithm.

C++

```
BOOL WINAPI DecryptFileAESW(LPWSTR szSource, LPWSTR szDestination, LPWSTR szPassword);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR szSource	The source file name
LPWSTR szDestination	The decrypted file name
LPWSTR szPassword	The password

Returns

Returns TRUE if the file is successfully decrypted, otherwise returns FALSE.

Description

Use this function to decrypt the file using AES algorithm

1.1.56 DeleteCardBuffer Function

Deletes XML buffer

C++

```
void WINAPI DeleteCardBuffer(void* buffer);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
void* buffer	The memory buffer to store information

Description

Use this function to delete the buffer

1.1.57 DeleteToRecycleBinA Function

Deletes file to the Windows Recycle Bin

C++

```
void WINAPI DeleteToRecycleBinA(LPSTR fileName, BOOL silent);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
BOOL silent	Do not display a progress dialog box

Description

Use this function to delete the file to Windows Recycle Bin

1.1.58 DeleteToRecycleBinW Function

Deletes file to WIndows Recycle Bin

C++

```
void WINAPI DeleteToRecycleBinW(LPWSTR fileName, BOOL silent);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
BOOL silent	Do not display a progress dialog box

Description

Use this function to delete the file to Windows Recycle Bin

1.1.59 DestroyFont Function

This is function DestroyFont.

C++

```
void WINAPI DestroyFont(HFONT hFont);
```

File

Graphics.h (see page 183)

1.1.60 DestroyImageBuffer Function

Destroys the memory buffer

C++

```
void WINAPI DestroyImageBuffer(void* buffer);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
void* buffer	The memory buffer

Description

Destroys the memory buffer created to hold the image returned by GetPNGA (see page 67) (Ansi) or GetPNGW (see page 68) (Unicode) functions

1.1.61 DirectoryExistsA Function

Determines whether a specified directory exists.

C++

```
BOOL WINAPI DirectoryExistsA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the directory

Returns

Returns TRUE if the directory exists, otherwise returns FALSE

Description

Call DirectoryExists to determine whether the directory specified by the Directory parameter exists. If the directory exists, the function returns True. If the directory does not exist, the function returns False. If a full path name is entered, DirectoryExists searches for the directory along the designated path. Otherwise, the Directory parameter is interpreted as a relative path name from the current directory.

1.1.62 DirectoryExistsW Function

Determines whether a specified directory exists.

C++

```
BOOL WINAPI DirectoryExistsW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the directory

Returns

Returns TRUE if the directory exists, otherwise returns FALSE

Description

Call DirectoryExists to determine whether the directory specified by the Directory parameter exists. If the directory exists, the function returns True. If the directory does not exist, the function returns False. If a full path name is entered, DirectoryExists searches for the directory along the designated path. Otherwise, the Directory parameter is interpreted as a relative path name from the current directory.

1.1.63 DisplayCertificate Function

Displays the dialog window with certificate information

C++

```
void WINAPI DisplayCertificate(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidCertificate certificate	The certificate data

Description

Use this function to show the certificate for the user

1.1.64 DpiY Function

This is function DpiY.

C++

```
int WINAPI DpiY();
```

File

Graphics.h (see page 183)

1.1.65 DrawAlphaText Function

This is function DrawAlphaText.

C++

```
void WINAPI DrawAlphaText(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF color, int width, int
height, UINT flags);
```

File

Graphics.h (see page 183)

1.1.66 DrawAlphaTextRect Function

This is function DrawAlphaTextRect.

C++

```
void WINAPI DrawAlphaTextRect(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF color, LPRECT
lpRect, UINT flags);
```

File

Graphics.h (see page 183)

1.1.67 DrawLayeredWindow Function

Repaints the surface of the layered window

C++

```
void WINAPI DrawLayeredWindow(HWND handle, int left, int top, int width, int height, HDC
buffer, COLORREF colorKey, byte alpha, BOOL redrawOnly);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND handle	The handle of the window
int left	The horizontal coordinate of the window
int top	The vertical coordinate of the window
int width	The width of the window
int height	The height of the window
HDC buffer	Handle to a DC for the surface that defines the layered window

COLORREF colorKey	COLORREF structure that specifies the color key to be used when composing the layered window.
byte alpha	Specifies an alpha transparency value to be used on the entire source bitmap
BOOL redrawOnly	Only redraw and do not update the window position

1.1.68 DrawNativeBitmap Function

This is function DrawNativeBitmap.

C++

```
void WINAPI DrawNativeBitmap(HBITMAP src, HBITMAP dst, int width, int height);
```

File

Graphics.h (see page 183)

1.1.69 DrawTextDirect Function

This is function DrawTextDirect.

C++

```
void WINAPI DrawTextDirect(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF color, int left, int
top);
```

File

Graphics.h (see page 183)

1.1.70 DrawTextDirectEx Function

This is function DrawTextDirectEx.

C++

```
void WINAPI DrawTextDirectEx(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF color, COLORREF
background, int left, int top);
```

File

Graphics.h (see page 183)

1.1.71 DrawTextGlow Function

This is function DrawTextGlow.

C++

```
void WINAPI DrawTextGlow(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF foreColor, int left, int
top);
```

File

Graphics.h (see page 183)

1.1.72 DrawTextLine Function

This is function DrawTextLine.

C++

```
void WINAPI DrawTextLine(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF color, int left, int top);
```

File

Graphics.h (see page 183)

1.1.73 DrawTextOutline Function

This is function DrawTextOutline.

C++

```
void WINAPI DrawTextOutline(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF foreColor, COLORREF backColor, int left, int top);
```

File

Graphics.h (see page 183)

1.1.74 DrawTextRect Function

This is function DrawTextRect.

C++

```
void WINAPI DrawTextRect(HDC hDC, LPCWSTR s, HFONT hFont, COLORREF color, COLORREF background, LPRECT lpRect, UINT flags);
```

File

Graphics.h (see page 183)

1.1.75 EmptyRecycleBin Function

Empties the recycle bin

C++

```
void WINAPI EmptyRecycleBin();
```

File

System.h (see page 189)

Description

Removes all files from the Windows recycle bin

1.1.76 EmToPixels Function

This is function EmToPixels.

C++

```
int WINAPI EmToPixels(int em);
```

File

Graphics.h (see page 183)

1.1.77 EncodeCertificate Function

Performs Base64 encoding of the certificate

C++

```
int WINAPI EncodeCertificate(PEidCertificate certificate, BYTE* buffer, int bufferSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidCertificate certificate	The certificate data
BYTE* buffer	The Base64 encoded certificate buffer
int bufferSize	The size of the buffer

Returns

Returns the size of the buffer needed to hold the encoded certificate

1.1.78 EncodePhoto Function

Performs Base64 encoding of the photo

C++

```
int WINAPI EncodePhoto(PeidPicture photo, BYTE* buffer, int bufferSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure
BYTE* buffer	The Base64 encoded photo buffer
int bufferSize	The size of the buffer

Returns

Returns the size of the buffer needed to hold the encoded photo

Description

Use this function for Base64 encoding of the photo

1.1.79 EncryptFileAESA Function

Encrypts file using AES algorithm.

C++

```
BOOL WINAPI EncryptFileAESA(LPSTR szSource, LPSTR szDestination, LPSTR szPassword);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR szSource	The source file name
LPSTR szDestination	The encrypted file name
LPSTR szPassword	The password

Returns

Returns TRUE if the file is successfully encrypted, otherwise returns FALSE

Description

Use this function to encrypt the file using AES algorithm

1.1.80 EncryptFileAESW Function

Encrypts file using AES algorithm.

C++

```
BOOL WINAPI EncryptFileAESW(LPWSTR szSource, LPWSTR szDestination, LPWSTR szPassword);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR szSource	The source file name
LPWSTR szDestination	The encrypted file name
LPWSTR szPassword	The password

Returns

Returns TRUE if the file is successfully encrypted, otherwise returns FALSE

Description

Use this function to encrypt the file using AES algorithm

1.1.81 FileCloseA Function

Concludes input/output (I/O) to a file opened using the FileCreateRewrite function.

C++

```
void WINAPI FileCloseA(HANDLE handle);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file

Description

Closes the file handle of the specified file when its not in use anymore

1.1.82 FileCloseW Function

Concludes input/output (I/O) to a file opened using the FileCreateRewrite function.

C++

```
void WINAPI FileCloseW(HANDLE handle);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file

Description

Closes the file handle of the specified file when its not in use anymore

1.1.83 FileCopyA Function

The CopyFile function copies an existing file to a new file.

C++

```
BOOL WINAPI FileCopyA(LPSTR oldName, LPSTR newName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR oldName	The name of the source file
LPSTR newName	The name of the destination file

Returns

The result of the function is TRUE when the file is successfully copied to the new location, otherwise the result is FALSE.

Description

This function makes a copy of the file with the new name or path.

1.1.84 FileCopyW Function

The CopyFile function copies an existing file to a new file.

C++

```
BOOL WINAPI FileCopyW(LPWSTR oldName, LPWSTR newName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR oldName	The name of the source file
LPWSTR newName	The name of the destination file

Returns

The result of the function is TRUE when the file is successfully copied to the new location, otherwise the result is FALSE.

Description

This function makes a copy of the file with the new name or path.

1.1.85 FileCreateRewriteA Function

Creates new or overwrites existing file

C++

```
HANDLE WINAPI FileCreateRewriteA(LPCSTR fileName);
```

File

FileOperations.h (see page 181)

Returns

The result of the function is the handle of the file

Description

This function creates the new file with provided file name if the file with given name does not exists. If the file exists, it will be overwritten and the current content of the file will be lost

1.1.86 FileCreateRewriteW Function

Creates new or overwrites existing file

C++

```
HANDLE WINAPI FileCreateRewriteW(LPCWSTR fileName);
```

File

FileOperations.h (see page 181)

Returns

The result of the function is the handle of the file

Description

This function creates the new file with provided file name if the file with given name does not exists. If the file exists, it will be overwritten and the current content of the file will be lost

1.1.87 FileDeleteA Function

Deletes a file from disk.

C++

```
void WINAPI FileDeleteA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file

Description

DeleteFile deletes the file named by fileName from the disk.

1.1.88 FileDeleteW Function

Deletes a file from disk.

C++

```
void WINAPI FileDeleteW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file

Description

DeleteFile deletes the file named by fileName from the disk.

1.1.89 FileExistsA Function

Tests whether a specified file exists.

C++

```
BOOL WINAPI FileExistsA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Returns

FileExists returns TRUE if the file specified by FileName exists. If the file does not exist, FileExists returns FALSE.

Description

Use this function to check if the file with provided name exists.

1.1.90 FileExistsW Function

Tests whether a specified file exists.

C++

```
BOOL WINAPI FileExistsW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Returns

FileExists returns TRUE if the file specified by FileName exists. If the file does not exist, FileExists returns FALSE.

Description

Use this function to check if the file with provided name exists.

1.1.91 FileExtensionIsA Function

Checks the file extension

C++

```
BOOL WINAPI FileExtensionIsA(LPCSTR fileName, LPCSTR ext);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCSTR fileName	The name of the file
LPCSTR ext	The file name extension

Returns

Returns true if the file has a specified extension, otherwise returns false.

Description

This function checks if the file has a given extension

1.1.92 FileExtensionIsW Function

Checks the file extension

C++

```
BOOL WINAPI FileExtensionIsW(LPCWSTR fileName, LPCWSTR ext);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The name of the file
LPCWSTR ext	The file name extension

Returns

Returns true if the file has a specified extension, otherwise returns false.

Description

This function checks if the file has a given extension

1.1.93 FileGetSizeA Function

Retrieves the size of a specified file.

C++

```
DWORD WINAPI FileGetSizeA(LPCSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCSTR fileName	The name of the file

Returns

The size of the file in bytes.

Description

This function determines the size of the file specified by the file name.

1.1.94 FileGetSizeW Function

Retrieves the size of a specified file.

C++

```
DWORD WINAPI FileGetSizeW(LPCWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The name of the file

Returns

The size of the file in bytes.

Description

This function determines the size of the file specified by the name of the file

1.1.95 FileIsExeA Function

Checks if the file is a Windows executable

C++

```
BOOL WINAPI FileIsExeA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file

Returns

Returns TRUE if the file is a Windows executable, otherwise returns FALSE.

1.1.96 FileIsExeW Function

Checks if the file is a Windows executable

C++

```
BOOL WINAPI FileIsExeW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file

Returns

Returns TRUE if the file is a Windows executable, otherwise returns FALSE.

1.1.97 FileIsIconA Function

Checks if the file is a Windows icon (.ico) file

C++

```
BOOL WINAPI FileIsIconA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file

Returns

Returns TRUE if the file is a Windows icon (.ico) file, otherwise returns FALSE.

1.1.98 FileIsIconW Function

Checks if the file is a Windows icon (.ico) file

C++

```
BOOL WINAPI FileIsIconW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file

Returns

Returns TRUE if the file is a Windows icon (.ico) file, otherwise returns FALSE.

1.1.99 FileIsImageA Function

Checks if the file is an image file

C++

```
BOOL WINAPI FileIsImageA(LPCSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCSTR fileName	The name of the file

Returns

Returns TRUE if the file is an image file, otherwise returns FALSE.

1.1.100 FileIsImageW Function

Checks if the file is an image file

C++

```
BOOL WINAPI FileIsImageW(LPCWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The name of the file

Returns

Returns TRUE if the file is an image file, otherwise returns FALSE.

1.1.101 FileIsLink Function

Checks to see if the file specified by file name is a Microsoft Windows shortcut (.Lnk) file (and is neither a file nor a folder).

C++

```
BOOL WINAPI FileIsLink(LPCWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The name of the file to check

Returns

Returns TRUE if the file is a Microsoft Windows shortcut, otherwise returns FALSE

Description

This function is used to check if the file with provided file name is a Windows shortcut or not

1.1.102 FileOrFolderExistsA Function

Checks if the file or folder with the given name exists

C++

```
BOOL WINAPI FileOrFolderExistsA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The file or folder name

Returns

Returns TRUE if the file or folder exists, otherwise returns FALSE.

1.1.103 FileOrFolderExistsW Function

Checks if the file or folder with the given name exists

C++

```
BOOL WINAPI FileOrFolderExistsW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The file or folder name

Returns

Returns TRUE if the file or folder exists, otherwise returns FALSE.

1.1.104 FileRenameA Function

Renames the file

C++

```
BOOL WINAPI FileRenameA(LPSTR oldName, LPSTR newName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR oldName	File to be renamed
LPSTR newName	New name of the file

Returns

Returns TRUE if the file was successfully renamed, otherwise returns FALSE

1.1.105 FileRenameW Function

Renames the file

C++

```
BOOL WINAPI FileRenameW(LPWSTR oldName, LPWSTR newName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR oldName	File to be renamed
LPWSTR newName	New name of the file

Returns

Returns TRUE if the file was successfully renamed, otherwise returns FALSE

1.1.106 FileWriteA Function

Writes string to the file

C++

```
void WINAPI FileWriteA(HANDLE handle, LPSTR text);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file
LPSTR text	The text to write

1.1.107 FileWriteCharA Function

Writes one character to the file

C++

```
void WINAPI FileWriteCharA(HANDLE handle, CHAR text);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file
CHAR text	The character to write

1.1.108 FileWriteCharW Function

Writes one character to the file

C++

```
void WINAPI FileWriteCharW(HANDLE handle, WCHAR text);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file
WCHAR text	The character to write

1.1.109 FileWriteNewLineA Function

Writes new line sequence to the file

C++

```
void WINAPI FileWriteNewLineA(HANDLE handle);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file

1.1.110 FileWriteNewLineW Function

Writes new line sequence to the file

C++

```
void WINAPI FileWriteNewLineW(HANDLE handle);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file

1.1.111 FileWriteW Function

Writes string to the file

C++

```
void WINAPI FileWriteW(HANDLE handle, LPWSTR text);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
HANDLE handle	The handle of the file

LPWSTR text	The text to write

1.1.112 fpreset Function

This is function fpreset.

C++

```
void fpreset();
```

File

System.h (see page 189)

1.1.113 FreeContainer Function

Deallocates ASIC container

C++

```
void WINAPI FreeContainer(LPVOID container);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPVOID container	Container handle, which is allocated by calling InitializeContainer (see page 75) function

Returns

None

Description

Call this function to deallocate container memory and release container handle.

1.1.114 FullPathA Function

Gets the full path to the file based on file name

C++

```
BOOL WINAPI FullPathA(LPSTR fileName, LPSTR fullName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
LPSTR fullName	The full path to the file

1.1.115 FullPathW Function

Gets the full path to the file based on file name

C++

```
BOOL WINAPI FullPathW(LPWSTR fileName, LPWSTR fullName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
LPWSTR fullName	The full path to the file

1.1.116 GenerateAuthenticationSignatureA Function

Generate authentication signature

C++

```
BOOL WINAPI GenerateAuthenticationSignatureA(LPSTR pinCode, BYTE* dataHash, int hashSize,
BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate authentication signature using provided hash value

1.1.117 GenerateAuthenticationSignatureExA Function

Generate authentication signature

C++

```
BOOL WINAPI GenerateAuthenticationSignatureExA(int readerNumber, LPSTR pinCode, BYTE*
dataHash, int hashSize, BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate authentication signature using provided hash value

1.1.118 GenerateAuthenticationSignatureExW Function

Generate authentication signature

C++

```
BOOL WINAPI GenerateAuthenticationSignatureExW(int readerNumber, LPWSTR pinCode, BYTE*
dataHash, int hashSize, BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate authentication signature using provided hash value

1.1.119 GenerateAuthenticationSignatureW Function

Generate authentication signature

C++

```
BOOL WINAPI GenerateAuthenticationSignatureW(LPWSTR pinCode, BYTE* dataHash, int hashSize,
BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate authentication signature using provided hash value

1.1.120 GenerateBMPA Function

Generates Windows Bitmap file with QR Code image

C++

```
void WINAPI GenerateBMPA(LPSTR fileName, LPSTR text, int margin, int size, int level);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
LPSTR text	The text to encode
int margin	The margin from the border in points
int size	The size of the one point in pixels
int level	The error correction level

Description

Generate Windows Bitmap file with encoded text as QR Code image

1.1.121 GenerateBMPW Function

Generates Windows Bitmap file with QR Code image

C++

```
void WINAPI GenerateBMPW(LPWSTR fileName, LPWSTR text, int margin, int size, int level);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
LPWSTR text	The text to encode
int margin	The margin from the border in points

int size	The size of the one point in pixels
int level	The error correction level

Description

Generate Windows Bitmap file with encoded text as QR Code image

1.1.122 GenerateNonRepudiationSignatureA Function

Generate non repudiation signature

C++

```
BOOL WINAPI GenerateNonRepudiationSignatureA(LPSTR pinCode, BYTE* dataHash, int hashSize,
BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate non repudiation signature using provided hash value

1.1.123 GenerateNonRepudiationSignatureExA Function

Generate non repudiation signature

C++

```
BOOL WINAPI GenerateNonRepudiationSignatureExA(int readerNumber, LPSTR pinCode, BYTE*
dataHash, int hashSize, BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate non repudiation signature using provided hash value

1.1.124 GenerateNonRepudiationSignatureExW Function

Generate non repudiation signature

C++

```
BOOL WINAPI GenerateNonRepudiationSignatureExW(int readerNumber, LPWSTR pinCode, BYTE*
dataHash, int hashSize, BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate non repudiation signature using provided hash value

1.1.125 GenerateNonRepudiationSignatureW Function

Generate non repudiation signature

C++

```
BOOL WINAPI GenerateNonRepudiationSignatureW(LPWSTR pinCode, BYTE* dataHash, int hashSize,
BYTE* signature, LPDWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR pinCode	The card PIN code value
BYTE* dataHash	The hash value buffer
int hashSize	The size of the hash data buffer
BYTE* signature	The output buffer that contains the generated signature
LPDWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is successfully generated, otherwise returns FALSE

Description

Generate non repudiation signature using provided hash value

1.1.126 GeneratePNGA Function

Generates PNG file with QR Code image

C++

```
void WINAPI GeneratePNGA(LPSTR fileName, LPSTR text, int margin, int size, int level);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
LPSTR text	The text to encode
int margin	The margin from the border in points
int size	The size of the one point in pixels
int level	The error correction level

Description

Generates PNG file with encoded text as QR Code image

1.1.127 GeneratePNGW Function

Generates PNG file with QR Code image

C++

```
void WINAPI GeneratePNGW(LPWSTR fileName, LPWSTR text, int margin, int size, int level);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
LPWSTR text	The text to encode
int margin	The margin from the border in points
int size	The size of the one point in pixels
int level	The error correction level

Description

Generates PNG file with encoded text as QR Code image

1.1.128 GenerateQRCodeA Function

Read eID card and save the identity information and address to PNG QR Code file

C++

```
BOOL WINAPI GenerateQRCodeA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and generate the QR Code PNG image

1.1.129 GenerateQRCodeExA Function

Read eID card and save the identity information and address to PNG QR Code file

C++

```
BOOL WINAPI GenerateQRCodeExA(int readerNumber, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and generate the QR Code PNG image

1.1.130 GenerateQRCodeExW Function

Read eID card and save the identity information and address to PNG QR Code file

C++

```
BOOL WINAPI GenerateQRCodeExW(int readerNumber, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and generate the QR Code PNG image

1.1.131 GenerateQRCodeW Function

Read eID card and save the identity information and address to PNG QR Code file

C++

```
BOOL WINAPI GenerateQRCodeW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and generate the QR Code PNG image

1.1.132 GetAllFiles Function

Returns the names of files in a specified directory.

C++

```
void WINAPI GetAllFiles(FOLDERENUMPROC lpEnumProc, LPWSTR folderName, LPWSTR searchMask,
LPARAM lParam);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
FOLDERENUMPROC lpEnumProc	Callback function
LPWSTR folderName	The name of the folder
LPWSTR searchMask	The search string to match against the names of files in path.
LPARAM lParam	Specifies an application-defined value to be passed to the callback function

Description

This function enumerates all files in the specified folder which names match the searchMask parameter and calls the callback function passing the name of the file to it.

1.1.133 GetCardBufferA Function

Gets XML or CSV information from the memory buffer

C++

```
void WINAPI GetCardBufferA(void* buffer, void* strDest, int count);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
void* buffer	Memory buffer with information
void* strDest	Destination buffer
int count	Destination buffer size

Returns

None

Description

Use this function to get the card information in CSV or XML format from the memory buffer

1.1.134 GetCardBufferSize Function

This function returns the size of the buffer needed to hold the information from the eID card in the XML or CSV format

C++

```
int WINAPI GetCardBufferSize(void* buffer);
```

File

Swelio.h (see page 184)

Returns

The size of the XML or CSV buffer

Description

Use this function to get the size of the XML buffer before allocating the buffer in memory

1.1.135 GetCardBufferW Function

Gets XML or CSV information from the memory buffer

C++

```
void WINAPI GetCardBufferW(void* buffer, void* strDest, int count);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
void* buffer	Memory buffer with information
void* strDest	Destination buffer
int count	Destination buffer size

Returns

None

Description

Use this function to get the card information in CSV or XML format from the memory buffer

1.1.136 GetCardSerialNumber Function

Get the serial number of EID card

C++

```
BOOL WINAPI GetCardSerialNumber(int readerNumber, BYTE* serialNumber, LPDWORD
serialNumberSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
BYTE* serialNumber	The memory buffer for getting the card serial number
LPDWORD serialNumberSize	the size of the memory buffer in bytes

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to get the serial number of EID card into the memory buffer

1.1.137 GetCardVersion Function

Get the applet version number for card in the reader with specified number

C++

```
int WINAPI GetCardVersion(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The reader index, starting from 0

Returns

The card applet version number

Description

Get the version number of eid card applet using the zero-based reader index. The first reader has number 0, second - 1, etc... You can read the information only from one selected reader at once.

1.1.138 GetEncodedCertificateSize Function

Returns the size of the Base64 encoded certificate

C++

```
int WINAPI GetEncodedCertificateSize(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidCertificate certificate	The certificate data

Returns

Returns the size of the buffer needed to hold the encoded certificate

Description

Use this function to calculate the size of the buffer needed to encode the certificate

1.1.139 GetEncodedPhotoSize Function

Calculates buffer size for Base64 encoded photo

C++

```
int WINAPI GetEncodedPhotoSize(PeidPicture photo);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure

Returns

The desired size of the buffer

Description

Use this function to calculate the size of the buffer needed for Base64 encoding of the photo This can be useful for including the photo data to the text document, for example to XML file

1.1.140 GetFileMD5A Function

Gets the MD5 hash value for the file

C++

```
BOOL WINAPI GetFileMD5A(LPSTR fileName, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
BYTE* buffer	The buffer to store the hash value
int bufferSize	The size of the buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates MD5 hash value for the given file

1.1.141 GetFileMD5W Function

Gets the MD5 hash value for the file

C++

```
BOOL WINAPI GetFileMD5W(LPWSTR fileName, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
BYTE* buffer	The buffer to store the hash value
int bufferSize	The size of the buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates MD5 hash value for the given file

1.1.142 GetFilesCountA Function

Calculates the number of files in the given folder

C++

```
int WINAPI GetFilesCountA(LPSTR folderName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR folderName	The name of the folder

Returns

The number of files in the given folder

1.1.143 GetFilesCountW Function

Calculates the number of files in the given folder

C++

```
int WINAPI GetFilesCountW(LPWSTR folderName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR folderName	The name of the folder

Returns

The number of files in the given folder

1.1.144 GetFileSHA1A Function

Gets the SHA1 hash value for the file

C++

```
BOOL WINAPI GetFileSHA1A(LPSTR fileName, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
BYTE* buffer	The buffer to store the hash value
int bufferSize	The size of the buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates SHA1 hash value for the given file

1.1.145 GetFileSHA1W Function

Gets the SHA1 hash value for the file

C++

```
BOOL WINAPI GetFileSHA1W(LPWSTR fileName, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
BYTE* buffer	The buffer to store the hash value
int bufferSize	The size of the buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates SHA1 hash value for the given file

1.1.146 GetFileSHA256A Function

Gets the SHA256 hash value for the file

C++

```
BOOL WINAPI GetFileSHA256A(LPSTR fileName, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
BYTE* buffer	The buffer to store the hash value
int bufferSize	The size of the buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates SHA256 hash value for the given file

1.1.147 GetFileSHA256W Function

Gets the SHA256 hash value for the file

C++

```
BOOL WINAPI GetFileSHA256W(LPWSTR fileName, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
BYTE* buffer	The buffer to store the hash value
int bufferSize	The size of the buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates SHA256 hash value for the given file

1.1.148 GetHBitmapA Function

Generates Windows Bitmap in memory with QR Code image

C++

```
HBITMAP WINAPI GetHBitmapA(LPSTR text, int margin, int size, int level);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPSTR text	The text to encode
int margin	The margin from the border in points
int size	The size of the one point in pixels
int level	The error correction level

Returns

The result of the function is HBITMAP handle. You have to destroy it by yourself when its not needed anymore.

Description

Generates Windows Bitmap in memory file with encoded text as QR Code image

1.1.149 GetHBitmapW Function

Generates Windows Bitmap in memory with QR Code image

C++

```
HBITMAP WINAPI GetHBitmapW(LPWSTR text, int margin, int size, int level);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPWSTR text	The text to encode
int margin	The margin from the border in points
int size	The size of the one point in pixels
int level	The error correction level

Returns

The result of the function is HBITMAP handle. You have to destroy it by yourself when its not needed anymore.

Description

Generates Windows Bitmap in memory file with encoded text as QR Code image

1.1.150 GetISOCodeA Function

Returns the country ISO code based on the nationality string

C++

```
BOOL WINAPI GetISOCodeA(LPCSTR nationality, LPSTR iso, int bufferSize);
```

File

NationalityConverter.h (see page 184)

Parameters

Parameters	Description
LPCSTR nationality	The nationality string
LPSTR iso	The ISO code memory buffer
int bufferSize	The size if the memory buffer

Returns

Returns TRUE if the ISO code is successfully obtained; FALSE otherwise

Description

This function converts the nationality string stored on ID card to the country ISO code

1.1.151 GetISOCodeW Function

Returns the country ISO code based on the nationality string

C++

```
BOOL WINAPI GetISOCodeW(LPCWSTR nationality, LPWSTR iso, int bufferSize);
```

File

NationalityConverter.h (see page 184)

Parameters

Parameters	Description
LPCWSTR nationality	The nationality string
LPWSTR iso	The ISO code memory buffer
int bufferSize	The size if the memory buffer

Returns

Returns TRUE if the ISO code is successfully obtained; FALSE otherwise

Description

This function converts the nationality string stored on ID card to the country ISO code

1.1.152 GetMD5 Function

Gets the MD5 hash value for the content of the memory buffer

C++

```
BOOL WINAPI GetMD5(BYTE* source, int sourceSize, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
BYTE* source	The source memory block
int sourceSize	The size of the source memory block
BYTE* buffer	The buffer for the hash value
int bufferSize	The size of the destination buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates MD5 hash value for the given memory buffer

1.1.153 GetPNGA Function

Writes PNG image to the memory buffer.

C++

```
void WINAPI GetPNGA(LPSTR text, int margin, int size, int level, LPINT bufSize,
__deref_opt_out void ** ppvBits);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPSTR text	The text to encode
int margin	The margin from the image border in points
int size	The size of the point in pixels
int level	The error correction level
LPINT bufSize	The size of the output buffer
__deref_opt_out void ** ppvBits	The buffer when the resulting image is stored

Description

Writes PNG image to the memory buffer. Can be useful for web development.

1.1.154 GetPNGW Function

Writes PNG image to the memory buffer.

C++

```
void WINAPI GetPNGW(LPWSTR text, int margin, int size, int level, LPINT bufSize,
__deref_opt_out void ** ppvBits);
```

File

quricol.h (see page 184)

Parameters

Parameters	Description
LPWSTR text	The text to encode
int margin	The margin from the image border in points
int size	The size of the point in pixels
int level	The error correction level
LPINT bufSize	The size of the output buffer
__deref_opt_out void ** ppvBits	The buffer when the resulting image is stored

Description

Writes PNG image to the memory buffer. Can be useful for web development.

1.1.155 GetReaderIndexA Function

Returns the zero-based reader index with specified name

C++

```
int WINAPI GetReaderIndexA(LPSTR readerName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR readerName	The name of the reader

Returns

The zero-based reader index

Description

Use this function to get the zero-based index of the card reader with specified name

1.1.156 GetReaderIndexW Function

Returns the zero-based reader index with specified name

C++

```
int WINAPI GetReaderIndexW(LPWSTR readerName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR readerName	The name of the reader

Returns

The zero-based reader index

Description

Use this function to get the zero-based index of the card reader with specified name

1.1.157 GetReaderNameA Function

Returns the name of the card reader

C++

```
int WINAPI GetReaderNameA(int readerNumber, LPSTR strDest, int count);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
LPSTR strDest	Destination string
int count	The number of characters to be copied

Returns

Returns the reader name length

Description

Returns the name of the card reader with the specified zero-based index

1.1.158 GetReaderNameLenA Function

Returns the length of the reader name

C++

```
int WINAPI GetReaderNameLenA(int readerNumber);
```

File

Swelio.h (see page 184)

Returns

The length of the reader name

Description

Returns the length of the reader name for the smart card reader with specified zero-based index

1.1.159 GetReaderNameLenW Function

Returns the length of the reader name

C++

```
int WINAPI GetReaderNameLenW(int readerNumber);
```

File

Swelio.h (see page 184)

Returns

The length of the reader name

Description

Returns the length of the reader name for the smart card reader with specified zero-based index

1.1.160 GetReaderNameW Function

Returns the name of the card reader

C++

```
int WINAPI GetReaderNameW(int readerNumber, LPWSTR strDest, int count);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
LPWSTR strDest	Destination string
int count	Number of characters to be copied

Returns

The character count of the reader name

Description

Returns the name of the card reader with the specified zero-based index

1.1.161 GetReadersCount Function

Get number of card readers connected to PC

C++

```
int WINAPI GetReadersCount(VOID);
```

File

Swelio.h (see page 184)

Returns

The number of the connected smart card readers

Description

Checks how many smart card readers are connected to PC. If there is no readers connected then the usage of the Swelio Engine is not possible. The engine can control the change of the number of the card readers and can raise an event when the reader is connected or disconnected from PC

1.1.162 GetSelectedReaderIndex Function

Returns the index of the active smart card reader

C++

```
int WINAPI GetSelectedReaderIndex();
```

File

Swelio.h (see page 184)

Returns

The index of the selected card reader. The first reader has index 0.

Description

The zero-based index of the selected card reader. If there is only one reader is connected to PC then this reader has the index 0 and it's a default (selected) reader.

1.1.163 GetSHA1 Function

Gets the SHA1 hash value for the content of the memory buffer

C++

```
BOOL WINAPI GetSHA1(BYTE* source, int sourceSize, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
BYTE* source	The source memory block
int sourceSize	The size of the source memory block
BYTE* buffer	The buffer for the hash value
int bufferSize	The size of the destination buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates SHA1 hash value for the given memory buffer

1.1.164 GetSHA256 Function

Gets the SHA256 hash value for the content of the memory buffer

C++

```
BOOL WINAPI GetSHA256(BYTE* source, int sourceSize, BYTE* buffer, int bufferSize);
```

File

Encryption.h (see page 180)

Parameters

Parameters	Description
BYTE* source	The source memory block
int sourceSize	The size of the source memory block
BYTE* buffer	The buffer for the hash value
int bufferSize	The size of the destination buffer

Returns

The result of the function is equal to TRUE if operation is completed successfully, otherwise the result is FALSE

Description

Calculates SHA256 hash value for the given memory buffer

1.1.165 GetStartupA Function

Checks if the application is registered to run when Windows starts

C++

```
BOOL WINAPI GetStartupA(LPCSTR appName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCSTR appName	The name of the application

1.1.166 GetStartupW Function

Checks if the application is registered to run when Windows starts

C++

```
BOOL WINAPI GetStartupW(LPCWSTR appName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCWSTR appName	The name of the application

1.1.167 GetSupportSIS Function

Checks if the SIS cards are supported by the engine

C++

```
BOOL WINAPI GetSupportSIS();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE if SIS card support is activated, otherwise returns FALSE

Description

The SIS card reading operation takes more time than the reading of the eID card. By default when the card is inserted in the reader the engine will try to detect the card type and the card insertion event will be raised for eID cards only. If you want to support the SIS cards in your application then you have to activate it using SetSupportSIS (see page 124) function. Use GetSupportSIS function to check if the SIS card support is activated.

1.1.168 GetTextLineSize Function

This is function GetTextLineSize.

C++

```
SIZE WINAPI GetTextLineSize(LPCWSTR s, HFONT hFont);
```

File

Graphics.h (see page 183)

1.1.169 GetTextSize Function

This is function GetTextSize.

C++

```
SIZE WINAPI GetTextSize(LPCWSTR s, HFONT hFont, UINT flags);
```

File

Graphics.h (see page 183)

1.1.170 GetTextSizeEx Function

This is function GetTextSizeEx.

C++

```
SIZE WINAPI GetTextSizeEx(LPCWSTR s, HFONT hFont, UINT proposedWidth, UINT flags, BOOL
margins);
```

File

Graphics.h (see page 183)

1.1.171 HibernateWindows Function

Hibernates Windows

C++

```
BOOL WINAPI HibernateWindows();
```

File

System.h (see page 189)

1.1.172 IgnoreHardwareEvents Function

Ignore USB reader insert / remove events

C++

```
void WINAPI IgnoreHardwareEvents(BOOL value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
BOOL value	true - to ignore reader remove / insert events

Returns

None

1.1.173 IgnoreServiceEvents Function

Ignore smartcard service stop events when reporting readers list change

C++

```
void WINAPI IgnoreServiceEvents(BOOL value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
BOOL value	To ignore or not the service event

Returns

None

1.1.174 InitializeContainer Function

Initializes ASIC container

C++

```
LPVOID WINAPI InitializeContainer();
```

File

Swelio.h (see page 184)

Returns

Retrns container handle pointer

Description

This functions initializes container handle needed for all container operations. Must be called first prior to other container-related calls

1.1.175 IsAnimatedGIFA Function

Checks if the file is an animated GIF image file

C++

```
BOOL WINAPI IsAnimatedGIFA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file

Returns

Returns TRUE if the file is an animated GIF image file, otherwise returns FALSE.

1.1.176 IsAnimatedGIFW Function

Checks if the file is an animated GIF image file

C++

```
BOOL WINAPI IsAnimatedGIFW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file

Returns

Returns TRUE if the file is an animated GIF image file, otherwise returns FALSE.

1.1.177 IsCardActivated Function

Checks the connection between a smart card and a reader

C++

```
BOOL WINAPI IsCardActivated();
```

File

Swelio.h (see page 184)

Description

This function checks the connection between the calling application and a smart card in the target reader.

1.1.178 IsCardActivatedEx Function

Checks the connection between a smart card and a reader

C++

```
BOOL WINAPI IsCardActivatedEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader

Description

This function checks the connection between the calling application and a smart card in the target reader.

1.1.179 IsCardPresent Function

Checks if the card is present in the card reader

C++

```
BOOL WINAPI IsCardPresent();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE if the card is inserted in the reader, otherwise returns FALSE

Description

Use IsCardPresent function to check if the card is inserted in the card reader or not

1.1.180 IsCardPresentEx Function

Checks if the card is present in the card reader

C++

```
BOOL WINAPI IsCardPresentEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader

Returns

Returns TRUE if the card is inserted in the reader, otherwise returns FALSE

Description

Use isCardPresent function to check if the card is inserted in the card reader or not

1.1.181 IsCardStillInserted Function

Checks if the card is still inserted in the card reader

C++

```
BOOL WINAPI IsCardStillInserted();
```

File

Swelio.h (see page 184)

Description

This function checks if the card is still present in the card reader

1.1.182 IsCardStillInsertedEx Function

Checks if the card is still inserted in the card reader

C++

```
BOOL WINAPI IsCardStillInsertedEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader

Description

This function checks if the card is still present in the card reader

1.1.183 IsCitrixSession Function

Checks if application is running in Citrix session

C++

```
BOOL WINAPI IsCitrixSession();
```

File

SystemInfo.h (see page 190)

1.1.184 IsConnectedToInternet Function

Checks if PC is connected to Internet

C++

```
BOOL WINAPI IsConnectedToInternet();
```

File

SystemInfo.h (see page 190)

1.1.185 IsDirectoryA Function

Verifies that a path is a valid directory.

C++

```
BOOL WINAPI IsDirectoryA(LPSTR folderName);
```

File

FileOperations.h (see page 181)

Returns

Returns TRUE if the path is a valid directory, or FALSE otherwise.

Description

This function verifies if provided value is the name of the folder

1.1.186 IsDirectoryW Function

Verifies that a path is a valid directory.

C++

```
BOOL WINAPI IsDirectoryW(LPWSTR folderName);
```

File

FileOperations.h (see page 181)

Returns

Returns TRUE if the path is a valid directory, or FALSE otherwise.

Description

This function verifies if provided value is the name of the folder

1.1.187 IsEIDCard Function

Check if Belgian EID card is inserted into card reader

C++

```
BOOL WINAPI IsEIDCard();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE, if Belgian eID card is inserted in the reader. If there is no card in the reader or the card of other type is inserted, returns FALSE

Description

If the card is inserted in the reader, this function performs the card type check.

1.1.188 IsEIDCardEx Function

Check if Belgian EID card is inserted into card reader

C++

```
BOOL WINAPI IsEIDCardEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.

Returns

Returns TRUE, if Belgian eID card is inserted in the reader. If there is no card in the reader or the card of other type is inserted, returns FALSE

Description

If the card is inserted in the reader, this function performs the card type check.

1.1.189 IsEngineActive Function

Checks if the Swelio Engine is activated

C++

```
BOOL WINAPI IsEngineActive();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE if the Swelio Engine is active, otherwise returns FALSE.

Description

This function checks if the Engine already activated using the StartEngine (see page 126) function.

1.1.190 IsFemaleA Function

Checks if the card owner is female

C++

```
BOOL WINAPI IsFemaleA(PEidIdentityA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidIdentityA identity	The person identity information structure

Returns

Returns TRUE if the card owner is female, otherwise returns FALSE

Description

Use this function to check the gender of the card owner

1.1.191 IsFemaleW Function

Checks if the card owner is female

C++

```
BOOL WINAPI IsFemaleW(PEidIdentityW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidIdentityW identity	The person identity information structure

Returns

Returns TRUE if the card owner is female, otherwise returns FALSE

Description

Use this function to check the gender of the card owner

1.1.192 IsMaleA Function

Checks if the card owner is male

C++

```
BOOL WINAPI IsMaleA(PEidIdentityA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidIdentityA identity	The person identity information structure

Returns

Returns TRUE if the card owner is male, otherwise returns FALSE

Description

Use this function to check the gender of the card owner

1.1.193 IsMaleW Function

Checks if the card owner is male

C++

```
BOOL WINAPI IsMaleW(PEidIdentityW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidIdentityW identity	The person identity information structure

Returns

Returns TRUE if the card owner is male, otherwise returns FALSE

Description

Use this function to check the gender of the card owner

1.1.194 IsMediaCenter Function

Checks if the Media Center version of Windows is installed

C++

```
BOOL WINAPI IsMediaCenter();
```

File

SystemInfo.h (see page 190)

1.1.195 IsMetroActive Function

Checks if metro interface is active

C++

```
BOOL WINAPI IsMetroActive();
```

File

SystemInfo.h (see page 190)

1.1.196 IsMultiTouchReady Function

Checks if the system is multi touch ready

C++

```
BOOL WINAPI IsMultiTouchReady();
```

File

SystemInfo.h (see page 190)

1.1.197 IsNativeWin64 Function

Checks if the application is native 64 bit executable

C++

```
BOOL WINAPI IsNativeWin64();
```

File

SystemInfo.h (see page 190)

1.1.198 IsRemoteSession Function

Checks if application is running in RDP session

C++

```
BOOL WINAPI IsRemoteSession();
```

File

SystemInfo.h (see page 190)

1.1.199 IsSISCard Function

Check if Belgian SIS card is inserted into card reader

C++

```
BOOL WINAPI IsSISCard();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE, if Belgian SIS card is inserted in the reader. If there is no card in the reader or the card of other type is

inserted, returns FALSE

Description

If the card is inserted in the reader, this function performs the card type check.

1.1.200 IsSISCardEx Function

Check if Belgian SIS card is inserted into card reader

C++

```
BOOL WINAPI IsSISCardEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.

Returns

Returns TRUE, if Belgian SIS card is inserted in the reader. If there is no card in the reader or the card of other type is inserted, returns FALSE

Description

If the card is inserted in the reader, this function performs the card type check.

1.1.201 IsTabletPC Function

Checks if the application is running on the Tablet PC

C++

```
BOOL WINAPI IsTabletPC();
```

File

SystemInfo.h (see page 190)

1.1.202 IsUnicodeFileA Function

Checks if the file is UNICODE file

C++

```
BOOL WINAPI IsUnicodeFileA(LPCSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCSTR fileName	The name of the file

Returns

Returns TRUE if file is stored in UNICODE format, otherwise returns FALSE.

Description

This function checks the file encoding based on BOM (Byte Order Mark).

1.1.203 IsUnicodeFileW Function

Checks if the file is UNICODE file

C++

```
BOOL WINAPI IsUnicodeFileW(LPCWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The name of the file

Returns

Returns TRUE if file is stored in UNICODE format, otherwise returns FALSE.

Description

This function checks the file encoding based on BOM (Byte Order Mark).

1.1.204 IsValidFileNameA Function

Checks if provided string is a valid file name

C++

```
BOOL WINAPI IsValidFileNameA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file

Returns

Returns TRUE if provided string is valid file name, otherwise returns FALSE

Description

Checks if provided string is a valid file name and does not contain any illegal characters

1.1.205 IsValidFileNameW Function

Checks if provided string is a valid file name

C++

```
BOOL WINAPI IsValidFileNameW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file

Returns

Returns TRUE if provided string is valid file name, otherwise returns FALSE

Description

Checks if provided string is a valid file name and does not contain any illegal characters

1.1.206 IsValidPathNameA Function

Checks if provided string is a valid file path

C++

```
BOOL WINAPI IsValidPathNameA(LPSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The file path to check

Returns

Returns TRUE if provided string is valid file path, otherwise returns FALSE

Description

Checks if provided string is a valid file path and does not contain any illegal characters

1.1.207 IsValidPathNameW Function

Checks if provided string is a valid file path

C++

```
BOOL WINAPI IsValidPathNameW(LPWSTR fileName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The file path to check

Returns

Returns TRUE if provided string is valid file path, otherwise returns FALSE

Description

Checks if provided string is a valid file path and does not contain any illegal characters

1.1.208 IsWindows10 Function

Checks if PC is running Windows 10 or better

C++

```
BOOL WINAPI IsWindows10();
```

File

SystemInfo.h (see page 190)

1.1.209 IsWindows7 Function

Checks if PC is running Windows 7 or better

C++

```
BOOL WINAPI IsWindows7();
```

File

SystemInfo.h (see page 190)

1.1.210 IsWindows8 Function

Checks if PC is Running Windows 8 or better

C++

```
BOOL WINAPI IsWindows8();
```

File

SystemInfo.h (see page 190)

1.1.211 IsWindowsVista Function

Checks if PC is running Windows Vista or better

C++

```
BOOL WINAPI IsWindowsVista();
```

File

SystemInfo.h (see page 190)

1.1.212 IsWindowsXP Function

Checks if PC is running Windows XP

C++

```
BOOL WINAPI IsWindowsXP();
```

File

SystemInfo.h (see page 190)

1.1.213 IsWindowsXPSP2 Function

Checks if PC is running Windows XP with Service Pack 2 installed

C++

```
BOOL WINAPI IsWindowsXPSP2();
```

File

SystemInfo.h (see page 190)

1.1.214 IsWow64 Function

Checks if the 32 bit application runs on 64 bit Windows

C++

```
BOOL WINAPI IsWow64();
```

File

SystemInfo.h (see page 190)

1.1.215 LayeredWndProcA Function

The default window procedure for the layered window

C++

```
LRESULT CALLBACK LayeredWndProcA(HWND hWnd, UINT message, WPARAM wParam, LPARAM lParam);
```

File

System.h (see page 189)

1.1.216 LayeredWndProcW Function

The default window procedure for the layered window

C++

```
LRESULT CALLBACK LayeredWndProcW(HWND hWnd, UINT message, WPARAM wParam, LPARAM lParam);
```

File

System.h (see page 189)

1.1.217 LoadBitmapJPG Function

This is function LoadBitmapJPG.

C++

```
HBITMAP WINAPI LoadBitmapJPG(LPCWSTR szFile, LPINT lpiWidth, LPINT lpiHeight,
__deref_opt_out void ** ppvBits);
```

File

Graphics.h (see page 183)

1.1.218 LoadBitmapPNG Function

This is function LoadBitmapPNG.

C++

```
HBITMAP WINAPI LoadBitmapPNG(LPCWSTR szFile, LPINT lpiWidth, LPINT lpiHeight,
__deref_opt_out void ** ppvBits);
```

File

Graphics.h (see page 183)

1.1.219 LoadCertificateA Function

Reads the certificate from a file

C++

```
void WINAPI LoadCertificateA(LPSTR fileName, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	The source file name
PEidCertificate certificate	The pointer to EidCertificate (see page 143) structure

Description

Use this function to read the certificate from the file

1.1.220 LoadCertificateW Function

Reads the certificate from a file

C++

```
void WINAPI LoadCertificateW(LPWSTR fileName, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	The source file name
PEidCertificate certificate	The pointer to EidCertificate (see page 143) structure

Description

Use this function to read the certificate from the file

1.1.221 LoadIdentityA Function

Reads the raw identity information from a file

C++

```
void WINAPI LoadIdentityA(LPSTR fileName, PEidIdentityA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	The name of the source file
PEidIdentityA identity	The pointer to EidIdentityA (see page 143) structure

Description

Use this function to read back the identity information stored to the file using SaveIdentityA (see page 111) function

1.1.222 LoadIdentityW Function

Reads the raw identity information from a file

C++

```
void WINAPI LoadIdentityW(LPWSTR fileName, PEidIdentityW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	The name of the source file

PEidIdentityW identity	The pointer to EidIdentityW (see page 148) structure

Description

Use this function to read back the identity information stored to the file using SaveIdentityW (see page 111) function

1.1.223 LoadPhotoA Function

Loads photo from a file

C++

```
void WINAPI LoadPhotoA(PeidPicture photo, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure
LPSTR fileName	Destination file name

Description

Loads raw picture data from a file

1.1.224 LoadPhotoW Function

Loads photo from a file

C++

```
void WINAPI LoadPhotoW(PeidPicture photo, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure
LPWSTR fileName	Destination file name

Description

Loads raw picture data from a file

1.1.225 LoadPNGResource Function

This is function LoadPNGResource.

C++

```
HBITMAP WINAPI LoadPNGResource(HMODULE handle, LPCWSTR szName, LPINT lpiWidth, LPINT
lpiHeight, __deref_opt_out void ** ppvBits);
```

File

Graphics.h (see page 183)

1.1.226 MakeCompatibleBitmap Function

This is function MakeCompatibleBitmap.

C++

```
HBITMAP WINAPI MakeCompatibleBitmap(HBITMAP source, int width, int height);
```

File

Graphics.h (see page 183)

1.1.227 MakeSoundFromFileA Function

Plays the wave sound from the file

C++

```
void WINAPI MakeSoundFromFileA(LPCSTR soundName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCSTR soundName	The name of the file

Description

This function plays a sound specified by the given file name.

1.1.228 MakeSoundFromFileW Function

Plays the wave sound from the file

C++

```
void WINAPI MakeSoundFromFileW(LPCWSTR soundName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCWSTR soundName	The name of the file

Description

This function plays a sound specified by the given file name.

1.1.229 MakeSoundFromResourceA Function

Plays the wave sound from the resource

C++

```
void WINAPI MakeSoundFromResourceA(HMODULE hModule, LPCSTR soundName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HMODULE hModule	Handle to the executable file that contains the resource to be loaded.
LPCSTR soundName	A string that specifies the sound to play.

Description

This function plays a sound specified by the given resource name.

1.1.230 MakeSoundFromResourceW Function

Plays the wave sound from the resource

C++

```
void WINAPI MakeSoundFromResourceW(HMODULE hModule, LPCWSTR soundName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HMODULE hModule	Handle to the executable file that contains the resource to be loaded.
LPCWSTR soundName	A string that specifies the sound to play.

Description

This function plays a sound specified by the given resource name.

1.1.231 PointsToPixels Function

This is function PointsToPixels.

C++

```
int WINAPI PointsToPixels(int points);
```

File

Graphics.h (see page 183)

1.1.232 PortAvailable Function

Checks if the port with specified number is available

C++

```
BOOL WINAPI PortAvailable(int port);
```

File

SystemInfo.h (see page 190)

1.1.233 ReadAddressA Function

Read address information from Belgian eID card

C++

```
BOOL WINAPI ReadAddressA(PEidAddressA address);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidAddressA address	The pointer to the address information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.234 ReadAddressExA Function

Read address information from Belgian eID card

C++

```
BOOL WINAPI ReadAddressExA(int readerNumber, PEidAddressA address);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PEidAddressA address	The pointer to the address information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.235 ReadAddressExW Function

Read address information from Belgian eID card

C++

```
BOOL WINAPI ReadAddressExW(int readerNumber, PEidAddressW address);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PEidAddressW address	The pointer to the address information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.236 ReadAddressW Function

Read address information from Belgian eID card

C++

```
BOOL WINAPI ReadAddressW(PEidAddressW address);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidAddressW address	the pointer to the address information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.237 ReadAuthenticationCertificate Function

Read Authentication Certificate to memory

C++

```
BOOL WINAPI ReadAuthenticationCertificate(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Authentication Certificate from the card to EidCertificate (see page 143) structure

1.1.238 ReadAuthenticationCertificateEx Function

Read Authentication Certificate to memory

C++

```
BOOL WINAPI ReadAuthenticationCertificateEx(int readerNumber, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Authentication Certificate from the card to EidCertificate (see page 143) structure

1.1.239 ReadBufferFromFileA Function

Reads the content of the file to the memory buffer

C++

```
void WINAPI ReadBufferFromFileA(LPSTR fileName, BYTE* buffer, int bufferSize);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
BYTE* buffer	The address of the memory block
int bufferSize	The size of the memory block

Description

Use this function to retrieve the content of the file to the memory block

1.1.240 ReadBufferFromFileW Function

Reads the content of the file to the memory buffer

C++

```
void WINAPI ReadBufferFromFileW(LPWSTR fileName, BYTE* buffer, int bufferSize);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
BYTE* buffer	The address of the memory block
int bufferSize	The size of the memory block

Description

Use this function to retrieve the content of the file to the memory block

1.1.241 ReadCaCertificate Function

Read Ca Certificate to memory

C++

```
BOOL WINAPI ReadCaCertificate(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Ca Certificate to EidCertificate (see page 143) structure

1.1.242 ReadCaCertificateEx Function

Read Ca Certificate to memory

C++

```
BOOL WINAPI ReadCaCertificateEx(int readerNumber, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Ca Certificate to EidCertificate (see page 143) structure

1.1.243 ReadIdentityA Function

Read identity information from Belgian eID card

C++

```
BOOL WINAPI ReadIdentityA(PEidIdentityA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidIdentityA identity	The pointer to the identity information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.244 ReadIdentityExA Function

Read identity information from Belgian eID card

C++

```
BOOL WINAPI ReadIdentityExA(int readerNumber, PEidIdentityA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PEidIdentityA identity	The pointer to the identity information structure

Returns

Returns TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.245 ReadIdentityExW Function

Read identity information from Belgian eID card

C++

```
BOOL WINAPI ReadIdentityExW(int readerNumber, PEidIdentityW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PEidIdentityW identity	The pointer to the identity information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.246 ReadIdentityW Function

Read identity information from Belgian eID card

C++

```
BOOL WINAPI ReadIdentityW(PEidIdentityW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidIdentityW identity	The pointer to the identity information structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

1.1.247 ReadNonRepudiationCertificate Function

Read Non Repudiation Certificate to memory

C++

```
BOOL WINAPI ReadNonRepudiationCertificate(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Non Repudiation Certificate to EidCertificate (see page 143) structure

1.1.248 ReadNonRepudiationCertificateEx Function

Read Non Repudiation Certificate to memory

C++

```
BOOL WINAPI ReadNonRepudiationCertificateEx(int readerNumber, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Non Repudiation Certificate to EidCertificate (see page 143) structure

1.1.249 ReadPhoto Function

Reads a photo from a card

C++

```
BOOL WINAPI ReadPhoto(PeidPicture photo);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

Description

Reads a photo from Belgian eID card to EidPicture (see page 149) structure. This structure holds the raw image bytes and the length of the image bytes array

1.1.250 ReadPhotoAsBitmap Function

Reads the picture from the card, converts it to bitmap and returns the bitmap handle Decription: Reads the photo from the Belgian eID card and returns the bitmap handle Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
HBITMAP WINAPI ReadPhotoAsBitmap();
```

File

Swelio.h (see page 184)

Returns

A handle to a bitmap indicates success. NULL indicates failure.

1.1.251 ReadPhotoAsBitmapEx Function

Reads the picture from the card, converts it to bitmap and returns the bitmap handle Decription: Reads the photo from the Belgian eID card and returns the Windows bitmap handle Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
HBITMAP WINAPI ReadPhotoAsBitmapEx(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.

Returns

A handle to a bitmap indicates success. NULL indicates failure.

1.1.252 ReadPhotoEx Function

Reads a photo from a card

C++

```
BOOL WINAPI ReadPhotoEx(int readerNumber, PeidPicture photo);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PeidPicture photo	The pointer to EidPicture (see page 149) structure

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

Description

Reads a photo from Belgian eID card to EidPicture (see page 149) structure

1.1.253 ReadRootCaCertificate Function

Read Root Ca Certificate to memory

C++

```
BOOL WINAPI ReadRootCaCertificate(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Root Ca Certificate to EidCertificate (see page 143) structure

1.1.254 ReadRootCaCertificateEx Function

Read Root Ca Certificate to memory

C++

```
BOOL WINAPI ReadRootCaCertificateEx(int readerNumber, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Root Ca Certificate to EidCertificate (see page 143) structure

1.1.255 ReadRrnCertificate Function

Read Rrn Certificate to memory

C++

```
BOOL WINAPI ReadRrnCertificate(PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Rrn Certificate to EidCertificate (see page 143) structure

1.1.256 ReadRrnCertificateEx Function

Read Rrn Certificate to memory

C++

```
BOOL WINAPI ReadRrnCertificateEx(int readerNumber, PEidCertificate certificate);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader
cerificate	The pointer to EidCertificate (see page 143) structure

Returns

TRUE when certificate is successfully received from the card; otherwise returns FALSE

Description

Read Rrn Certificate to EidCertificate (see page 143) structure

1.1.257 ReadSISCardA Function

Read Belgian SIS card.

C++

```
BOOL WINAPI ReadSISCardA(PSISRecordA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PSISRecordA	The pointer to SISRecordA (see page 159) structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

Description

Read the public information from the Belgian SIS card. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

1.1.258 ReadSISCardExA Function

Read Belgian SIS card.

C++

```
BOOL WINAPI ReadSISCardExA(int readerNumber, PSISRecordA identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PSISRecordA	The pointer to SISRecordA (see page 159) structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

Description

Read the public information from the Belgian SIS card. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

1.1.259 ReadSISCardExW Function

Read Belgian SIS card.

C++

```
BOOL WINAPI ReadSISCardExW(int readerNumber, PSISRecordW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
PSISRecordW	The pointer to SISRecordW (see page 160) structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

Description

Read the public information from the Belgian SIS card. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

1.1.260 ReadSISCardW Function

Read Belgian SIS card.

C++

```
BOOL WINAPI ReadSISCardW(PSISRecordW identity);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PSISRecordW	The pointer to SISRecordW (see page 160) structure

Returns

TRUE when information is successfully received from the card; otherwise returns FALSE

Description

Read the public information from the Belgian SIS card. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

1.1.261 RecycleBinEmpty Function

Returns TRUE if Windows Recyce Bin is empty

C++

```
BOOL WINAPI RecycleBinEmpty();
```

File

SystemInfo.h (see page 190)

1.1.262 ReloadReadersList Function

Reloads the list of the available card readers

C++

```
void WINAPI ReloadReadersList();
```

File

Swelio.h (see page 184)

Description

When the card reader is inserted or removed you may need to reload the list of the available card readers

1.1.263 RemoveCallback Function

Remove callback procedure for card events

C++

```
void WINAPI RemoveCallback();
```

File

Swelio.h (see page 184)

Description

Use this function to deactivate card events callback procedure

1.1.264 RemoveStartupA Function

Removes the application from the list of the automatically started applications

C++

```
void WINAPI RemoveStartupA(LPCSTR appName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCSTR appName	The name of the application

Description

For application that starts automatically when Windows starts removes it from the automatically launching applications list

1.1.265 RemoveStartupW Function

Removes the application from the list of the automatically started applications

C++

```
void WINAPI RemoveStartupW(LPCWSTR appName);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCWSTR appName	The name of the application

Description

For application that starts automatically when Windows starts removes it from the automatically launching applications list

1.1.266 RestoreWindowSubclassA Function

Restores window standard procedure

C++

```
void WINAPI RestoreWindowSubclassA(HWND hwnd);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND hwnd	The window handle

1.1.267 RestoreWindowSubclassW Function

Restores window standard procedure

C++

```
void WINAPI RestoreWindowSubclassW(HWND hwnd);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND hwnd	The window handle

1.1.268 SaveAuthenticationCertificateA Function

Save Authentication Certificate to a file

C++

```
void WINAPI SaveAuthenticationCertificateA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the certificate

Description

Read Authentication Certificate from the card and save it to a file.

1.1.269 SaveAuthenticationCertificateExW Function

Save Authentication Certificate to a file

C++

```
void WINAPI SaveAuthenticationCertificateExW(LPWSTR fileName, int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate
int readerNumber	The reader index, starting from 0

Description

Read Authentication Certificate from the card and save it to a file.

1.1.270 SaveAuthenticationCertificateW Function

Save Authentication Certificate to a file

C++

```
void WINAPI SaveAuthenticationCertificateW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate

Description

Read Authentication Certificate from the card and save it to a file.

1.1.271 SaveCaCertificateA Function

Save Ca Certificate to a file

C++

```
void WINAPI SaveCaCertificateA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the certificate

Description

Read Ca Certificate from the card and save it to a file

1.1.272 SaveCaCertificateExW Function

Save Ca Certificate to a file

C++

```
void WINAPI SaveCaCertificateExW(LPWSTR fileName, int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate
int readerNumber	The reader index, starting from 0

Description

Read Ca Certificate from the card and save it to a file

1.1.273 SaveCaCertificateW Function

Save Ca Certificate to a file

C++

```
void WINAPI SaveCaCertificateW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate

Description

Read Ca Certificate from the card and save it to a file

1.1.274 SaveCardToToXMLStreamExA Function

Read eID card and save the information to XML buffer

C++

```
BOOL WINAPI SaveCardToToXMLStreamExA(int readerNumber, void* buffer);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
void* buffer	The memory buffer to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to XML buffer in the memory.

1.1.275 SaveCardToToXMLStreamExW Function

Read eID card and save the information to XML buffer

C++

```
BOOL WINAPI SaveCardToToXMLStreamExW(int readerNumber, void* buffer);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
void* buffer	The memory buffer to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to XML buffer in the memory.

1.1.276 SaveCardToXmlA Function

Read eID card and save the information to XML file

C++

```
BOOL WINAPI SaveCardToXmlA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to XML file.

1.1.277 SaveCardToXmlExA Function

Read eID card and save the information to XML file

C++

```
BOOL WINAPI SaveCardToXmlExA(int readerNumber, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to XML file.

1.1.278 SaveCardToXmlExW Function

Read eID card and save the information to XML file

C++

```
BOOL WINAPI SaveCardToXmlExW(int readerNumber, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to XML file.

1.1.279 SaveCardToXmlW Function

Read eID card and save the information to XML file

C++

```
BOOL WINAPI SaveCardToXmlW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to XML file.

1.1.280 SaveContainer Function

Save container to the file

C++

```
BOOL WINAPI SaveContainer(LPVOID container, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPVOID container	Container handle, which is allocated by calling InitializeContainer (see page 75) function
LPWSTR fileName	Desired name of the container file

Returns

Returns true if the operation is successful, otherwise returns false

Description

After adding all necessary files to the container call this function to save container to the file

1.1.281 SaveIdentityA Function

Saves indentity information to a file

C++

```
void WINAPI SaveIdentityA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	The name of the destination file

Description

Use this function to store the raw identity information from the Belgian eID card to a file. You can use LoadIdentityA (see page 89) to read this information from the file to EidIdentityA (see page 143) structure

1.1.282 SaveIdentityW Function

Saves indentity information to a file

C++

```
void WINAPI SaveIdentityW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	The name of the destination file

Description

Use this function to store the raw identity information from the Belgian eID card to a file. You can use LoadIdentityW (see page 89) to read this information from the file to EidIdentityW (see page 148) structure

1.1.283 SaveNonRepudiationCertificateA Function

Save Non Repudiation Certificate to a file

C++

```
void WINAPI SaveNonRepudiationCertificateA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the certificate

Description

Read Non Repudiation Certificate from the card and save it to a file

1.1.284 SaveNonRepudiationCertificateExW Function

Save Non Repudiation Certificate to a file

C++

```
void WINAPI SaveNonRepudiationCertificateExW(LPWSTR fileName, int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate
int readerNumber	The reader index, starting from 0

Description

Read Non Repudiation Certificate from the card and save it to a file

1.1.285 SaveNonRepudiationCertificateW Function

Save Non Repudiation Certificate to a file

C++

```
void WINAPI SaveNonRepudiationCertificateW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate

Description

Read Non Repudiation Certificate from the card and save it to a file

1.1.286 SavePersonCsvToStreamA Function

Read eID card and save the identity information to CSV memory buffer

C++

```
BOOL WINAPI SavePersonCsvToStreamA(int readerNumber, void* buffer);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	the card reader index
void* buffer	Memory buffer

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to CSV memory buffer.

1.1.287 SavePersonCsvToStreamW Function

Read eID card and save the identity information to CSV memory buffer

C++

```
BOOL WINAPI SavePersonCsvToStreamW(int readerNumber, void* buffer);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	the card reader index
void* buffer	Memory buffer

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to CSV memory buffer.

1.1.288 SavePersonToCsvA Function

Read eID card and save the identity information and address to CSV file

C++

```
BOOL WINAPI SavePersonToCsvA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to CSV file.

1.1.289 SavePersonToCsvExA Function

Read eID card and save the identity information and address to CSV file

C++

```
BOOL WINAPI SavePersonToCsvExA(int readerNumber, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to CSV file.

1.1.290 SavePersonToCsvExW Function

Read eID card and save the identity information and address to CSV file

C++

```
BOOL WINAPI SavePersonToCsvExW(int readerNumber, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to CSV file.

1.1.291 SavePersonToCsvW Function

Read eID card and save the identity information and address to CSV file

C++

```
BOOL WINAPI SavePersonToCsvW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store information

Returns

Returns TRUE if the information is retrieved from the card, otherwise returns FALSE

Description

Use this function to read the information about the owner of the card and save it to CSV file.

1.1.292 SavePhotoA Function

Save photo to a file

C++

```
void WINAPI SavePhotoA(PeidPicture photo, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure
LPSTR fileName	Destination file name

Description

Save the raw picture data to a file

1.1.293 SavePhotoAsBitmapA Function

Save the picture from the card to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsBitmapA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.294 SavePhotoAsBitmapExA Function

Reads the picture from the card and saves it to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsBitmapExA(int readerNumber, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.295 SavePhotoAsBitmapExW Function

Reads the picture from the card and saves it to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsBitmapExW(int readerNumber, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	the zero-based index of the card reader.
LPWSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.296 SavePhotoAsBitmapW Function

Save the picture from the card to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsBitmapW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.297 SavePhotoAsJpegA Function

Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsJpegA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.298 SavePhotoAsJpegExA Function

Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsJpegExA(int readerNumber, LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	the zero-based index of the card reader.
LPSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.299 SavePhotoAsJpegExW Function

Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsJpegExW(int readerNumber, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.300 SavePhotoAsJpegW Function

Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.

C++

```
BOOL WINAPI SavePhotoAsJpegW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the photo

Returns

Returns TRUE if the photo is retrieved from the card, otherwise return FALSE

1.1.301 SavePhotoW Function

Saves photo to a file

C++

```
void WINAPI SavePhotoW(PeidPicture photo, LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PeidPicture photo	The pointer to EidPicture (see page 149) structure
LPWSTR fileName	Destination file name

Description

Saves the raw picture data to a file

1.1.302 SaveRootCaCertificateA Function

Save Root Ca Certificate to a file

C++

```
void WINAPI SaveRootCaCertificateA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the certificate

Description

Read Root CA certificate from the card and save it to a file

1.1.303 SaveRootCaCertificateExW Function

Save Root Ca Certificate to a file

C++

```
void WINAPI SaveRootCaCertificateExW(LPWSTR fileName, int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate
int readerNumber	The reader index, starting from 0

Description

Read Root CA certificate from the card and save it to a file

1.1.304 SaveRootCaCertificateW Function

Save Root Ca Certificate to a file

C++

```
void WINAPI SaveRootCaCertificateW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate

Description

Read Root CA certificate from the card and save it to a file

1.1.305 SaveRrnCertificateA Function

Save RRN Certificate to a file

C++

```
void WINAPI SaveRrnCertificateA(LPSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR fileName	File name to store the certificate

Description

Read RRN certificate from the card and save it to a file

1.1.306 SaveRrnCertificateExW Function

Save RRN Certificate to a file

C++

```
void WINAPI SaveRrnCertificateExW(LPWSTR fileName, int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate
int readerNumber	The reader index, starting from 0

Description

Read RRN certificate from the card and save it to a file

1.1.307 SaveRrnCertificateW Function

Save RRN Certificate to a file

C++

```
void WINAPI SaveRrnCertificateW(LPWSTR fileName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR fileName	File name to store the certificate

Description

Read RRN certificate from the card and save it to a file

1.1.308 SelectReader Function

When more than 1 reader connected, select the reader with specified number

C++

```
BOOL WINAPI SelectReader(int readerNumber);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The reader index, starting from 0

Returns

TRUE if the reader is selected, FALSE if the reader with specified number does not exist

Description

Selects the default card reader using the zero-based reader index. The first reader has number 0, second - 1, etc... You can read the information only from one selected reader at once.

1.1.309 SelectReaderByNameA Function

Select active smart card reader by providing the reader name

C++

```
BOOL WINAPI SelectReaderByNameA(LPSTR readerName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR readerName	The name of the card reader

Returns

TRUE if the reader is selected. If the reader with specified name is not found - returns FALSE

Description

Activates the reader with specified name

1.1.310 SelectReaderByNameW Function

Select active smart card reader by providing the reader name

C++

```
BOOL WINAPI SelectReaderByNameW(LPWSTR readerName);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR readerName	The name of the card reader

Returns

Returns TRUE if the reader is selected. If the reader with specified name is not found - returns FALSE

Description

Activates the reader with specified name

1.1.311 SendAPDU Function

This is function SendAPDU.

C++

```
BOOL WINAPI SendAPDU(int readerNumber, LPCBYTE apdu, DWORD apduLen, PCHAR result, LPDWORD
len);
```

File

Swelio.h (see page 184)

1.1.312 SetCallback Function

Activates callback procedure for card status change event

C++

```
void WINAPI SetCallback(CALLBACK_HANDLER callback, LPVOID userContext);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
CALLBACK_HANDLER callback	The pointer to callback procedure
LPVOID userContext	The user defined value passed to the callback procedure

Description

Your application can be notified about insertion or removal of the card from the card reader and the changes of the available card readers list (the reader is connected or disconnected from PC) Use this function to install the callback procedure

1.1.313 SetMWCompatibility Function

Set the compatibility mode with the old version of the oficial EID MiddleWare

C++

```
void WINAPI SetMWCompatibility();
```

File

Swelio.h (see page 184)

Description

The compatibility mode can be useful when the MiddleWare version 1.x or 2.x is installed on the target PC. Usually the more recent MiddleWare is used and this function is provided for backward compatibility only

1.1.314 SetStartupA Function

Register application to run when Windows starts

C++

```
void WINAPI SetStartupA(LPCSTR appName, LPCWSTR appPath);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCSTR appName	The name of the application
LPCWSTR appPath	The path to the application executable

1.1.315 SetStartupW Function

Register application to run when Windows starts

C++

```
void WINAPI SetStartupW(LPCWSTR appName, LPCWSTR appPath);
```

File

System.h (see page 189)

Parameters

Parameters	Description
LPCWSTR appName	The name of the application
LPCWSTR appPath	The path to the application executable

1.1.316 SetSupportSIS Function

Activates or deactivates SIS card support by engine

C++

```
void WINAPI SetSupportSIS(BOOL value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
BOOL value	The SIS card support status

Description

Use SetSupportSIS to activate or deactivate the SIS card detection and reading. Even if SIS card support is activated it can be used only with ACR38U card readers Other card readers are not supported.

1.1.317 ShellCopyFileA Function

Copies file to the new location

C++

```
void WINAPI ShellCopyFileA(LPSTR oldName, LPSTR newName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR oldName	The source file name
LPSTR newName	The destination file name

Description

Copies file to the new location using Windows shell copy routine

1.1.318 ShellCopyFileW Function

Copies file to the new location

C++

```
void WINAPI ShellCopyFileW(LPWSTR oldName, LPWSTR newName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR oldName	The source file name
LPWSTR newName	The destination file name

Description

Copies file to the new location using Windows shell copy routine

1.1.319 ShutdownWindows Function

Logs off the interactive user, shuts down the system.

C++

```
BOOL WINAPI ShutdownWindows(UINT flags);
```

File

System.h (see page 189)

Returns

If the function succeeds returns TRUE, otherwise returns FALSE

Description

Logs off the interactive user, shuts down the system, or shuts down and restarts the system. It sends the WM_QUERYENDSESSION message to all applications to determine if they can be terminated.

This function accepts the following parameter:

flags : The shutdown type. This parameter must include one of the following values:

Value	Meaning
EWX_LOGOFF	Shuts down all processes running in the logon session of the process that called the ExitWindowsEx function. Then it logs the user off.
EWX_POWEROFF	Shuts down the system and turns off the power. The system must support the power-off feature.
EWX_REBOOT	Shuts down the system and then restarts the system.
EWX_RESTARTAPPS	Shuts down the system and then restarts it
EWX_SHUTDOWN	Shuts down the system to a point at which it is safe to turn off the power.

1.1.320 StartEngine Function

Activates the Swelio Engine.

C++

```
BOOL WINAPI StartEngine();
```

File

Swelio.h (see page 184)

Returns

Returns TRUE if the Swelio Engine is successfully started; otherwize returns FALSE

Description

This procedure must be called first before any other functions from Swelio library can be used.

1.1.321 StopEngine Function

Deactivates the Swelio Engine

C++

```
void WINAPI StopEngine();
```

File

Swelio.h (see page 184)

Description

Deactivates the Swelio Engine and clean up the used memory. Call this procedure at the end of you application once to finalize the usage of the Swelio Engine.

1.1.322 StretchNativeBitmap Function

This is function StretchNativeBitmap.

C++

```
BOOL WINAPI StretchNativeBitmap(HBITMAP src, HBITMAP dst, int srcWidth, int srcHeight, int
dstWidth, int dstHeight);
```

File

Graphics.h (see page 183)

1.1.323 StripFileNameA Function

Replaces environment variable names with values

C++

```
void WINAPI StripFileNameA(LPCSTR fileName, LPSTR fullName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCSTR fileName	The source file name
LPSTR fullName	The expanded file name

Description

This function expands environment-variable strings and replaces them with their defined values in the file name.

1.1.324 StripFileNameW Function

Replaces environment variable names with values

C++

```
void WINAPI StripFileNameW(LPCWSTR fileName, LPWSTR fullName);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPCWSTR fileName	The source file name
LPWSTR fullName	The expanded file name

Description

This function expands environment-variable strings and replaces them with their defined values in the file name.

1.1.325 SuspendWindows Function

Suspends Windows

C++

```
BOOL WINAPI SuspendWindows();
```

File

System.h (see page 189)

1.1.326 TurnMonitorOff Function

Turns the monitor off

C++

```
void WINAPI TurnMonitorOff();
```

File

System.h (see page 189)

1.1.327 TurnMonitorOn Function

Turns the monitor on

C++

```
void WINAPI TurnMonitorOn();
```

File

System.h (see page 189)

1.1.328 UpdateWindowPosition Function

Updated the window position

C++

```
void WINAPI UpdateWindowPosition(HWND handle, int x, int y);
```

File

System.h (see page 189)

Parameters

Parameters	Description
HWND handle	The handle of the window
int x	New horizontal coordinate
int y	New vertical coordinate

1.1.329 VerifyPinA Function

Verify PIN code

C++

```
BOOL WINAPI VerifyPinA(LPSTR value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPSTR value	PIN code to verify

Returns

TRUE when the correct PIN code is provided; otherwise returns FALSE

1.1.330 VerifyPinExA Function

Verify PIN code

C++

```
BOOL WINAPI VerifyPinExA(int readerNumber, LPSTR value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPSTR value	PIN code to verify

Returns

TRUE when the correct PIN code is provided; otherwise returns FALSE

1.1.331 VerifyPinExW Function

Verify PIN code

C++

```
BOOL WINAPI VerifyPinExW(int readerNumber, LPWSTR value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
int readerNumber	The zero-based index of the card reader.
LPWSTR value	PIN code to verify

Returns

TRUE when the correct PIN code is provided; otherwise returns FALSE

1.1.332 VerifyPinW Function

Verify PIN code

C++

```
BOOL WINAPI VerifyPinW(LPWSTR value);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
LPWSTR value	PIN code to verify

Returns

TRUE when the correct PIN code is provided; otherwise returns FALSE

1.1.333 VerifySignature Function

Verifies the signature from the specified hash value.

C++

```
BOOL WINAPI VerifySignature(PEidCertificate certificate, BYTE* buffer, int bufferSize,
BYTE* signature, DWORD signatureSize);
```

File

Swelio.h (see page 184)

Parameters

Parameters	Description
PEidCertificate certificate	The public certificate
BYTE* buffer	The hash buffer
int bufferSize	The size of the hash buffer
BYTE* signature	The signature to be verified.
DWORD signatureSize	The size of the signature buffer

Returns

Returns TRUE if the signature is valid for the hash; otherwise, FALSE.

Description

Verify the signature using the public certificate of the signer

1.1.334 WriteBufferToFileA Function

Writes the memory buffer to file

C++

```
void WINAPI WriteBufferToFileA(LPSTR fileName, BYTE* buffer, int bufferSize);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPSTR fileName	The name of the file
BYTE* buffer	The address of the memory block
int bufferSize	The size of the memory block

Description

This function stores the content of the memory buffer to the file.

1.1.335 WriteBufferToFileW Function

Writes the memory buffer to file

C++

```
void WINAPI WriteBufferToFileW(LPWSTR fileName, BYTE* buffer, int bufferSize);
```

File

FileOperations.h (see page 181)

Parameters

Parameters	Description
LPWSTR fileName	The name of the file
BYTE* buffer	The address of the memory block
int bufferSize	The size of the memory block

Description

This function stores the content of the memory buffer to the file.

1.2 Structs, Records, Enums

The following table lists structs, records, enums in this documentation.

Enumerations

	Name	Description
	tagCardEventType (see page 132)	The type of the reader event
	CardEventType (see page 142)	The type of the reader event

Structures

	Name	Description
	structErrorInformation (see page 132)	Information about error when operation with the card or certificate is performed
	tagEidAddressA (see page 133)	EID address information, stored on the card - ANSI version
	tagEidAddressW (see page 133)	EID address information, stored on the card - UNICODE version
	tagEidCertificate (see page 133)	Certificate, stored on EID card
	tagEidIdentityA (see page 134)	Identity information stored on EID card - ANSI version
	tagEidIdentityExA (see page 135)	Identity information stored on EID card - ANSI version
	tagEidIdentityExW (see page 137)	Identity information stored on EID card - UNICODE version
	tagEidIdentityW (see page 138)	Identity information stored on EID card - UNICODE version
	tagEidPicture (see page 140)	Raw picture data from EID card
	tagSISRecordA (see page 140)	Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	tagSISRecordW (see page 141)	Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	EidAddressA (see page 142)	EID address information, stored on the card - ANSI version
	EidAddressW (see page 142)	EID address information, stored on the card - UNICODE version
	EidCertificate (see page 143)	Certificate, stored on EID card
	EidIdentityA (see page 143)	Identity information stored on EID card - ANSI version
	EidIdentityExA (see page 145)	Identity information stored on EID card - ANSI version
	EidIdentityExW (see page 146)	Identity information stored on EID card - UNICODE version
	EidIdentityW (see page 148)	Identity information stored on EID card - UNICODE version

	EidPicture (see page 149)	Raw picture data from EID card
	ErrorInformation (see page 149)	Information about error when operation with the card or certificate is performed
	PEidAddressA (see page 150)	EID address information, stored on the card - ANSI version
	PEidAddressW (see page 150)	EID address information, stored on the card - UNICODE version
	PEidCertificate (see page 150)	Certificate, stored on EID card
	PEidIdentityA (see page 151)	Identity information stored on EID card - ANSI version
	PEidIdentityExA (see page 152)	Identity information stored on EID card - ANSI version
	PEidIdentityExW (see page 154)	Identity information stored on EID card - UNICODE version
	PEidIdentityW (see page 155)	Identity information stored on EID card - UNICODE version
	PeidPicture (see page 157)	Raw picture data from EID card
	PErrorInformation (see page 157)	Information about error when operation with the card or certificate is performed
	PSISRecordA (see page 157)	Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	PSISRecordW (see page 158)	Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	SISRecordA (see page 159)	Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	SISRecordW (see page 160)	Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

1.2.1 structErrorInformation Structure

Information about error when operation with the card or certificate is performed

C++

```
struct structErrorInformation {
  int Code;
  WCHAR Description[ERROR_MAX_DESCRIPTION + 1];
};
```

File

CardStructures.h (see page 178)

1.2.2 tagCardEventType Enumeration

The type of the reader event

C++

```
enum tagCardEventType {
  ewtUnknownEvent,
  ewtCardInsert,
  ewtCardRemove,
  ewtReadersChange
};
```

File

CardEvents.h (see page 177)

Members

Members	Description
ewtUnknownEvent	Unknown event
ewtCardInsert	The card was inserted in the reader
ewtCardRemove	The card was removed from the reader
ewtReadersChange	The readers list changed

1.2.3 tagEidAddressA Structure

EID address information, stored on the card - ANSI version

C++

```
struct tagEidAddressA {
  char street[EID_MAX_STREET_LEN+1];
  char zip[EID_MAX_ZIP_LEN+1];
  char municipality[EID_MAX_MUNICIPALITY_LEN+1];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
char street[EID_MAX_STREET_LEN+1];	Street name
char zip[EID_MAX_ZIP_LEN+1];	ZIP code
char municipality[EID_MAX_MUNICIPALITY_LEN+1];	Municipality

1.2.4 tagEidAddressW Structure

EID address information, stored on the card - UNICODE version

C++

```
struct tagEidAddressW {
  WCHAR street[EID_MAX_STREET_LEN+1];
  WCHAR zip[EID_MAX_ZIP_LEN+1];
  WCHAR municipality[EID_MAX_MUNICIPALITY_LEN+1];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR street[EID_MAX_STREET_LEN+1];	Street name
WCHAR zip[EID_MAX_ZIP_LEN+1];	ZIP code
WCHAR municipality[EID_MAX_MUNICIPALITY_LEN+1];	Municipality

1.2.5 tagEidCertificate Structure

Certificate, stored on EID card

C++

```
struct tagEidCertificate {
  BYTE certificate[EID_MAX_CERT_LEN+1];
  int certificateLength;
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
BYTE certificate[EID_MAX_CERT_LEN+1];	Certificate raw data buffer
int certificateLength;	Certificate data length

1.2.6 tagEidIdentityA Structure

Identity information stored on EID card - ANSI version

C++

```
struct tagEidIdentityA {
  char cardNumber[EID_MAX_CARD_NUMBER_LEN+1];
  char chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];
  char validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];
  char validityDateEnd[EID_MAX_DATE_END_LEN +1];
  char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];
  char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];
  char name[EID_MAX_NAME_LEN+1];
  char firstName1[EID_MAX_FIRST_NAME1_LEN+1];
  char firstName2[EID_MAX_FIRST_NAME2_LEN+1];
  char nationality[EID_MAX_NATIONALITY_LEN+1];
  char birthLocation[EID_MAX_BIRTHPLACE_LEN+1];
  char birthDate[EID_MAX_BIRTHDATE_LEN+1];
  char sex[EID_MAX_SEX_LEN+1];
  char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  char duplicate[EID_MAX_DUPLICATE_LEN + 1];
  char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  char vat1[EID_MAX_VAT1_LEN + 1];
  char vat2[EID_MAX_VAT2_LEN + 1];
  char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
char cardNumber[EID_MAX_CARD_NUMBER_LEN+1];	Electronic ID card number
char chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];	Electronic ID card physical chip number
char validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];	Card validity start date
char validityDateEnd[EID_MAX_DATE_END_LEN +1];	Card validity end date

char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];	Card delivery municipality
char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];	National number
char name[EID_MAX_NAME_LEN+1];	Surname
char firstName1[EID_MAX_FIRST_NAME1_LEN+1];	First name (2 first given names)
char firstName2[EID_MAX_FIRST_NAME2_LEN+1];	First name part 2 (first letter of the 3rd given name).
char nationality[EID_MAX_NATIONALITY_LEN+1];	Nationality
char birthLocation[EID_MAX_BIRTHPLACE_LEN+1];	Birth location
char birthDate[EID_MAX_BIRTHDATE_LEN+1];	Birth date
char sex[EID_MAX_SEX_LEN+1];	Sex
char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
char duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work Permit type
char vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
char vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.7 tagEidIdentityExA Structure

Identity information stored on EID card - ANSI version

C++

```
struct tagEidIdentityExA {
  WORD nSize;
  WORD nVersion;
  char cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];
  char chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];
  char validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];
  char validityDateEnd[EID_MAX_DATE_END_LEN + 1];
  char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];
  char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];
  char name[EID_MAX_NAME_LEN + 1];
  char firstName1[EID_MAX_FIRST_NAME1_LEN + 1];
  char firstName2[EID_MAX_FIRST_NAME2_LEN + 1];
  char nationality[EID_MAX_NATIONALITY_LEN + 1];
  char birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];
```

```
    char birthDate[EID_MAX_BIRTHDATE_LEN + 1];
    char sex[EID_MAX_SEX_LEN + 1];
    char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];
    int documentType;
    BOOL whiteCane;
    BOOL yellowCane;
    BOOL extendedMinority;
    char duplicate[EID_MAX_DUPLICATE_LEN + 1];
    char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
    BOOL memberOfFamily;
    char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
    char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
    char vat1[EID_MAX_VAT1_LEN + 1];
    char vat2[EID_MAX_VAT2_LEN + 1];
    char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
    char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
    char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
WORD nSize;	The size of the structure
WORD nVersion;	The version of the structure
char cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];	Electronic ID card number
char chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];	Electronic ID card physical chip number
char validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];	Card validity start date
char validityDateEnd[EID_MAX_DATE_END_LEN + 1];	Card validity end date
char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];	Card delivery municipality
char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];	National number
char name[EID_MAX_NAME_LEN + 1];	Surname
char firstName1[EID_MAX_FIRST_NAME1_LEN + 1];	First name (2 first given names)
char firstName2[EID_MAX_FIRST_NAME2_LEN + 1];	First name part 2 (first letter of the 3rd given name).
char nationality[EID_MAX_NATIONALITY_LEN + 1];	Nationality
char birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];	Birth location
char birthDate[EID_MAX_BIRTHDATE_LEN + 1];	Birth date
char sex[EID_MAX_SEX_LEN + 1];	Sex
char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
char duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family

char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work Permit type
char vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
char vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.8 tagEidIdentityExW Structure

Identity information stored on EID card - UNICODE version

C++

```
struct tagEidIdentityExW {
  WORD nSize;
  WORD nVersion;
  WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];
  WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];
  WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];
  WCHAR validityDateEnd[EID_MAX_DATE_END_LEN + 1];
  WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];
  WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];
  WCHAR name[EID_MAX_NAME_LEN + 1];
  WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN + 1];
  WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN + 1];
  WCHAR nationality[EID_MAX_NATIONALITY_LEN + 1];
  WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];
  WCHAR birthDate[EID_MAX_BIRTHDATE_LEN + 1];
  WCHAR sex[EID_MAX_SEX_LEN + 1];
  WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];
  WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  WCHAR vat1[EID_MAX_VAT1_LEN + 1];
  WCHAR vat2[EID_MAX_VAT2_LEN + 1];
  WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
WORD nSize;	The size of the structure
WORD nVersion;	The version of the structure
WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];	Electronic ID card number
WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];	Electronic ID card physical chip number
WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];	Card validity start date

WCHAR validityDateEnd[EID_MAX_DATE_END_LEN + 1];	Card validity end date
WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];	Card delivery municipality
WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];	National number
WCHAR name[EID_MAX_NAME_LEN + 1];	Surname
WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN + 1];	First name (2 first given names)
WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN + 1];	First name part 2 (first letter of the 3rd given name).
WCHAR nationality[EID_MAX_NATIONALITY_LEN + 1];	Nationality
WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];	Birth location
WCHAR birthDate[EID_MAX_BIRTHDATE_LEN + 1];	Birth date
WCHAR sex[EID_MAX_SEX_LEN + 1];	Sex
WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work permit type
WCHAR vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
WCHAR vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.9 tagEidIdentityW Structure

Identity information stored on EID card - UNICODE version

C++

```
struct tagEidIdentityW {
  WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN+1];
  WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];
  WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];
  WCHAR validityDateEnd[EID_MAX_DATE_END_LEN +1];
  WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];
  WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];
  WCHAR name[EID_MAX_NAME_LEN+1];
  WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN+1];
  WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN+1];
  WCHAR nationality[EID_MAX_NATIONALITY_LEN+1];
```

```
  WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN+1];
  WCHAR birthDate[EID_MAX_BIRTHDATE_LEN+1];
  WCHAR sex[EID_MAX_SEX_LEN+1];
  WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];
  WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  WCHAR vat1[EID_MAX_VAT1_LEN + 1];
  WCHAR vat2[EID_MAX_VAT2_LEN + 1];
  WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN+1];	Electronic ID card number
WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];	Electronic ID card physical chip number
WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];	Card validity start date
WCHAR validityDateEnd[EID_MAX_DATE_END_LEN +1];	Card validity end date
WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];	Card delivery municipality
WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];	National number
WCHAR name[EID_MAX_NAME_LEN+1];	Surname
WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN+1];	First name (2 first given names)
WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN+1];	First name part 2 (first letter of the 3rd given name).
WCHAR nationality[EID_MAX_NATIONALITY_LEN+1];	Nationality
WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN+1];	Birth location
WCHAR birthDate[EID_MAX_BIRTHDATE_LEN+1];	Birth date
WCHAR sex[EID_MAX_SEX_LEN+1];	Sex
WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family

WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work permit type
WCHAR vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
WCHAR vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.10 tagEidPicture Structure

Raw picture data from EID card

C++

```
struct tagEidPicture {
  BYTE picture[EID_MAX_PICTURE_LEN+1];
  int pictureLength;
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
BYTE picture[EID_MAX_PICTURE_LEN+1];	Picture raw data buffer
int pictureLength;	Picture raw data buffer length

1.2.11 tagSISRecordA Structure

Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

C++

```
struct tagSISRecordA {
  char Name[SIS_MAX_NAME_LEN + 1];
  char FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];
  char Initial[SIS_MAX_INITIAL_LEN+ 1];
  char Sex[SIS_MAX_SEX_LEN + 1];
  char BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];
  char SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];
  char CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];
  char ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];
  char ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];
  char CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];
  char CardName[SIS_MAX_CARDNAME_LEN +1 ];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
char Name[SIS_MAX_NAME_LEN + 1];	Family name of the card owner

char FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];	First name of the card owner
char Initial[SIS_MAX_INITIAL_LEN+ 1];	Initial of the card owner
char Sex[SIS_MAX_SEX_LEN + 1];	Sex of the card owner
char BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];	Birth date of the card owner
char SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];	Social security number of the card owner
char CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];	Date of issue
char ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];	Card validity begin date
char ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];	Card validity end date
char CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];	Card number
char CardName[SIS_MAX_CARDNAME_LEN +1];	Name of the card

1.2.12 tagSISRecordW Structure

Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

C++

```
struct tagSISRecordW {
  WCHAR Name[SIS_MAX_NAME_LEN + 1];
  WCHAR FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];
  WCHAR Initial[SIS_MAX_INITIAL_LEN+ 1];
  WCHAR Sex[SIS_MAX_SEX_LEN + 1];
  WCHAR BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];
  WCHAR SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];
  WCHAR CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];
  WCHAR ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];
  WCHAR ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];
  WCHAR CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];
  WCHAR CardName[SIS_MAX_CARDNAME_LEN +1 ];
};
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR Name[SIS_MAX_NAME_LEN + 1];	Family name of the card owner
WCHAR FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];	First name of the card owner
WCHAR Initial[SIS_MAX_INITIAL_LEN+ 1];	Initial of the card owner
WCHAR Sex[SIS_MAX_SEX_LEN + 1];	Sex of the card owner
WCHAR BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];	Birth date of the card owner
WCHAR SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];	Social security number of the card owner
WCHAR CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];	Date of issue
WCHAR ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];	Card validity begin date
WCHAR ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];	Card validity end date
WCHAR CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];	Card number
WCHAR CardName[SIS_MAX_CARDNAME_LEN +1];	Name of the card

1.2.13 CardEventType Enumeration

The type of the reader event

C++

```
typedef enum tagCardEventType {
  ewtUnknownEvent,
  ewtCardInsert,
  ewtCardRemove,
  ewtReadersChange
} CardEventType;
```

File

CardEvents.h (see page 177)

Members

Members	Description
ewtUnknownEvent	Unknown event
ewtCardInsert	The card was inserted in the reader
ewtCardRemove	The card was removed from the reader
ewtReadersChange	The readers list changed

1.2.14 EidAddressA Structure

EID address information, stored on the card - ANSI version

C++

```
typedef struct tagEidAddressA {
  char street[EID_MAX_STREET_LEN+1];
  char zip[EID_MAX_ZIP_LEN+1];
  char municipality[EID_MAX_MUNICIPALITY_LEN+1];
} EidAddressA, * PEidAddressA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
char street[EID_MAX_STREET_LEN+1];	Street name
char zip[EID_MAX_ZIP_LEN+1];	ZIP code
char municipality[EID_MAX_MUNICIPALITY_LEN+1];	Municipality

1.2.15 EidAddressW Structure

EID address information, stored on the card - UNICODE version

C++

```
typedef struct tagEidAddressW {
  WCHAR street[EID_MAX_STREET_LEN+1];
  WCHAR zip[EID_MAX_ZIP_LEN+1];
  WCHAR municipality[EID_MAX_MUNICIPALITY_LEN+1];
} EidAddressW, * PEidAddressW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR street[EID_MAX_STREET_LEN+1];	Street name
WCHAR zip[EID_MAX_ZIP_LEN+1];	ZIP code
WCHAR municipality[EID_MAX_MUNICIPALITY_LEN+1];	Municipality

1.2.16 EidCertificate Structure

Certificate, stored on EID card

C++

```
typedef struct tagEidCertificate {
  BYTE certificate[EID_MAX_CERT_LEN+1];
  int certificateLength;
} EidCertificate, * PEidCertificate;
```

File

CardStructures.h (see page 178)

Members

Members	Description
BYTE certificate[EID_MAX_CERT_LEN+1];	Certificate raw data buffer
int certificateLength;	Certificate data length

1.2.17 EidIdentityA Structure

Identity information stored on EID card - ANSI version

C++

```
typedef struct tagEidIdentityA {
  char cardNumber[EID_MAX_CARD_NUMBER_LEN+1];
  char chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];
  char validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];
  char validityDateEnd[EID_MAX_DATE_END_LEN +1];
  char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];
  char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];
  char name[EID_MAX_NAME_LEN+1];
  char firstName1[EID_MAX_FIRST_NAME1_LEN+1];
  char firstName2[EID_MAX_FIRST_NAME2_LEN+1];
  char nationality[EID_MAX_NATIONALITY_LEN+1];
  char birthLocation[EID_MAX_BIRTHPLACE_LEN+1];
  char birthDate[EID_MAX_BIRTHDATE_LEN+1];
  char sex[EID_MAX_SEX_LEN+1];
  char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  char duplicate[EID_MAX_DUPLICATE_LEN + 1];
  char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  char vat1[EID_MAX_VAT1_LEN + 1];
```

```
    char vat2[EID_MAX_VAT2_LEN + 1];
    char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
    char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
    char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityA, * PEidIdentityA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
char cardNumber[EID_MAX_CARD_NUMBER_LEN+1];	Electronic ID card number
char chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];	Electronic ID card physical chip number
char validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];	Card validity start date
char validityDateEnd[EID_MAX_DATE_END_LEN +1];	Card validity end date
char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];	Card delivery municipality
char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];	National number
char name[EID_MAX_NAME_LEN+1];	Surname
char firstName1[EID_MAX_FIRST_NAME1_LEN+1];	First name (2 first given names)
char firstName2[EID_MAX_FIRST_NAME2_LEN+1];	First name part 2 (first letter of the 3rd given name).
char nationality[EID_MAX_NATIONALITY_LEN+1];	Nationality
char birthLocation[EID_MAX_BIRTHPLACE_LEN+1];	Birth location
char birthDate[EID_MAX_BIRTHDATE_LEN+1];	Birth date
char sex[EID_MAX_SEX_LEN+1];	Sex
char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
char duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work Permit type
char vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
char vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.18 EidIdentityExA Structure

Identity information stored on EID card - ANSI version

C++

```
typedef struct tagEidIdentityExA {
  WORD nSize;
  WORD nVersion;
  char cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];
  char chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];
  char validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];
  char validityDateEnd[EID_MAX_DATE_END_LEN + 1];
  char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];
  char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];
  char name[EID_MAX_NAME_LEN + 1];
  char firstName1[EID_MAX_FIRST_NAME1_LEN + 1];
  char firstName2[EID_MAX_FIRST_NAME2_LEN + 1];
  char nationality[EID_MAX_NATIONALITY_LEN + 1];
  char birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];
  char birthDate[EID_MAX_BIRTHDATE_LEN + 1];
  char sex[EID_MAX_SEX_LEN + 1];
  char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  char duplicate[EID_MAX_DUPLICATE_LEN + 1];
  char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  char vat1[EID_MAX_VAT1_LEN + 1];
  char vat2[EID_MAX_VAT2_LEN + 1];
  char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityExA, * PEidIdentityExA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WORD nSize;	The size of the structure
WORD nVersion;	The version of the structure
char cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];	Electronic ID card number
char chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];	Electronic ID card physical chip number
char validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];	Card validity start date
char validityDateEnd[EID_MAX_DATE_END_LEN + 1];	Card validity end date
char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];	Card delivery municipality
char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];	National number
char name[EID_MAX_NAME_LEN + 1];	Surname
char firstName1[EID_MAX_FIRST_NAME1_LEN + 1];	First name (2 first given names)
char firstName2[EID_MAX_FIRST_NAME2_LEN + 1];	First name part 2 (first letter of the 3rd given name).

char nationality[EID_MAX_NATIONALITY_LEN + 1];	Nationality
char birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];	Birth location
char birthDate[EID_MAX_BIRTHDATE_LEN + 1];	Birth date
char sex[EID_MAX_SEX_LEN + 1];	Sex
char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
char duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work Permit type
char vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
char vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.19 EidIdentityExW Structure

Identity information stored on EID card - UNICODE version

C++

```
typedef struct tagEidIdentityExW {
  WORD nSize;
  WORD nVersion;
  WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];
  WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];
  WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];
  WCHAR validityDateEnd[EID_MAX_DATE_END_LEN + 1];
  WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];
  WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];
  WCHAR name[EID_MAX_NAME_LEN + 1];
  WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN + 1];
  WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN + 1];
  WCHAR nationality[EID_MAX_NATIONALITY_LEN + 1];
  WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];
  WCHAR birthDate[EID_MAX_BIRTHDATE_LEN + 1];
  WCHAR sex[EID_MAX_SEX_LEN + 1];
  WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];
  WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
```

```
  WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  WCHAR vat1[EID_MAX_VAT1_LEN + 1];
  WCHAR vat2[EID_MAX_VAT2_LEN + 1];
  WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityExW, * PEidIdentityExW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WORD nSize;	The size of the structure
WORD nVersion;	The version of the structure
WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];	Electronic ID card number
WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];	Electronic ID card physical chip number
WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];	Card validity start date
WCHAR validityDateEnd[EID_MAX_DATE_END_LEN + 1];	Card validity end date
WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];	Card delivery municipality
WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];	National number
WCHAR name[EID_MAX_NAME_LEN + 1];	Surname
WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN + 1];	First name (2 first given names)
WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN + 1];	First name part 2 (first letter of the 3rd given name).
WCHAR nationality[EID_MAX_NATIONALITY_LEN + 1];	Nationality
WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];	Birth location
WCHAR birthDate[EID_MAX_BIRTHDATE_LEN + 1];	Birth date
WCHAR sex[EID_MAX_SEX_LEN + 1];	Sex
WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work permit type
WCHAR vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
WCHAR vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number

WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.20 EidIdentityW Structure

Identity information stored on EID card - UNICODE version

C++

```
typedef struct tagEidIdentityW {
  WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN+1];
  WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];
  WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];
  WCHAR validityDateEnd[EID_MAX_DATE_END_LEN +1];
  WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];
  WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];
  WCHAR name[EID_MAX_NAME_LEN+1];
  WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN+1];
  WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN+1];
  WCHAR nationality[EID_MAX_NATIONALITY_LEN+1];
  WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN+1];
  WCHAR birthDate[EID_MAX_BIRTHDATE_LEN+1];
  WCHAR sex[EID_MAX_SEX_LEN+1];
  WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];
  WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  WCHAR vat1[EID_MAX_VAT1_LEN + 1];
  WCHAR vat2[EID_MAX_VAT2_LEN + 1];
  WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityW, * PEidIdentityW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN+1];	Electronic ID card number
WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];	Electronic ID card physical chip number
WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];	Card validity start date
WCHAR validityDateEnd[EID_MAX_DATE_END_LEN +1];	Card validity end date
WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];	Card delivery municipality
WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];	National number
WCHAR name[EID_MAX_NAME_LEN+1];	Surname
WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN+1];	First name (2 first given names)
WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN+1];	First name part 2 (first letter of the 3rd given name).
WCHAR nationality[EID_MAX_NATIONALITY_LEN+1];	Nationality

WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN+1];	Birth location
WCHAR birthDate[EID_MAX_BIRTHDATE_LEN+1];	Birth date
WCHAR sex[EID_MAX_SEX_LEN+1];	Sex
WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work permit type
WCHAR vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
WCHAR vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.21 EidPicture Structure

Raw picture data from EID card

C++

```
typedef struct tagEidPicture {
  BYTE picture[EID_MAX_PICTURE_LEN+1];
  int pictureLength;
} EidPicture, * PeidPicture;
```

File

CardStructures.h (see page 178)

Members

Members	Description
BYTE picture[EID_MAX_PICTURE_LEN+1];	Picture raw data buffer
int pictureLength;	Picture raw data buffer length

1.2.22 ErrorInformation Structure

Information about error when operation with the card or certificate is performed

C++

```
typedef struct structErrorInformation {
```

```
  int Code;
  WCHAR Description[ERROR_MAX_DESCRIPTION + 1];
} ErrorInformation, * PErrorInformation;
```

File

CardStructures.h (see page 178)

1.2.23 PEidAddressA Structure

EID address information, stored on the card - ANSI version

C++

```
typedef struct tagEidAddressA {
  char street[EID_MAX_STREET_LEN+1];
  char zip[EID_MAX_ZIP_LEN+1];
  char municipality[EID_MAX_MUNICIPALITY_LEN+1];
} EidAddressA, * PEidAddressA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
char street[EID_MAX_STREET_LEN+1];	Street name
char zip[EID_MAX_ZIP_LEN+1];	ZIP code
char municipality[EID_MAX_MUNICIPALITY_LEN+1];	Municipality

1.2.24 PEidAddressW Structure

EID address information, stored on the card - UNICODE version

C++

```
typedef struct tagEidAddressW {
  WCHAR street[EID_MAX_STREET_LEN+1];
  WCHAR zip[EID_MAX_ZIP_LEN+1];
  WCHAR municipality[EID_MAX_MUNICIPALITY_LEN+1];
} EidAddressW, * PEidAddressW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR street[EID_MAX_STREET_LEN+1];	Street name
WCHAR zip[EID_MAX_ZIP_LEN+1];	ZIP code
WCHAR municipality[EID_MAX_MUNICIPALITY_LEN+1];	Municipality

1.2.25 PEidCertificate Structure

Certificate, stored on EID card

C++

```
typedef struct tagEidCertificate {
  BYTE certificate[EID_MAX_CERT_LEN+1];
  int certificateLength;
} EidCertificate, * PEidCertificate;
```

File

CardStructures.h (see page 178)

Members

Members	Description
BYTE certificate[EID_MAX_CERT_LEN+1];	Certificate raw data buffer
int certificateLength;	Certificate data length

1.2.26 PEidIdentityA Structure

Identity information stored on EID card - ANSI version

C++

```
typedef struct tagEidIdentityA {
  char cardNumber[EID_MAX_CARD_NUMBER_LEN+1];
  char chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];
  char validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];
  char validityDateEnd[EID_MAX_DATE_END_LEN +1];
  char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];
  char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];
  char name[EID_MAX_NAME_LEN+1];
  char firstName1[EID_MAX_FIRST_NAME1_LEN+1];
  char firstName2[EID_MAX_FIRST_NAME2_LEN+1];
  char nationality[EID_MAX_NATIONALITY_LEN+1];
  char birthLocation[EID_MAX_BIRTHPLACE_LEN+1];
  char birthDate[EID_MAX_BIRTHDATE_LEN+1];
  char sex[EID_MAX_SEX_LEN+1];
  char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  char duplicate[EID_MAX_DUPLICATE_LEN + 1];
  char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  char vat1[EID_MAX_VAT1_LEN + 1];
  char vat2[EID_MAX_VAT2_LEN + 1];
  char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityA, * PEidIdentityA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
char cardNumber[EID_MAX_CARD_NUMBER_LEN+1];	Electronic ID card number
char chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];	Electronic ID card physical chip number
char validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];	Card validity start date
char validityDateEnd[EID_MAX_DATE_END_LEN +1];	Card validity end date

char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];	Card delivery municipality
char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];	National number
char name[EID_MAX_NAME_LEN+1];	Surname
char firstName1[EID_MAX_FIRST_NAME1_LEN+1];	First name (2 first given names)
char firstName2[EID_MAX_FIRST_NAME2_LEN+1];	First name part 2 (first letter of the 3rd given name).
char nationality[EID_MAX_NATIONALITY_LEN+1];	Nationality
char birthLocation[EID_MAX_BIRTHPLACE_LEN+1];	Birth location
char birthDate[EID_MAX_BIRTHDATE_LEN+1];	Birth date
char sex[EID_MAX_SEX_LEN+1];	Sex
char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
char duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work Permit type
char vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
char vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.27 PEidIdentityExA Structure

Identity information stored on EID card - ANSI version

C++

```
typedef struct tagEidIdentityExA {
  WORD nSize;
  WORD nVersion;
  char cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];
  char chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];
  char validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];
  char validityDateEnd[EID_MAX_DATE_END_LEN + 1];
  char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];
  char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];
  char name[EID_MAX_NAME_LEN + 1];
  char firstName1[EID_MAX_FIRST_NAME1_LEN + 1];
  char firstName2[EID_MAX_FIRST_NAME2_LEN + 1];
  char nationality[EID_MAX_NATIONALITY_LEN + 1];
  char birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];
```

```
    char birthDate[EID_MAX_BIRTHDATE_LEN + 1];
    char sex[EID_MAX_SEX_LEN + 1];
    char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];
    int documentType;
    BOOL whiteCane;
    BOOL yellowCane;
    BOOL extendedMinority;
    char duplicate[EID_MAX_DUPLICATE_LEN + 1];
    char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
    BOOL memberOfFamily;
    char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
    char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
    char vat1[EID_MAX_VAT1_LEN + 1];
    char vat2[EID_MAX_VAT2_LEN + 1];
    char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
    char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
    char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityExA, * PEidIdentityExA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WORD nSize;	The size of the structure
WORD nVersion;	The version of the structure
char cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];	Electronic ID card number
char chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];	Electronic ID card physical chip number
char validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];	Card validity start date
char validityDateEnd[EID_MAX_DATE_END_LEN + 1];	Card validity end date
char municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];	Card delivery municipality
char nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];	National number
char name[EID_MAX_NAME_LEN + 1];	Surname
char firstName1[EID_MAX_FIRST_NAME1_LEN + 1];	First name (2 first given names)
char firstName2[EID_MAX_FIRST_NAME2_LEN + 1];	First name part 2 (first letter of the 3rd given name).
char nationality[EID_MAX_NATIONALITY_LEN + 1];	Nationality
char birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];	Birth location
char birthDate[EID_MAX_BIRTHDATE_LEN + 1];	Birth date
char sex[EID_MAX_SEX_LEN + 1];	Sex
char nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
char duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
char specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family

char dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
char workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work Permit type
char vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
char vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
char regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
char brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
char brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.28 PEidIdentityExW Structure

Identity information stored on EID card - UNICODE version

C++

```
typedef struct tagEidIdentityExW {
  WORD nSize;
  WORD nVersion;
  WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];
  WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];
  WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];
  WCHAR validityDateEnd[EID_MAX_DATE_END_LEN + 1];
  WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];
  WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];
  WCHAR name[EID_MAX_NAME_LEN + 1];
  WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN + 1];
  WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN + 1];
  WCHAR nationality[EID_MAX_NATIONALITY_LEN + 1];
  WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];
  WCHAR birthDate[EID_MAX_BIRTHDATE_LEN + 1];
  WCHAR sex[EID_MAX_SEX_LEN + 1];
  WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];
  WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  WCHAR vat1[EID_MAX_VAT1_LEN + 1];
  WCHAR vat2[EID_MAX_VAT2_LEN + 1];
  WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityExW, * PEidIdentityExW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WORD nSize;	The size of the structure
WORD nVersion;	The version of the structure
WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN + 1];	Electronic ID card number
WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN + 1];	Electronic ID card physical chip number
WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN + 1];	Card validity start date

WCHAR validityDateEnd[EID_MAX_DATE_END_LEN + 1];	Card validity end date
WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN + 1];	Card delivery municipality
WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN + 1];	National number
WCHAR name[EID_MAX_NAME_LEN + 1];	Surname
WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN + 1];	First name (2 first given names)
WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN + 1];	First name part 2 (first letter of the 3rd given name).
WCHAR nationality[EID_MAX_NATIONALITY_LEN + 1];	Nationality
WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN + 1];	Birth location
WCHAR birthDate[EID_MAX_BIRTHDATE_LEN + 1];	Birth date
WCHAR sex[EID_MAX_SEX_LEN + 1];	Sex
WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN + 1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family
WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work permit type
WCHAR vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
WCHAR vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.29 PEidIdentityW Structure

Identity information stored on EID card - UNICODE version

C++

```
typedef struct tagEidIdentityW {
  WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN+1];
  WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];
  WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];
  WCHAR validityDateEnd[EID_MAX_DATE_END_LEN +1];
  WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];
  WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];
  WCHAR name[EID_MAX_NAME_LEN+1];
  WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN+1];
  WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN+1];
  WCHAR nationality[EID_MAX_NATIONALITY_LEN+1];
```

```
  WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN+1];
  WCHAR birthDate[EID_MAX_BIRTHDATE_LEN+1];
  WCHAR sex[EID_MAX_SEX_LEN+1];
  WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];
  int documentType;
  BOOL whiteCane;
  BOOL yellowCane;
  BOOL extendedMinority;
  WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];
  WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];
  BOOL memberOfFamily;
  WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];
  WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];
  WCHAR vat1[EID_MAX_VAT1_LEN + 1];
  WCHAR vat2[EID_MAX_VAT2_LEN + 1];
  WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];
  WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];
  WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];
} EidIdentityW, * PEidIdentityW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR cardNumber[EID_MAX_CARD_NUMBER_LEN+1];	Electronic ID card number
WCHAR chipNumber[EID_MAX_CHIP_NUMBER_LEN+1];	Electronic ID card physical chip number
WCHAR validityDateBegin[EID_MAX_DATE_BEGIN_LEN+1];	Card validity start date
WCHAR validityDateEnd[EID_MAX_DATE_END_LEN +1];	Card validity end date
WCHAR municipality[EID_MAX_DELIVERY_MUNICIPALITY_LEN+1];	Card delivery municipality
WCHAR nationalNumber[EID_MAX_NATIONAL_NUMBER_LEN+1];	National number
WCHAR name[EID_MAX_NAME_LEN+1];	Surname
WCHAR firstName1[EID_MAX_FIRST_NAME1_LEN+1];	First name (2 first given names)
WCHAR firstName2[EID_MAX_FIRST_NAME2_LEN+1];	First name part 2 (first letter of the 3rd given name).
WCHAR nationality[EID_MAX_NATIONALITY_LEN+1];	Nationality
WCHAR birthLocation[EID_MAX_BIRTHPLACE_LEN+1];	Birth location
WCHAR birthDate[EID_MAX_BIRTHDATE_LEN+1];	Birth date
WCHAR sex[EID_MAX_SEX_LEN+1];	Sex
WCHAR nobleCondition[EID_MAX_NOBLE_CONDITION_LEN+1];	Noble condition
int documentType;	Document type code (Belgian citizen card, Kids Card, Foreigner card)
BOOL whiteCane;	White cane (blind people)
BOOL yellowCane;	Yellow cane (partially sighted people)
BOOL extendedMinority;	Extended minority
WCHAR duplicate[EID_MAX_DUPLICATE_LEN + 1];	Duplicata
WCHAR specialOrganization[EID_MAX_SPECIALORGANIZATION_LEN + 1];	Special Organization
BOOL memberOfFamily;	Member of family

WCHAR dateAndCountryOfProtection[EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN + 1];	Date and country of protection
WCHAR workPermitType[EID_MAX_WORKPERMITTYPE_LEN + 1];	Work permit type
WCHAR vat1[EID_MAX_VAT1_LEN + 1];	Employer VAT1
WCHAR vat2[EID_MAX_VAT2_LEN + 1];	Employer VAT2
WCHAR regionalFileNumber[EID_MAX_REGIONALFILENUMBER_LEN + 1];	Regional file number
WCHAR brexitMention1[EID_MAX_BREXITMENTION1_LEN + 1];	BREXIT
WCHAR brexitMention2[EID_MAX_BREXITMENTION2_LEN + 1];	BREXIT

1.2.30 PeidPicture Structure

Raw picture data from EID card

C++

```
typedef struct tagEidPicture {
  BYTE picture[EID_MAX_PICTURE_LEN+1];
  int pictureLength;
} EidPicture, * PeidPicture;
```

File

CardStructures.h (see page 178)

Members

Members	Description
BYTE picture[EID_MAX_PICTURE_LEN+1];	Picture raw data buffer
int pictureLength;	Picture raw data buffer length

1.2.31 PErrorInformation Structure

Information about error when operation with the card or certificate is performed

C++

```
typedef struct structErrorInformation {
  int Code;
  WCHAR Description[ERROR_MAX_DESCRIPTION + 1];
} ErrorInformation, * PErrorInformation;
```

File

CardStructures.h (see page 178)

1.2.32 PSISRecordA Structure

Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

C++

```
typedef struct tagSISRecordA {
  char Name[SIS_MAX_NAME_LEN + 1];
  char FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];
  char Initial[SIS_MAX_INITIAL_LEN+ 1];
```

```
  char Sex[SIS_MAX_SEX_LEN + 1];
  char BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];
  char SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];
  char CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];
  char ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];
  char ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];
  char CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];
  char CardName[SIS_MAX_CARDNAME_LEN +1 ];
} SISRecordA, * PSISRecordA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
char Name[SIS_MAX_NAME_LEN + 1];	Family name of the card owner
char FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];	First name of the card owner
char Initial[SIS_MAX_INITIAL_LEN+ 1];	Initial of the card owner
char Sex[SIS_MAX_SEX_LEN + 1];	Sex of the card owner
char BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];	Birth date of the card owner
char SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];	Social security number of the card owner
char CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];	Date of issue
char ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];	Card validity begin date
char ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];	Card validity end date
char CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];	Card number
char CardName[SIS_MAX_CARDNAME_LEN +1];	Name of the card

1.2.33 PSISRecordW Structure

Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

C++

```
typedef struct tagSISRecordW {
  WCHAR Name[SIS_MAX_NAME_LEN + 1];
  WCHAR FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];
  WCHAR Initial[SIS_MAX_INITIAL_LEN+ 1];
  WCHAR Sex[SIS_MAX_SEX_LEN + 1];
  WCHAR BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];
  WCHAR SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];
  WCHAR CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];
  WCHAR ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];
  WCHAR ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];
  WCHAR CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];
  WCHAR CardName[SIS_MAX_CARDNAME_LEN +1 ];
} SISRecordW, * PSISRecordW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR Name[SIS_MAX_NAME_LEN + 1];	Family name of the card owner
WCHAR FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];	First name of the card owner
WCHAR Initial[SIS_MAX_INITIAL_LEN+ 1];	Initial of the card owner

WCHAR Sex[SIS_MAX_SEX_LEN + 1];	Sex of the card owner
WCHAR BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];	Birth date of the card owner
WCHAR SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];	Social security number of the card owner
WCHAR CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];	Date of issue
WCHAR ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];	Card validity begin date
WCHAR ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];	Card validity end date
WCHAR CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];	Card number
WCHAR CardName[SIS_MAX_CARDNAME_LEN +1];	Name of the card

1.2.34 SISRecordA Structure

Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

C++

```
typedef struct tagSISRecordA {
  char Name[SIS_MAX_NAME_LEN + 1];
  char FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];
  char Initial[SIS_MAX_INITIAL_LEN+ 1];
  char Sex[SIS_MAX_SEX_LEN + 1];
  char BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];
  char SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];
  char CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];
  char ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];
  char ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];
  char CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];
  char CardName[SIS_MAX_CARDNAME_LEN +1 ];
} SISRecordA, * PSISRecordA;
```

File

CardStructures.h (see page 178)

Members

Members	Description
char Name[SIS_MAX_NAME_LEN + 1];	Family name of the card owner
char FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];	First name of the card owner
char Initial[SIS_MAX_INITIAL_LEN+ 1];	Initial of the card owner
char Sex[SIS_MAX_SEX_LEN + 1];	Sex of the card owner
char BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];	Birth date of the card owner
char SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];	Social security number of the card owner
char CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];	Date of issue
char ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];	Card validity begin date
char ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];	Card validity end date
char CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];	Card number
char CardName[SIS_MAX_CARDNAME_LEN +1];	Name of the card

1.2.35 SISRecordW Structure

Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

C++

```
typedef struct tagSISRecordW {
  WCHAR Name[SIS_MAX_NAME_LEN + 1];
  WCHAR FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];
  WCHAR Initial[SIS_MAX_INITIAL_LEN+ 1];
  WCHAR Sex[SIS_MAX_SEX_LEN + 1];
  WCHAR BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];
  WCHAR SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];
  WCHAR CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];
  WCHAR ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];
  WCHAR ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];
  WCHAR CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];
  WCHAR CardName[SIS_MAX_CARDNAME_LEN +1 ];
} SISRecordW, * PSISRecordW;
```

File

CardStructures.h (see page 178)

Members

Members	Description
WCHAR Name[SIS_MAX_NAME_LEN + 1];	Family name of the card owner
WCHAR FirstName[SIS_MAX_FIRSTNAMES_LEN + 1];	First name of the card owner
WCHAR Initial[SIS_MAX_INITIAL_LEN+ 1];	Initial of the card owner
WCHAR Sex[SIS_MAX_SEX_LEN + 1];	Sex of the card owner
WCHAR BirthDate[SIS_FIELD_MAX_BIRTHDATE_LEN + 1];	Birth date of the card owner
WCHAR SocialSecurityNumber[SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN + 1];	Social security number of the card owner
WCHAR CaptureDate[SIS_FIELD_MAX_CAPTUREDATE_LEN + 1];	Date of issue
WCHAR ValidityDateBegin[SIS_FIELD_MAX_VALIDBEGIN_LEN + 1];	Card validity begin date
WCHAR ValidityDateEnd[SIS_FIELD_MAX_VALIDEND_LEN + 1];	Card validity end date
WCHAR CardNumber[SIS_FIELD_MAX_CARDNUMBER_LEN + 1];	Card number
WCHAR CardName[SIS_MAX_CARDNAME_LEN +1];	Name of the card

1.3 Macros

The following table lists macros in this documentation.

Macros

Name	Description
EID_MAX_BIRTHDATE_LEN (see page 162)	Maximum length of the birthdate
EID_MAX_BIRTHPLACE_LEN (see page 162)	Maximum length of the birthplace
EID_MAX_BREXITMENTION1_LEN (see page 163)	Maximum length of the BREXIT mention field
EID_MAX_BREXITMENTION2_LEN (see page 163)	Maximum length of the BREXIT mention field

EID_MAX_CARD_NUMBER_LEN (see page 163)	Maximum length of the card number field
EID_MAX_CERT_LEN (see page 163)	Maximum length of the certificate data
EID_MAX_CHIP_NUMBER_LEN (see page 163)	Maximum length of the chip number field
EID_MAX_DATE_BEGIN_LEN (see page 164)	Maximum length of the begin date field
EID_MAX_DATE_END_LEN (see page 164)	Maximum length of the end date field
EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN (see page 164)	Maximum length of theDate and country of protection field
EID_MAX_DELIVERY_MUNICIPALITY_LEN (see page 164)	Maximum length of the name of the devivery municipality
EID_MAX_DOCUMENT_TYPE_LEN (see page 165)	Maximum length of the document type field
EID_MAX_DUPLICATE_LEN (see page 165)	Maximum length of the Duplicate field
EID_MAX_FIRST_NAME1_LEN (see page 165)	Maximum length of the first name
EID_MAX_FIRST_NAME2_LEN (see page 165)	Maximum length of the first name
EID_MAX_MEMBEROFFAMILY_LEN (see page 166)	Maximum length of the Member of family field
EID_MAX_MUNICIPALITY_LEN (see page 166)	Maximum length of the municipality name field
EID_MAX_NAME_LEN (see page 166)	Maximum length of the surname
EID_MAX_NATIONAL_NUMBER_LEN (see page 166)	Maximum length of the national number
EID_MAX_NATIONALITY_LEN (see page 166)	Maximum length of the nationality
EID_MAX_NOBLE_CONDITION_LEN (see page 167)	Maximum length of the noble condition field
EID_MAX_PICTURE_LEN (see page 167)	Maximum length of the picture data
EID_MAX_REGIONALFILENUMBER_LEN (see page 167)	Maximum length of regional file number field
EID_MAX_SEX_LEN (see page 167)	Maximum length of the sex field
EID_MAX_SPECIAL_STATUS_LEN (see page 168)	Maximum length of the special status field
EID_MAX_SPECIALORGANIZATION_LEN (see page 168)	Maximum length of the Special organization field
EID_MAX_STREET_LEN (see page 168)	Maximum length of the street name field
EID_MAX_VAT1_LEN (see page 168)	Maximum length of the VAT1 field
EID_MAX_VAT2_LEN (see page 168)	Maximum length of the VAT2 field
EID_MAX_WORKPERMITTYPE_LEN (see page 169)	Maximum length of the type of the workpermit
EID_MAX_ZIP_LEN (see page 169)	Maximum length of the ZIP code field
ERR_CARD_ERROR (see page 169)	This is macro ERR_CARD_ERROR.
ERR_CERTIFICATE_ATTR (see page 169)	This is macro ERR_CERTIFICATE_ATTR.
ERR_MESSAGE_DECODE (see page 170)	This is macro ERR_MESSAGE_DECODE.
ERR_MESSAGE_ENCODE (see page 170)	This is macro ERR_MESSAGE_ENCODE.
ERR_MESSAGE_PARAM (see page 170)	This is macro ERR_MESSAGE_PARAM.
ERR_MESSAGE_UPDATE (see page 170)	This is macro ERR_MESSAGE_UPDATE.
ERR_NO_CERTIFICATE (see page 170)	This is macro ERR_NO_CERTIFICATE.
ERR_NO_DATA (see page 171)	This is macro ERR_NO_DATA.
ERR_NO_PRIVATE_KEY (see page 171)	This is macro ERR_NO_PRIVATE_KEY.
ERR_OPEN_CERTIFICATE (see page 171)	This is macro ERR_OPEN_CERTIFICATE.
ERR_SIGN_ERROR (see page 171)	This is macro ERR_SIGN_ERROR.
ERR_TIMESTAMP (see page 172)	This is macro ERR_TIMESTAMP.
ERROR_MAX_DESCRIPTION (see page 172)	Maximum error description length
FONT_BOLD (see page 172)	This is macro FONT_BOLD.
FONT_ITALIC (see page 172)	This is macro FONT_ITALIC.
FONT_NORMAL (see page 172)	This is macro FONT_NORMAL.
FONT_STRIKEOUT (see page 173)	This is macro FONT_STRIKEOUT.

FONT_UNDERLINE (see page 173)	This is macro FONT_UNDERLINE.
GetFileSHA256 (see page 173)	This is macro GetFileSHA256.
IID_PPV_ARG (see page 173)	IID_PPV_ARG(IType, ppType) IType is the type of pType ppType is the variable of type IType that will be filled RESULTS in: IID_IType, ppvType will create a compiler error if wrong level of indirection is used. macro for QueryInterface and related functions that require a IID and a (void **) this will insure that the cast is safe and appropriate on C
SaveCardToToXMLStreamEx (see page 174)	This is macro SaveCardToToXMLStreamEx.
SavePersonCsvToStream (see page 174)	This is macro SavePersonCsvToStream.
SIS_FIELD_MAX_BIRTHDATE_LEN (see page 174)	Maximum length of the birth date field
SIS_FIELD_MAX_CAPTUREDATE_LEN (see page 174)	Maximum length of the capture date field
SIS_FIELD_MAX_CARDNUMBER_LEN (see page 175)	Maximum length of the car number field
SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN (see page 175)	Maximum length of the social security number field
SIS_FIELD_MAX_VALIDBEGIN_LEN (see page 175)	Maximum length of the start validity date field
SIS_FIELD_MAX_VALIDEND_LEN (see page 175)	Maximum length of the end validity date field
SIS_MAX_CARDNAME_LEN (see page 176)	Maximum length of the card name field
SIS_MAX_FIRSTNAMES_LEN (see page 176)	Maximum length of the first name field
SIS_MAX_INITIAL_LEN (see page 176)	Maximum length of the initial field
SIS_MAX_NAME_LEN (see page 176)	Maximum length of the surname field
SIS_MAX_SEX_LEN (see page 176)	Maximum length of the sex field
WIDTHBYTES (see page 177)	This is macro WIDTHBYTES.

1.3.1 EID_MAX_BIRTHDATE_LEN Macro

Maximum length of the birthdate

C++

```
#define EID_MAX_BIRTHDATE_LEN 0xc
```

File

CardStructures.h (see page 178)

1.3.2 EID_MAX_BIRTHPLACE_LEN Macro

Maximum length of the birthplace

C++

```
#define EID_MAX_BIRTHPLACE_LEN 0x50
```

File

CardStructures.h (see page 178)

1.3.3 EID_MAX_BREXITMENTION1_LEN Macro

Maximum length of the BREXIT mention field

C++

```
#define EID_MAX_BREXITMENTION1_LEN 1
```

File

CardStructures.h (see page 178)

1.3.4 EID_MAX_BREXITMENTION2_LEN Macro

Maximum length of the BREXIT mention field

C++

```
#define EID_MAX_BREXITMENTION2_LEN 1
```

File

CardStructures.h (see page 178)

1.3.5 EID_MAX_CARD_NUMBER_LEN Macro

Maximum length of the card number field

C++

```
#define EID_MAX_CARD_NUMBER_LEN 0xc
```

File

CardStructures.h (see page 178)

1.3.6 EID_MAX_CERT_LEN Macro

Maximum length of the certificate data

C++

```
#define EID_MAX_CERT_LEN 0x800
```

File

CardStructures.h (see page 178)

1.3.7 EID_MAX_CHIP_NUMBER_LEN Macro

Maximum length of the chip number field

C++

```
#define EID_MAX_CHIP_NUMBER_LEN 0x20
```

File

CardStructures.h (see page 178)

1.3.8 EID_MAX_DATE_BEGIN_LEN Macro

Maximum length of the begin date field

C++

```
#define EID_MAX_DATE_BEGIN_LEN 0xa
```

File

CardStructures.h (see page 178)

1.3.9 EID_MAX_DATE_END_LEN Macro

Maximum length of the end date field

C++

```
#define EID_MAX_DATE_END_LEN 0xa
```

File

CardStructures.h (see page 178)

1.3.10 EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN Macro

Maximum length of theDate and country of protection field

C++

```
#define EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN 13
```

File

CardStructures.h (see page 178)

1.3.11 EID_MAX_DELIVERY_MUNICIPALITY_LEN Macro

Maximum length of the name of the devivery municipality

C++

```
#define EID_MAX_DELIVERY_MUNICIPALITY_LEN 0x50
```

File

CardStructures.h (see page 178)

1.3.12 EID_MAX_DOCUMENT_TYPE_LEN Macro

Maximum length of the document type field

C++

```
#define EID_MAX_DOCUMENT_TYPE_LEN 0x2
```

File

CardStructures.h (see page 178)

1.3.13 EID_MAX_DUPLICATE_LEN Macro

Maximum length of the Duplicate field

C++

```
#define EID_MAX_DUPLICATE_LEN 2
```

File

CardStructures.h (see page 178)

1.3.14 EID_MAX_FIRST_NAME1_LEN Macro

Maximum length of the first name

C++

```
#define EID_MAX_FIRST_NAME1_LEN 0x5f
```

File

CardStructures.h (see page 178)

1.3.15 EID_MAX_FIRST_NAME2_LEN Macro

Maximum length of the first name

C++

```
#define EID_MAX_FIRST_NAME2_LEN 0x3
```

File

CardStructures.h (see page 178)

1.3.16 EID_MAX_MEMBEROFFAMILY_LEN Macro

Maximum length of the Member of family field

C++

```
#define EID_MAX_MEMBEROFFAMILY_LEN 1
```

File

CardStructures.h (see page 178)

1.3.17 EID_MAX_MUNICIPALITY_LEN Macro

Maximum length of the municipality name field

C++

```
#define EID_MAX_MUNICIPALITY_LEN 0x43
```

File

CardStructures.h (see page 178)

1.3.18 EID_MAX_NAME_LEN Macro

Maximum length of the surname

C++

```
#define EID_MAX_NAME_LEN 0x6e
```

File

CardStructures.h (see page 178)

1.3.19 EID_MAX_NATIONAL_NUMBER_LEN Macro

Maximum length of the national number

C++

```
#define EID_MAX_NATIONAL_NUMBER_LEN 0xb
```

File

CardStructures.h (see page 178)

1.3.20 EID_MAX_NATIONALITY_LEN Macro

Maximum length of the nationality

C++

```
#define EID_MAX_NATIONALITY_LEN 0x55
```

File

CardStructures.h (see page 178)

1.3.21 EID_MAX_NOBLE_CONDITION_LEN Macro

Maximum length of the noble condition field

C++

```
#define EID_MAX_NOBLE_CONDITION_LEN 0x32
```

File

CardStructures.h (see page 178)

1.3.22 EID_MAX_PICTURE_LEN Macro

Maximum length of the picture data

C++

```
#define EID_MAX_PICTURE_LEN 0x1000
```

File

CardStructures.h (see page 178)

1.3.23 EID_MAX_REGIONALFILENUMBER_LEN Macro

Maximum length of regional file number field

C++

```
#define EID_MAX_REGIONALFILENUMBER_LEN 18
```

File

CardStructures.h (see page 178)

1.3.24 EID_MAX_SEX_LEN Macro

Maximum length of the sex field

C++

```
#define EID_MAX_SEX_LEN 0x1
```

File

CardStructures.h (see page 178)

1.3.25 EID_MAX_SPECIAL_STATUS_LEN Macro

Maximum length of the special status field

C++

```
#define EID_MAX_SPECIAL_STATUS_LEN 0x2
```

File

CardStructures.h (see page 178)

1.3.26 EID_MAX_SPECIALORGANIZATION_LEN Macro

Maximum length of the Special organization field

C++

```
#define EID_MAX_SPECIALORGANIZATION_LEN 1
```

File

CardStructures.h (see page 178)

1.3.27 EID_MAX_STREET_LEN Macro

Maximum length of the street name field

C++

```
#define EID_MAX_STREET_LEN 0x50
```

File

CardStructures.h (see page 178)

1.3.28 EID_MAX_VAT1_LEN Macro

Maximum length of the VAT1 field

C++

```
#define EID_MAX_VAT1_LEN 13
```

File

CardStructures.h (see page 178)

1.3.29 EID_MAX_VAT2_LEN Macro

Maximum length of the VAT2 field

C++

```
#define EID_MAX_VAT2_LEN 13
```

File

CardStructures.h (see page 178)

1.3.30 EID_MAX_WORKPERMITTYPE_LEN Macro

Maximum length of the type of the workpermit

C++

```
#define EID_MAX_WORKPERMITTYPE_LEN 1
```

File

CardStructures.h (see page 178)

1.3.31 EID_MAX_ZIP_LEN Macro

Maximum length of the ZIP code field

C++

```
#define EID_MAX_ZIP_LEN 0x4
```

File

CardStructures.h (see page 178)

1.3.32 ERR_CARD_ERROR Macro

This is macro ERR_CARD_ERROR.

C++

```
#define ERR_CARD_ERROR 9
```

File

CardStructures.h (see page 178)

1.3.33 ERR_CERTIFICATE_ATTR Macro

This is macro ERR_CERTIFICATE_ATTR.

C++

```
#define ERR_CERTIFICATE_ATTR 4
```

File

CardStructures.h (see page 178)

1.3.34 ERR_MESSAGE_DECODE Macro

This is macro ERR_MESSAGE_DECODE.

C++

```
#define ERR_MESSAGE_DECODE 5
```

File

CardStructures.h (see page 178)

1.3.35 ERR_MESSAGE_ENCODE Macro

This is macro ERR_MESSAGE_ENCODE.

C++

```
#define ERR_MESSAGE_ENCODE 10
```

File

CardStructures.h (see page 178)

1.3.36 ERR_MESSAGE_PARAM Macro

This is macro ERR_MESSAGE_PARAM.

C++

```
#define ERR_MESSAGE_PARAM 6
```

File

CardStructures.h (see page 178)

1.3.37 ERR_MESSAGE_UPDATE Macro

This is macro ERR_MESSAGE_UPDATE.

C++

```
#define ERR_MESSAGE_UPDATE 7
```

File

CardStructures.h (see page 178)

1.3.38 ERR_NO_CERTIFICATE Macro

This is macro ERR_NO_CERTIFICATE.

C++

```
#define ERR_NO_CERTIFICATE 2
```

File

CardStructures.h (see page 178)

1.3.39 ERR_NO_DATA Macro

This is macro ERR_NO_DATA.

C++

```
#define ERR_NO_DATA 1
```

File

CardStructures.h (see page 178)

1.3.40 ERR_NO_PRIVATE_KEY Macro

This is macro ERR_NO_PRIVATE_KEY.

C++

```
#define ERR_NO_PRIVATE_KEY 3
```

File

CardStructures.h (see page 178)

1.3.41 ERR_OPEN_CERTIFICATE Macro

This is macro ERR_OPEN_CERTIFICATE.

C++

```
#define ERR_OPEN_CERTIFICATE 8
```

File

CardStructures.h (see page 178)

1.3.42 ERR_SIGN_ERROR Macro

This is macro ERR_SIGN_ERROR.

C++

```
#define ERR_SIGN_ERROR 12
```

File

CardStructures.h (see page 178)

1.3.43 ERR_TIMESTAMP Macro

This is macro ERR_TIMESTAMP.

C++

```
#define ERR_TIMESTAMP 11
```

File

CardStructures.h (see page 178)

1.3.44 ERROR_MAX_DESCRIPTION Macro

Maximum error description length

C++

```
#define ERROR_MAX_DESCRIPTION 254
```

File

CardStructures.h (see page 178)

1.3.45 FONT_BOLD Macro

This is macro FONT_BOLD.

C++

```
#define FONT_BOLD 0x01
```

File

Graphics.h (see page 183)

1.3.46 FONT_ITALIC Macro

This is macro FONT_ITALIC.

C++

```
#define FONT_ITALIC 0x02
```

File

Graphics.h (see page 183)

1.3.47 FONT_NORMAL Macro

This is macro FONT_NORMAL.

C++

```
#define FONT_NORMAL 0x00
```

File

Graphics.h (see page 183)

1.3.48 FONT_STRIKEOUT Macro

This is macro FONT_STRIKEOUT.

C++

```
#define FONT_STRIKEOUT 0x08
```

File

Graphics.h (see page 183)

1.3.49 FONT_UNDERLINE Macro

This is macro FONT_UNDERLINE.

C++

```
#define FONT_UNDERLINE 0x04
```

File

Graphics.h (see page 183)

1.3.50 GetFileSHA256 Macro

This is macro GetFileSHA256.

C++

```
#define GetFileSHA256 GetFileSHA256A
```

File

Encryption.h (see page 180)

1.3.51 IID_PPV_ARG Macro

IID_PPV_ARG(IType, ppType) IType is the type of pType ppType is the variable of type IType that will be filled

RESULTS in: IID_IType, ppvType will create a compiler error if wrong level of indirection is used.

macro for QueryInterface and related functions that require a IID and a (void **) this will insure that the cast is safe and appropriate on C

C++

```
#define IID_PPV_ARG(IType, ppType) &IID_##IType, (void**)(ppType))
```

File

FileOperations.h (see page 181)

1.3.52 SaveCardToToXMLStreamEx Macro

This is macro SaveCardToToXMLStreamEx.

C++

```
#define SaveCardToToXMLStreamEx SaveCardToToXMLStreamExA
```

File

Swelio.h (see page 184)

1.3.53 SavePersonCsvToStream Macro

This is macro SavePersonCsvToStream.

C++

```
#define SavePersonCsvToStream SavePersonCsvToStreamA
```

File

Swelio.h (see page 184)

1.3.54 SIS_FIELD_MAX_BIRTHDATE_LEN Macro

Maximum length of the birth date field

C++

```
#define SIS_FIELD_MAX_BIRTHDATE_LEN 0x8
```

File

CardStructures.h (see page 178)

1.3.55 SIS_FIELD_MAX_CAPTUREDATE_LEN Macro

Maximum length of the capture date field

C++

```
#define SIS_FIELD_MAX_CAPTUREDATE_LEN 0x8
```

File

CardStructures.h (see page 178)

1.3.56 SIS_FIELD_MAX_CARDNUMBER_LEN Macro

Maximum length of the car number field

C++

```
#define SIS_FIELD_MAX_CARDNUMBER_LEN 0xa
```

File

CardStructures.h (see page 178)

1.3.57 SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN Macro

Maximum length of the social security number field

C++

```
#define SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN 0xb
```

File

CardStructures.h (see page 178)

1.3.58 SIS_FIELD_MAX_VALIDBEGIN_LEN Macro

Maximum length of the start validity date field

C++

```
#define SIS_FIELD_MAX_VALIDBEGIN_LEN 0x8
```

File

CardStructures.h (see page 178)

1.3.59 SIS_FIELD_MAX_VALIDEND_LEN Macro

Maximum length of the end validity date field

C++

```
#define SIS_FIELD_MAX_VALIDEND_LEN 0x8
```

File

CardStructures.h (see page 178)

1.3.60 SIS_MAX_CARDNAME_LEN Macro

Maximum length of the card name field

C++

```
#define SIS_MAX_CARDNAME_LEN 0x6
```

File

CardStructures.h (see page 178)

1.3.61 SIS_MAX_FIRSTNAMES_LEN Macro

Maximum length of the first name field

C++

```
#define SIS_MAX_FIRSTNAMES_LEN 0x18
```

File

CardStructures.h (see page 178)

1.3.62 SIS_MAX_INITIAL_LEN Macro

Maximum length of the initial field

C++

```
#define SIS_MAX_INITIAL_LEN 0x1
```

File

CardStructures.h (see page 178)

1.3.63 SIS_MAX_NAME_LEN Macro

Maximum length of the surname field

C++

```
#define SIS_MAX_NAME_LEN 0x30
```

File

CardStructures.h (see page 178)

1.3.64 SIS_MAX_SEX_LEN Macro

Maximum length of the sex field

C++

```
#define SIS_MAX_SEX_LEN 0x1
```

File

CardStructures.h (see page 178)

1.3.65 WIDTHBYTES Macro

This is macro WIDTHBYTES.

C++

```
#define WIDTHBYTES(i) (((i) + 31) / 32 * 4)
```

File

Graphics.h (see page 183)

1.4 Files

The following table lists files in this documentation.

Files

Name	Description
CardEvents.h (see page 177)	The definition of the possible card events
CardStructures.h (see page 178)	The definition of the information storage structures
Encryption.h (see page 180)	Encryption and decryption operations
FileOperations.h (see page 181)	Files and folders manipulations
Graphics.h (see page 183)	This is file Graphics.h.
NationalityConverter.h (see page 184)	Nationality to ISO code cenverter
quricol.h (see page 184)	QR Code generator
Swelio.h (see page 184)	Belgian electronic Id card access engine
System.h (see page 189)	Windows related routines
SystemInfo.h (see page 190)	Routines for querying information about operating system

1.4.1 CardEvents.h

The definition of the possible card events

Enumerations

	Name	Description
	tagCardEventType (see page 132)	The type of the reader event
	CardEventType (see page 142)	The type of the reader event

1.4.2 CardStructures.h

The definition of the information storage structures

Macros

Name	Description
EID_MAX_BIRTHDATE_LEN (see page 162)	Maximum length of the birthdate
EID_MAX_BIRTHPLACE_LEN (see page 162)	Maximum length of the birthplace
EID_MAX_BREXITMENTION1_LEN (see page 163)	Maximum length of the BREXIT mention field
EID_MAX_BREXITMENTION2_LEN (see page 163)	Maximum length of the BREXIT mention field
EID_MAX_CARD_NUMBER_LEN (see page 163)	Maximum length of the card number field
EID_MAX_CERT_LEN (see page 163)	Maximum length of the certificate data
EID_MAX_CHIP_NUMBER_LEN (see page 163)	Maximum length of the chip number field
EID_MAX_DATE_BEGIN_LEN (see page 164)	Maximum length of the begin date field
EID_MAX_DATE_END_LEN (see page 164)	Maximum length of the end date field
EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN (see page 164)	Maximum length of theDate and country of protection field
EID_MAX_DELIVERY_MUNICIPALITY_LEN (see page 164)	Maximum length of the name of the devivery municipality
EID_MAX_DOCUMENT_TYPE_LEN (see page 165)	Maximum length of the document type field
EID_MAX_DUPLICATE_LEN (see page 165)	Maximum length of the Duplicate field
EID_MAX_FIRST_NAME1_LEN (see page 165)	Maximum length of the first name
EID_MAX_FIRST_NAME2_LEN (see page 165)	Maximum length of the first name
EID_MAX_MEMBEROFFAMILY_LEN (see page 166)	Maximum length of the Member of family field
EID_MAX_MUNICIPALITY_LEN (see page 166)	Maximum length of the municipality name field
EID_MAX_NAME_LEN (see page 166)	Maximum length of the surname
EID_MAX_NATIONAL_NUMBER_LEN (see page 166)	Maximum length of the national number
EID_MAX_NATIONALITY_LEN (see page 166)	Maximum length of the nationality
EID_MAX_NOBLE_CONDITION_LEN (see page 167)	Maximum length of the noble condition field
EID_MAX_PICTURE_LEN (see page 167)	Maximum length of the picture data
EID_MAX_REGIONALFILENUMBER_LEN (see page 167)	Maximum length of regional file number field
EID_MAX_SEX_LEN (see page 167)	Maximum length of the sex field
EID_MAX_SPECIAL_STATUS_LEN (see page 168)	Maximum length of the special status field
EID_MAX_SPECIALORGANIZATION_LEN (see page 168)	Maximum length of the Special organization field
EID_MAX_STREET_LEN (see page 168)	Maximum length of the street name field
EID_MAX_VAT1_LEN (see page 168)	Maximum length of the VAT1 field
EID_MAX_VAT2_LEN (see page 168)	Maximum length of the VAT2 field
EID_MAX_WORKPERMITTYPE_LEN (see page 169)	Maximum length of the type of the workpermit
EID_MAX_ZIP_LEN (see page 169)	Maximum length of the ZIP code field
ERR_CARD_ERROR (see page 169)	This is macro ERR_CARD_ERROR.
ERR_CERTIFICATE_ATTR (see page 169)	This is macro ERR_CERTIFICATE_ATTR.
ERR_MESSAGE_DECODE (see page 170)	This is macro ERR_MESSAGE_DECODE.

ERR_MESSAGE_ENCODE (see page 170)	This is macro ERR_MESSAGE_ENCODE.
ERR_MESSAGE_PARAM (see page 170)	This is macro ERR_MESSAGE_PARAM.
ERR_MESSAGE_UPDATE (see page 170)	This is macro ERR_MESSAGE_UPDATE.
ERR_NO_CERTIFICATE (see page 170)	This is macro ERR_NO_CERTIFICATE.
ERR_NO_DATA (see page 171)	This is macro ERR_NO_DATA.
ERR_NO_PRIVATE_KEY (see page 171)	This is macro ERR_NO_PRIVATE_KEY.
ERR_OPEN_CERTIFICATE (see page 171)	This is macro ERR_OPEN_CERTIFICATE.
ERR_SIGN_ERROR (see page 171)	This is macro ERR_SIGN_ERROR.
ERR_TIMESTAMP (see page 172)	This is macro ERR_TIMESTAMP.
ERROR_MAX_DESCRIPTION (see page 172)	Maximum error description length
SIS_FIELD_MAX_BIRTHDATE_LEN (see page 174)	Maximum length of the birth date field
SIS_FIELD_MAX_CAPTUREDATE_LEN (see page 174)	Maximum length of the capture date field
SIS_FIELD_MAX_CARDNUMBER_LEN (see page 175)	Maximum length of the car number field
SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN (see page 175)	Maximum length of the social security number field
SIS_FIELD_MAX_VALIDBEGIN_LEN (see page 175)	Maximum length of the start validity date field
SIS_FIELD_MAX_VALIDEND_LEN (see page 175)	Maximum length of the end validity date field
SIS_MAX_CARDNAME_LEN (see page 176)	Maximum length of the card name field
SIS_MAX_FIRSTNAMES_LEN (see page 176)	Maximum length of the first name field
SIS_MAX_INITIAL_LEN (see page 176)	Maximum length of the initial field
SIS_MAX_NAME_LEN (see page 176)	Maximum length of the surname field
SIS_MAX_SEX_LEN (see page 176)	Maximum length of the sex field

Structures

	Name	Description
	structErrorInformation (see page 132)	Information about error when operation with the card or certificate is performed
	tagEidAddressA (see page 133)	EID address information, stored on the card - ANSI version
	tagEidAddressW (see page 133)	EID address information, stored on the card - UNICODE version
	tagEidCertificate (see page 133)	Certificate, stored on EID card
	tagEidIdentityA (see page 134)	Identity information stored on EID card - ANSI version
	tagEidIdentityExA (see page 135)	Identity information stored on EID card - ANSI version
	tagEidIdentityExW (see page 137)	Identity information stored on EID card - UNICODE version
	tagEidIdentityW (see page 138)	Identity information stored on EID card - UNICODE version
	tagEidPicture (see page 140)	Raw picture data from EID card
	tagSISRecordA (see page 140)	Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	tagSISRecordW (see page 141)	Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	EidAddressA (see page 142)	EID address information, stored on the card - ANSI version
	EidAddressW (see page 142)	EID address information, stored on the card - UNICODE version
	EidCertificate (see page 143)	Certificate, stored on EID card
	EidIdentityA (see page 143)	Identity information stored on EID card - ANSI version
	EidIdentityExA (see page 145)	Identity information stored on EID card - ANSI version
	EidIdentityExW (see page 146)	Identity information stored on EID card - UNICODE version
	EidIdentityW (see page 148)	Identity information stored on EID card - UNICODE version
	EidPicture (see page 149)	Raw picture data from EID card

	ErrorInformation (see page 149)	Information about error when operation with the card or certificate is performed
	PEidAddressA (see page 150)	EID address information, stored on the card - ANSI version
	PEidAddressW (see page 150)	EID address information, stored on the card - UNICODE version
	PEidCertificate (see page 150)	Certificate, stored on EID card
	PEidIdentityA (see page 151)	Identity information stored on EID card - ANSI version
	PEidIdentityExA (see page 152)	Identity information stored on EID card - ANSI version
	PEidIdentityExW (see page 154)	Identity information stored on EID card - UNICODE version
	PEidIdentityW (see page 155)	Identity information stored on EID card - UNICODE version
	PeidPicture (see page 157)	Raw picture data from EID card
	PErrorInformation (see page 157)	Information about error when operation with the card or certificate is performed
	PSISRecordA (see page 157)	Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	PSISRecordW (see page 158)	Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	SISRecordA (see page 159)	Public information stored on Belgian SIS card - ANSI version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)
	SISRecordW (see page 160)	Public information stored on Belgian SIS card - UNICODE version. The SIS card is the social security card of each Belgian resident (Belgian or foreigner)

1.4.3 Encryption.h

Encryption and decryption operations

Functions

	Name	Description
	CardDecryptFileA (see page 15)	Decrypt file using Belgian Id card
	CardDecryptFileW (see page 16)	Decrypt file using Belgian Id card
	CardEncryptFileA (see page 16)	Encrypt file using Belgian Id card
	CardEncryptFileW (see page 16)	Encrypt file using Belgian Id card
	CheckMD5 (see page 22)	Checks the MD5 hash value of the memory buffer
	CheckSHA1 (see page 23)	Checks the SHA1 hash value of the memory buffer
	CheckSHA256 (see page 23)	Checks the SHA256 hash value of the memory buffer
	DecryptFileAESA (see page 30)	Decrypts file using AES algorithm.
	DecryptFileAESW (see page 30)	Decrypts file using AES algorithm.
	EncryptFileAESA (see page 38)	Encrypts file using AES algorithm.
	EncryptFileAESW (see page 38)	Encrypts file using AES algorithm.
	GetFileMD5A (see page 62)	Gets the MD5 hash value for the file
	GetFileMD5W (see page 62)	Gets the MD5 hash value for the file
	GetFileSHA1A (see page 63)	Gets the SHA1 hash value for the file
	GetFileSHA1W (see page 64)	Gets the SHA1 hash value for the file
	GetFileSHA256A (see page 64)	Gets the SHA256 hash value for the file
	GetFileSHA256W (see page 64)	Gets the SHA256 hash value for the file
	GetMD5 (see page 67)	Gets the MD5 hash value for the content of the memory buffer
	GetSHA1 (see page 71)	Gets the SHA1 hash value for the content of the memory buffer

	GetSHA256 (see page 71)	Gets the SHA256 hash value for the content of the memory buffer

Macros

Name	Description
GetFileSHA256 (see page 173)	This is macro GetFileSHA256.

1.4.4 FileOperations.h

Files and folders manipulations

Functions

	Name	Description
	AllocateBuffer (see page 11)	Allocates the buffer in memory
	ClearFileAttributesA (see page 23)	This function sets the file attributes to normal.
	ClearFileAttributesW (see page 24)	This function sets the file attributes to normal.
	CreateUnicodeFileA (see page 27)	Creates UNICODE file
	CreateUnicodeFileW (see page 27)	Creates UNICODE file
	DeallocateBuffer (see page 29)	Deallocates the memory buffer
	DeleteToRecycleBinA (see page 31)	Deletes file to the Windows Recycle Bin
	DeleteToRecycleBinW (see page 31)	Deletes file to WIndows Recycle Bin
	DirectoryExistsA (see page 32)	Determines whether a specified directory exists.
	DirectoryExistsW (see page 33)	Determines whether a specified directory exists.
	FileCloseA (see page 39)	Concludes input/output (I/O) to a file opened using the FileCreateRewrite function.
	FileCloseW (see page 39)	Concludes input/output (I/O) to a file opened using the FileCreateRewrite function.
	FileCopyA (see page 39)	The CopyFile function copies an existing file to a new file.
	FileCopyW (see page 40)	The CopyFile function copies an existing file to a new file.
	FileCreateRewriteA (see page 40)	Creates new or overwrites existing file
	FileCreateRewriteW (see page 40)	Creates new or overwrites existing file
	FileDeleteA (see page 41)	Deletes a file from disk.
	FileDeleteW (see page 41)	Deletes a file from disk.
	FileExistsA (see page 42)	Tests whether a specified file exists.
	FileExistsW (see page 42)	Tests whether a specified file exists.
	FileExtensionIsA (see page 42)	Checks the file extension
	FileExtensionIsW (see page 43)	Checks the file extension
	FileGetSizeA (see page 43)	Retrieves the size of a specified file.
	FileGetSizeW (see page 43)	Retrieves the size of a specified file.
	FileIsExeA (see page 44)	Checks if the file is a Windows executable
	FileIsExeW (see page 44)	Checks if the file is a Windows executable
	FileIsIconA (see page 45)	Checks if the file is a Windows icon (.ico) file
	FileIsIconW (see page 45)	Checks if the file is a Windows icon (.ico) file
	FileIsImageA (see page 45)	Checks if the file is an image file

	FileIsImageW (see page 46)	Checks if the file is an image file
	FileIsLink (see page 46)	Checks to see if the file specified by file name is a Microsoft Windows shortcut (.Lnk) file (and is neither a file nor a folder).
	FileOrFolderExistsA (see page 46)	Checks if the file or folder with the given name exists
	FileOrFolderExistsW (see page 47)	Checks if the file or folder with the given name exists
	FileRenameA (see page 47)	Renames the file
	FileRenameW (see page 47)	Renames the file
	FileWriteA (see page 48)	Writes string to the file
	FileWriteCharA (see page 48)	Writes one character to the file
	FileWriteCharW (see page 48)	Writes one character to the file
	FileWriteNewLineA (see page 49)	Writes new line sequence to the file
	FileWriteNewLineW (see page 49)	Writes new line sequence to the file
	FileWriteW (see page 49)	Writes string to the file
	FullPathA (see page 50)	Gets the full path to the file based on file name
	FullPathW (see page 51)	Gets the full path to the file based on file name
	GetAllFiles (see page 58)	Returns the names of files in a specified directory.
	GetFilesCountA (see page 62)	Calculates the number of files in the given folder
	GetFilesCountW (see page 63)	Calculates the number of files in the given folder
	IsAnimatedGIFA (see page 75)	Checks if the file is an animated GIF image file
	IsAnimatedGIFW (see page 75)	Checks if the file is an animated GIF image file
	IsDirectoryA (see page 78)	Verifies that a path is a valid directory.
	IsDirectoryW (see page 78)	Verifies that a path is a valid directory.
	IsUnicodeFileA (see page 83)	Checks if the file is UNICODE file
	IsUnicodeFileW (see page 84)	Checks if the file is UNICODE file
	IsValidFileNameA (see page 84)	Checks if provided string is a valid file name
	IsValidFileNameW (see page 84)	Checks if provided string is a valid file name
	IsValidPathNameA (see page 85)	Checks if provided string is a valid file path
	IsValidPathNameW (see page 85)	Checks if provided string is a valid file path
	ReadBufferFromFileA (see page 95)	Reads the content of the file to the memory buffer
	ReadBufferFromFileW (see page 95)	Reads the content of the file to the memory buffer
	ShellCopyFileA (see page 124)	Copies file to the new location
	ShellCopyFileW (see page 125)	Copies file to the new location
	StripFileNameA (see page 126)	Replaces environment variable names with values
	StripFileNameW (see page 127)	Replaces environment variable names with values
	WriteBufferToFileA (see page 130)	Writes the memory buffer to file
	WriteBufferToFileW (see page 130)	Writes the memory buffer to file

Macros

Name	Description
IID_PPV_ARG (see page 173)	IID_PPV_ARG(IType, ppType) IType is the type of pType ppType is the variable of type IType that will be filled RESULTS in: IID_IType, ppvType will create a compiler error if wrong level of indirection is used. macro for QueryInterface and related functions that require a IID and a (void **) this will insure that the cast is safe and appropriate on C

1.4.5 Graphics.h

This is file Graphics.h.

Functions

	Name	Description
	AlphaBlendBitmap (see page 15)	This is function AlphaBlendBitmap.
	AlphaBlendNative (see page 15)	This is function AlphaBlendNative.
	CloneFont (see page 24)	This is function CloneFont.
	CopyNativeBitmap (see page 26)	This is function CopyNativeBitmap.
	CreateNativeBitmap (see page 26)	This is function CreateNativeBitmap.
	CreateWindowsFont (see page 27)	This is function CreateWindowsFont.
	DestroyFont (see page 32)	This is function DestroyFont.
	DpiY (see page 33)	This is function DpiY.
	DrawAlphaText (see page 34)	This is function DrawAlphaText.
	DrawAlphaTextRect (see page 34)	This is function DrawAlphaTextRect.
	DrawNativeBitmap (see page 35)	This is function DrawNativeBitmap.
	DrawTextDirect (see page 35)	This is function DrawTextDirect.
	DrawTextDirectEx (see page 35)	This is function DrawTextDirectEx.
	DrawTextGlow (see page 35)	This is function DrawTextGlow.
	DrawTextLine (see page 36)	This is function DrawTextLine.
	DrawTextOutline (see page 36)	This is function DrawTextOutline.
	DrawTextRect (see page 36)	This is function DrawTextRect.
	EmToPixels (see page 37)	This is function EmToPixels.
	GetTextLineSize (see page 73)	This is function GetTextLineSize.
	GetTextSize (see page 73)	This is function GetTextSize.
	GetTextSizeEx (see page 73)	This is function GetTextSizeEx.
	LoadBitmapJPG (see page 88)	This is function LoadBitmapJPG.
	LoadBitmapPNG (see page 88)	This is function LoadBitmapPNG.
	LoadPNGResource (see page 90)	This is function LoadPNGResource.
	MakeCompatibleBitmap (see page 91)	This is function MakeCompatibleBitmap.
	PointsToPixels (see page 92)	This is function PointsToPixels.
	StretchNativeBitmap (see page 126)	This is function StretchNativeBitmap.

Macros

Name	Description
FONT_BOLD (see page 172)	This is macro FONT_BOLD.
FONT_ITALIC (see page 172)	This is macro FONT_ITALIC.
FONT_NORMAL (see page 172)	This is macro FONT_NORMAL.
FONT_STRIKEOUT (see page 173)	This is macro FONT_STRIKEOUT.
FONT_UNDERLINE (see page 173)	This is macro FONT_UNDERLINE.
WIDTHBYTES (see page 177)	This is macro WIDTHBYTES.

1.4.6 NationalityConverter.h

Nationality to ISO code cenverter

Functions

	Name	Description
	GetISOCodeA (see page 66)	Returns the country ISO code based on the nationality string
	GetISOCodeW (see page 66)	Returns the country ISO code based on the nationality string

1.4.7 quricol.h

QR Code generator

Functions

	Name	Description
	DestroyImageBuffer (see page 32)	Destroys the memory buffer
	GenerateBMPA (see page 53)	Generates Windows Bitmap file with QR Code image
	GenerateBMPW (see page 53)	Generates Windows Bitmap file with QR Code image
	GeneratePNGA (see page 56)	Generates PNG file with QR Code image
	GeneratePNGW (see page 56)	Generates PNG file with QR Code image
	GetHBitmapA (see page 65)	Generates Windows Bitmap in memory with QR Code image
	GetHBitmapW (see page 65)	Generates Windows Bitmap in memory with QR Code image
	GetPNGA (see page 67)	Writes PNG image to the memory buffer.
	GetPNGW (see page 68)	Writes PNG image to the memory buffer.

1.4.8 Swelio.h

Belgian electronic Id card access engine

Functions

	Name	Description
	ActivateCard (see page 10)	Established communication between the card and the reader
	ActivateCardEx (see page 10)	Established communication between the card and the reader
	AddFileToContainer (see page 10)	Add existing file to the container
	CardSignCadesT (see page 17)	Sign data with eID card according to CADES-T standard
	CardSignCadesTEx (see page 17)	Sign data with eID card according to CADES-T standard
	CardSignCMS (see page 18)	Sign data with eID card according to CMS standard
	CardSignCMSEx (see page 18)	Sign data with eID card according to CMS standard
	CertSignCadesT (see page 19)	Sign data with the certificate file according to CADES-T standard
	CertSignCadesTData (see page 19)	Sign data with the certificate according to CADES-T standard
	CertSignCadesTEx (see page 20)	Sign data with the certificate file according to CADES-T standard
	CertSignCMS (see page 21)	Sign data with the certificate file according to CMS standard

	CertSignCMSData (see page 21)	Sign data with the certificate according to CMS standard
	CertSignCMSEx (see page 22)	Sign data with the certificate file according to CMS standard
	ContainerCertificate (see page 25)	Assign certificate for signing ASIC container
	ContainerEidCertificate (see page 25)	Select EID card certificate to sign ASIC container
	ContainerPickCertificate (see page 25)	Pick certificate to sign ASIC container
	CreateCardBuffer (see page 26)	Creates XML buffer
	DeactivateCard (see page 28)	Terminates a cennection between a smart card and a reader
	DeactivateCardEx (see page 28)	Terminates a connection between a smart card and a reader
	DeleteCardBuffer (see page 31)	Deletes XML buffer
	DisplayCertificate (see page 33)	Displays the dialog window with certificate information
	EncodeCertificate (see page 37)	Performs Base64 encoding of the certificate
	EncodePhoto (see page 37)	Performs Base64 encoding of the photo
	FreeContainer (see page 50)	Deallocates ASIC container
	GenerateAuthenticationSignatureA (see page 51)	Generate authentication signature
	GenerateAuthenticationSignatureExA (see page 51)	Generate authentication signature
	GenerateAuthenticationSignatureExW (see page 52)	Generate authentication signature
	GenerateAuthenticationSignatureW (see page 52)	Generate authentication signature
	GenerateNonRepudiationSignatureA (see page 54)	Generate non repudiation signature
	GenerateNonRepudiationSignatureExA (see page 54)	Generate non repudiation signature
	GenerateNonRepudiationSignatureExW (see page 55)	Generate non repudiation signature
	GenerateNonRepudiationSignatureW (see page 55)	Generate non repudiation signature
	GenerateQRCodeA (see page 57)	Read eID card and save the identity information and address to PNG QR Code file
	GenerateQRCodeExA (see page 57)	Read eID card and save the identity information and address to PNG QR Code file
	GenerateQRCodeExW (see page 57)	Read eID card and save the identity information and address to PNG QR Code file
	GenerateQRCodeW (see page 58)	Read eID card and save the identity information and address to PNG QR Code file
	GetCardBufferA (see page 59)	Gets XML or CSV information from the memory buffer
	GetCardBufferSize (see page 59)	This function returns the size of the buffer needed to hold the information from the eID card in the XML or CSV format
	GetCardBufferW (see page 59)	Gets XML or CSV information from the memory buffer
	GetCardSerialNumber (see page 60)	Get the serial number of EID card
	GetCardVersion (see page 60)	Get the applet version number for card in the reader with specified number
	GetEncodedCertificateSize (see page 61)	Returns the size of the Base64 encoded certificate
	GetEncodedPhotoSize (see page 61)	Calculates buffer size for Base64 encoded photo
	GetReaderIndexA (see page 68)	Returns the zero-based reader index with specified name
	GetReaderIndexW (see page 68)	Returns the zero-based reader index with specified name
	GetReaderNameA (see page 69)	Returns the name of the card reader

	GetReaderNameLenA (see page 69)	Returns the length of the reader name
	GetReaderNameLenW (see page 70)	Returns the length of the reader name
	GetReaderNameW (see page 70)	Returns the name of the card reader
	GetReadersCount (see page 70)	Get number of card readers connected to PC
	GetSelectedReaderIndex (see page 71)	Returns the index of the active smart card reader
	GetSupportSIS (see page 73)	Checks if the SIS cards are supported by the engine
	IgnoreHardwareEvents (see page 74)	Ignore USB reader insert / remove events
	IgnoreServiceEvents (see page 74)	Ignore smartcard service stop events when reporting readers list change
	InitializeContainer (see page 75)	Initializes ASIC container
	IsCardActivated (see page 76)	Checks the connection between a smart card and a reader
	IsCardActivatedEx (see page 76)	Checks the connection between a smart card and a reader
	IsCardPresent (see page 76)	Checks if the card is present in the card reader
	IsCardPresentEx (see page 77)	Checks if the card is present in the card reader
	IsCardStillInserted (see page 77)	Checks if the card is still inserted in the card reader
	IsCardStillInsertedEx (see page 77)	Checks if the card is still inserted in the card reader
	IsEIDCard (see page 79)	Check if Belgian EID card is inserted into card reader
	IsEIDCardEx (see page 79)	Check if Belgian EID card is inserted into card reader
	IsEngineActive (see page 79)	Checks if the Swelio Engine is activated
	IsFemaleA (see page 80)	Checks if the card owner is female
	IsFemaleW (see page 80)	Checks if the card owner is female
	IsMaleA (see page 80)	Checks if the card owner is male
	IsMaleW (see page 81)	Checks if the card owner is male
	IsSISCard (see page 82)	Check if Belgian SIS card is inserted into card reader
	IsSISCardEx (see page 83)	Check if Belgian SIS card is inserted into card reader
	LoadCertificateA (see page 88)	Reads the certificate from a file
	LoadCertificateW (see page 89)	Reads the certificate from a file
	LoadIdentityA (see page 89)	Reads the raw identity information from a file
	LoadIdentityW (see page 89)	Reads the raw identity information from a file
	LoadPhotoA (see page 90)	Loads photo from a file
	LoadPhotoW (see page 90)	Loads photo from a file
	ReadAddressA (see page 93)	Read address information from Belgian eID card
	ReadAddressExA (see page 93)	Read address information from Belgian eID card
	ReadAddressExW (see page 94)	Read address information from Belgian eID card
	ReadAddressW (see page 94)	Read address information from Belgian eID card
	ReadAuthenticationCertificate (see page 94)	Read Authentication Certificate to memory
	ReadAuthenticationCertificateEx (see page 95)	Read Authentication Certificate to memory
	ReadCaCertificate (see page 96)	Read Ca Certificate to memory
	ReadCaCertificateEx (see page 96)	Read Ca Certificate to memory
	ReadIdentityA (see page 97)	Read identity information from Belgian eID card
	ReadIdentityExA (see page 97)	Read identity information from Belgian eID card
	ReadIdentityExW (see page 97)	Read identity information from Belgian eID card
	ReadIdentityW (see page 98)	Read identity information from Belgian eID card
	ReadNonRepudiationCertificate (see page 98)	Read Non Repudiation Certificate to memory
	ReadNonRepudiationCertificateEx (see page 99)	Read Non Repudiation Certificate to memory

	ReadPhoto (see page 99)	Reads a photo from a card
	ReadPhotoAsBitmap (see page 99)	Reads the picture from the card, converts it to bitmap and returns the bitmap handle Decription: Reads the photo from the Belgian eID card and returns the bitmap handle Reading the photo from the card is a time consuming operation. Do it only when needed.
	ReadPhotoAsBitmapEx (see page 100)	Reads the picture from the card, converts it to bitmap and returns the bitmap handle Decription: Reads the photo from the Belgian eID card and returns the Windows bitmap handle Reading the photo from the card is a time consuming operation. Do it only when needed.
	ReadPhotoEx (see page 100)	Reads a photo from a card
	ReadRootCaCertificate (see page 101)	Read Root Ca Certificate to memory
	ReadRootCaCertificateEx (see page 101)	Read Root Ca Certificate to memory
	ReadRrnCertificate (see page 101)	Read Rrn Certificate to memory
	ReadRrnCertificateEx (see page 102)	Read Rrn Certificate to memory
	ReadSISCardA (see page 102)	Read Belgian SIS card.
	ReadSISCardExA (see page 103)	Read Belgian SIS card.
	ReadSISCardExW (see page 103)	Read Belgian SIS card.
	ReadSISCardW (see page 103)	Read Belgian SIS card.
	ReloadReadersList (see page 104)	Reloads the list of the available card readers
	RemoveCallback (see page 104)	Remove callback procedure for card events
	SaveAuthenticationCertificateA (see page 106)	Save Authentication Certificate to a file
	SaveAuthenticationCertificateExW (see page 106)	Save Authentication Certificate to a file
	SaveAuthenticationCertificateW (see page 107)	Save Authentication Certificate to a file
	SaveCaCertificateA (see page 107)	Save Ca Certificate to a file
	SaveCaCertificateExW (see page 107)	Save Ca Certificate to a file
	SaveCaCertificateW (see page 108)	Save Ca Certificate to a file
	SaveCardToToXMLStreamExA (see page 108)	Read eID card and save the information to XML buffer
	SaveCardToToXMLStreamExW (see page 108)	Read eID card and save the information to XML buffer
	SaveCardToXmlA (see page 109)	Read eID card and save the information to XML file
	SaveCardToXmlExA (see page 109)	Read eID card and save the information to XML file
	SaveCardToXmlExW (see page 110)	Read eID card and save the information to XML file
	SaveCardToXmlW (see page 110)	Read eID card and save the information to XML file
	SaveContainer (see page 111)	Save container to the file
	SaveIdentityA (see page 111)	Saves indentity infornation to a file
	SaveIdentityW (see page 111)	Saves indentity infornation to a file
	SaveNonRepudiationCertificateA (see page 112)	Save Non Repudiation Certificate to a file
	SaveNonRepudiationCertificateExW (see page 112)	Save Non Repudiation Certificate to a file
	SaveNonRepudiationCertificateW (see page 112)	Save Non Repudiation Certificate to a file
	SavePersonCsvToStreamA (see page 113)	Read eID card and save the identity information to CSV memory buffer
	SavePersonCsvToStreamW (see page 113)	Read eID card and save the identity information to CSV memory buffer

	SavePersonToCsvA (see page 114)	Read eID card and save the identity information and address to CSV file
	SavePersonToCsvExA (see page 114)	Read eID card and save the identity information and address to CSV file
	SavePersonToCsvExW (see page 114)	Read eID card and save the identity information and address to CSV file
	SavePersonToCsvW (see page 115)	Read eID card and save the identity information and address to CSV file
	SavePhotoA (see page 115)	Save photo to a file
	SavePhotoAsBitmapA (see page 116)	Save the picture from the card to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsBitmapExA (see page 116)	Reads the picture from the card and saves it to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsBitmapExW (see page 116)	Reads the picture from the card and saves it to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsBitmapW (see page 117)	Save the picture from the card to Windows Bitmap file Decription: Reads the photo from the Belgian eID card and writes it to the file as bitmap image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegA (see page 117)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegExA (see page 117)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegExW (see page 118)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoAsJpegW (see page 118)	Save the picture from the card to JPG file Decription: Reads the photo from the Belgian eID card and writes it to the file as JPG image. Reading the photo from the card is a time consuming operation. Do it only when needed.
	SavePhotoW (see page 119)	Saves photo to a file
	SaveRootCaCertificateA (see page 119)	Save Root Ca Certificate to a file
	SaveRootCaCertificateExW (see page 119)	Save Root Ca Certificate to a file
	SaveRootCaCertificateW (see page 120)	Save Root Ca Certificate to a file
	SaveRrnCertificateA (see page 120)	Save RRN Certificate to a file
	SaveRrnCertificateExW (see page 120)	Save RRN Certificate to a file
	SaveRrnCertificateW (see page 121)	Save RRN Certificate to a file
	SelectReader (see page 121)	When more than 1 reader connected, select the reader with specified number
	SelectReaderByNameA (see page 121)	Select active smart card reader by providing the reader name

	SelectReaderByNameW (see page 122)	Select active smart card reader by providing the reader name
	SendAPDU (see page 122)	This is function SendAPDU.
	SetCallback (see page 122)	Activates callback procedure for card status change event
	SetMWCompatibility (see page 123)	Set the compatibility mode with the old version of the oficial EID MiddleWare
	SetSupportSIS (see page 124)	Activates or deactivates SIS card support by engine
	StartEngine (see page 126)	Activates the Swelio Engine.
	StopEngine (see page 126)	Deactivates the Swelio Engine
	VerifyPinA (see page 128)	Verify PIN code
	VerifyPinExA (see page 128)	Verify PIN code
	VerifyPinExW (see page 129)	Verify PIN code
	VerifyPinW (see page 129)	Verify PIN code
	VerifySignature (see page 130)	Verifies the signature from the specified hash value.

Macros

Name	Description
SaveCardToToXMLStreamEx (see page 174)	This is macro SaveCardToToXMLStreamEx.
SavePersonCsvToStream (see page 174)	This is macro SavePersonCsvToStream.

1.4.9 System.h

Windows related routines

Functions

	Name	Description
	AddRemoveMessageFilter (see page 11)	Adds or removes a message from the User Interface Privilege Isolation (UIPI) message filter.
	AllocateDefaultHWNDA (see page 12)	This function creates the invisible tool window
	AllocateDefaultHWNDW (see page 12)	This function creates the invisible tool window
	AllocateHWNDA (see page 13)	This function creates the invisible tool window using the provided window procedure
	AllocateHWNDW (see page 13)	This function creates the invisible tool window using the provided window procedure
	AllocateLayeredWindowA (see page 13)	This function creates the layered window using the provided window class name
	AllocateLayeredWindowW (see page 14)	This function creates the layered window using the provided window class name
	AllocateWindowClassA (see page 14)	This function creates the standard window using the provided window class name
	AllocateWindowClassW (see page 14)	This function creates the standard window using the provided window class name
	BringWindowToFront (see page 15)	This function brings the specified window to the top of the z-order.
	ClearUnusedMemory (see page 24)	Clears unused memory and minimized the application memory usage
	DeallocateHWNDA (see page 29)	This function destroys the specified window.
	DeallocateHWNDW (see page 29)	This function destroys the specified window.

	DrawLayeredWindow (see page 34)	Repaints the surface of the layered window
	EmptyRecycleBin (see page 36)	Empties the recycle bin
	fpreset (see page 50)	This is function fpreset.
	GetStartupA (see page 72)	Checks if the application is registered to run when Windows starts
	GetStartupW (see page 72)	Checks if the application is registered to run when Windows starts
	HibernateWindows (see page 74)	Hibernates Windows
	LayeredWndProcA (see page 87)	The default window procedure for the layered window
	LayeredWndProcW (see page 87)	The default window procedure for the layered window
	MakeSoundFromFileA (see page 91)	Plays the wave sound from the file
	MakeSoundFromFileW (see page 91)	Plays the wave sound from the file
	MakeSoundFromResourceA (see page 92)	Plays the wave sound from the resource
	MakeSoundFromResourceW (see page 92)	Plays the wave sound from the resource
	RemoveStartupA (see page 105)	Removes the application from the list of the automatically started applications
	RemoveStartupW (see page 105)	Removes the application from the list of the automatically started applications
	RestoreWindowSubclassA (see page 105)	Restores window standard procedure
	RestoreWindowSubclassW (see page 106)	Restores window standard procedure
	SetStartupA (see page 123)	Register application to run when Windows starts
	SetStartupW (see page 123)	Register application to run when Windows starts
	ShutdownWindows (see page 125)	Logs off the interactive user, shuts down the system.
	SuspendWindows (see page 127)	Suspends Windows
	TurnMonitorOff (see page 127)	Turns the monitor off
	TurnMonitorOn (see page 128)	Turns the monitor on
	UpdateWindowPosition (see page 128)	Updated the window position

1.4.10 SystemInfo.h

Routines for querying information about operating system

Functions

	Name	Description
	CurrentIPAddressA (see page 27)	Returns the IP address
	CurrentIPAddressW (see page 28)	Returns the IP address
	IsCitrixSession (see page 78)	Checks if application is running in Citrix session
	IsConnectedToInternet (see page 78)	Checks if PC is connected to Internet
	IsMediaCenter (see page 81)	Checks if the Media Center version of Windows is installed
	IsMetroActive (see page 81)	Checks if metro interface is active
	IsMultiTouchReady (see page 82)	Checks if the system is multi touch ready
	IsNativeWin64 (see page 82)	Checks if the application is native 64 bit executable

	IsRemoteSession (see page 82)	Checks if application is running in RDP session
	IsTabletPC (see page 83)	Checks if the application is running on the Tablet PC
	IsWindows10 (see page 86)	Checks if PC is running Windows 10 or better
	IsWindows7 (see page 86)	Checks if PC is running Windows 7 or better
	IsWindows8 (see page 86)	Checks if PC is Running Windows 8 or better
	IsWindowsVista (see page 86)	Checks if PC is running Windows Vista or better
	IsWindowsXP (see page 87)	Checks if PC is running Windows XP
	IsWindowsXPSP2 (see page 87)	Checks if PC is running Windows XP with Service Pack 2 installed
	IsWow64 (see page 87)	Checks if the 32 bit application runs on 64 bit Windows
	PortAvailable (see page 93)	Checks if the port with specified number is available
	RecycleBinEmpty (see page 104)	Returns TRUE if Windows Recyce Bin is empty

Index

A

ActivateCard 10
ActivateCard function 10
ActivateCardEx 10
ActivateCardEx function 10
AddFileToContainer 10
AddFileToContainer function 10
AddRemoveMessageFilter 11
AddRemoveMessageFilter function 11
AllocateBuffer 11
AllocateBuffer function 11
AllocateDefaultHWNDA 12
AllocateDefaultHWNDA function 12
AllocateDefaultHWNDW 12
AllocateDefaultHWNDW function 12
AllocateHWNDA 13
AllocateHWNDA function 13
AllocateHWNDW 13
AllocateHWNDW function 13
AllocateLayeredWindowA 13
AllocateLayeredWindowA function 13
AllocateLayeredWindowW 14
AllocateLayeredWindowW function 14
AllocateWindowClassA 14
AllocateWindowClassA function 14
AllocateWindowClassW 14
AllocateWindowClassW function 14
AlphaBlendBitmap 15
AlphaBlendBitmap function 15
AlphaBlendNative 15
AlphaBlendNative function 15

B

BringWindowToFront 15
BringWindowToFront function 15

C

CardDecryptFileA 15
CardDecryptFileA function 15
CardDecryptFileW 16
CardDecryptFileW function 16
CardEncryptFileA 16
CardEncryptFileA function 16
CardEncryptFileW 16
CardEncryptFileW function 16
CardEvents.h 177
CardEventType 142
CardEventType enumeration 142
CardSignCadesT 17
CardSignCadesT function 17
CardSignCadesTEx 17
CardSignCadesTEx function 17
CardSignCMS 18
CardSignCMS function 18
CardSignCMSEx 18
CardSignCMSEx function 18
CardStructures.h 178
CertSignCadesT 19
CertSignCadesT function 19
CertSignCadesTData 19
CertSignCadesTData function 19
CertSignCadesTEx 20
CertSignCadesTEx function 20
CertSignCMS 21
CertSignCMS function 21
CertSignCMSData 21
CertSignCMSData function 21
CertSignCMSEx 22
CertSignCMSEx function 22
CheckMD5 22
CheckMD5 function 22
CheckSHA1 23
CheckSHA1 function 23
CheckSHA256 23
CheckSHA256 function 23
ClearFileAttributesA 23
ClearFileAttributesA function 23
ClearFileAttributesW 24
ClearFileAttributesW function 24
ClearUnusedMemory 24
ClearUnusedMemory function 24

CloneFont 24
CloneFont function 24
ContainerCertificate 25
ContainerCertificate function 25
ContainerEidCertificate 25
ContainerEidCertificate function 25
ContainerPickCertificate 25
ContainerPickCertificate function 25
CopyNativeBitmap 26
CopyNativeBitmap function 26
CreateCardBuffer 26
CreateCardBuffer function 26
CreateNativeBitmap 26
CreateNativeBitmap function 26
CreateUnicodeFileA 27
CreateUnicodeFileA function 27
CreateUnicodeFileW 27
CreateUnicodeFileW function 27
CreateWindowsFont 27
CreateWindowsFont function 27
CurrentIPAddressA 27
CurrentIPAddressA function 27
CurrentIPAddressW 28
CurrentIPAddressW function 28

D

DeactivateCard 28
DeactivateCard function 28
DeactivateCardEx 28
DeactivateCardEx function 28
DeallocateBuffer 29
DeallocateBuffer function 29
DeallocateHWNDA 29
DeallocateHWNDA function 29
DeallocateHWNDW 29
DeallocateHWNDW function 29
DecryptFileAESA 30
DecryptFileAESA function 30
DecryptFileAESW 30
DecryptFileAESW function 30
DeleteCardBuffer 31
DeleteCardBuffer function 31
DeleteToRecycleBinA 31
DeleteToRecycleBinA function 31
DeleteToRecycleBinW 31
DeleteToRecycleBinW function 31
DestroyFont 32
DestroyFont function 32
DestroyImageBuffer 32
DestroyImageBuffer function 32
DirectoryExistsA 32
DirectoryExistsA function 32
DirectoryExistsW 33
DirectoryExistsW function 33
DisplayCertificate 33
DisplayCertificate function 33
DpiY 33
DpiY function 33
DrawAlphaText 34
DrawAlphaText function 34
DrawAlphaTextRect 34
DrawAlphaTextRect function 34
DrawLayeredWindow 34
DrawLayeredWindow function 34
DrawNativeBitmap 35
DrawNativeBitmap function 35
DrawTextDirect 35
DrawTextDirect function 35
DrawTextDirectEx 35
DrawTextDirectEx function 35
DrawTextGlow 35
DrawTextGlow function 35
DrawTextLine 36
DrawTextLine function 36
DrawTextOutline 36
DrawTextOutline function 36
DrawTextRect 36
DrawTextRect function 36

E

EID_MAX_BIRTHDATE_LEN 162
EID_MAX_BIRTHDATE_LEN macro 162
EID_MAX_BIRTHPLACE_LEN 162
EID_MAX_BIRTHPLACE_LEN macro 162

EID_MAX_BREXITMENTION1_LEN 163
EID_MAX_BREXITMENTION1_LEN macro 163
EID_MAX_BREXITMENTION2_LEN 163
EID_MAX_BREXITMENTION2_LEN macro 163
EID_MAX_CARD_NUMBER_LEN 163
EID_MAX_CARD_NUMBER_LEN macro 163
EID_MAX_CERT_LEN 163
EID_MAX_CERT_LEN macro 163
EID_MAX_CHIP_NUMBER_LEN 163
EID_MAX_CHIP_NUMBER_LEN macro 163
EID_MAX_DATE_BEGIN_LEN 164
EID_MAX_DATE_BEGIN_LEN macro 164
EID_MAX_DATE_END_LEN 164
EID_MAX_DATE_END_LEN macro 164
EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN 164
EID_MAX_DATEANDCOUNTRYOFPROTECTION_LEN macro 164
EID_MAX_DELIVERY_MUNICIPALITY_LEN 164
EID_MAX_DELIVERY_MUNICIPALITY_LEN macro 164
EID_MAX_DOCUMENT_TYPE_LEN 165
EID_MAX_DOCUMENT_TYPE_LEN macro 165
EID_MAX_DUPLICATE_LEN 165
EID_MAX_DUPLICATE_LEN macro 165
EID_MAX_FIRST_NAME1_LEN 165
EID_MAX_FIRST_NAME1_LEN macro 165
EID_MAX_FIRST_NAME2_LEN 165
EID_MAX_FIRST_NAME2_LEN macro 165
EID_MAX_MEMBEROFFAMILY_LEN 166
EID_MAX_MEMBEROFFAMILY_LEN macro 166
EID_MAX_MUNICIPALITY_LEN 166
EID_MAX_MUNICIPALITY_LEN macro 166
EID_MAX_NAME_LEN 166
EID_MAX_NAME_LEN macro 166
EID_MAX_NATIONAL_NUMBER_LEN 166
EID_MAX_NATIONAL_NUMBER_LEN macro 166
EID_MAX_NATIONALITY_LEN 166
EID_MAX_NATIONALITY_LEN macro 166
EID_MAX_NOBLE_CONDITION_LEN 167
EID_MAX_NOBLE_CONDITION_LEN macro 167
EID_MAX_PICTURE_LEN 167
EID_MAX_PICTURE_LEN macro 167
EID_MAX_REGIONALFILENUMBER_LEN 167
EID_MAX_REGIONALFILENUMBER_LEN macro 167
EID_MAX_SEX_LEN 167
EID_MAX_SEX_LEN macro 167
EID_MAX_SPECIAL_STATUS_LEN 168
EID_MAX_SPECIAL_STATUS_LEN macro 168
EID_MAX_SPECIALORGANIZATION_LEN 168
EID_MAX_SPECIALORGANIZATION_LEN macro 168
EID_MAX_STREET_LEN 168
EID_MAX_STREET_LEN macro 168
EID_MAX_VAT1_LEN 168
EID_MAX_VAT1_LEN macro 168
EID_MAX_VAT2_LEN 168
EID_MAX_VAT2_LEN macro 168
EID_MAX_WORKPERMITTYPE_LEN 169
EID_MAX_WORKPERMITTYPE_LEN macro 169
EID_MAX_ZIP_LEN 169
EID_MAX_ZIP_LEN macro 169
EidAddressA 142
EidAddressA structure 142
EidAddressW 142
EidAddressW structure 142
EidCertificate 143
EidCertificate structure 143
EidIdentityA 143
EidIdentityA structure 143
EidIdentityExA 145
EidIdentityExA structure 145
EidIdentityExW 146
EidIdentityExW structure 146
EidIdentityW 148
EidIdentityW structure 148
EidPicture 149
EidPicture structure 149
EmptyRecycleBin 36
EmptyRecycleBin function 36
EmToPixels 37
EmToPixels function 37
EncodeCertificate 37
EncodeCertificate function 37
EncodePhoto 37
EncodePhoto function 37
EncryptFileAESA 38
EncryptFileAESA function 38

EncryptFileAESW 38
EncryptFileAESW function 38
Encryption.h 180
ERR_CARD_ERROR 169
ERR_CARD_ERROR macro 169
ERR_CERTIFICATE_ATTR 169
ERR_CERTIFICATE_ATTR macro 169
ERR_MESSAGE_DECODE 170
ERR_MESSAGE_DECODE macro 170
ERR_MESSAGE_ENCODE 170
ERR_MESSAGE_ENCODE macro 170
ERR_MESSAGE_PARAM 170
ERR_MESSAGE_PARAM macro 170
ERR_MESSAGE_UPDATE 170
ERR_MESSAGE_UPDATE macro 170
ERR_NO_CERTIFICATE 170
ERR_NO_CERTIFICATE macro 170
ERR_NO_DATA 171
ERR_NO_DATA macro 171
ERR_NO_PRIVATE_KEY 171
ERR_NO_PRIVATE_KEY macro 171
ERR_OPEN_CERTIFICATE 171
ERR_OPEN_CERTIFICATE macro 171
ERR_SIGN_ERROR 171
ERR_SIGN_ERROR macro 171
ERR_TIMESTAMP 172
ERR_TIMESTAMP macro 172
ERROR_MAX_DESCRIPTION 172
ERROR_MAX_DESCRIPTION macro 172
ErrorInformation 149
ErrorInformation structure 149
ewtCardInsert enumeration member 132
ewtCardRemove enumeration member 132
ewtReadersChange enumeration member 132
ewtUnknownEvent enumeration member 132

F

FileCloseA 39
FileCloseA function 39
FileCloseW 39
FileCloseW function 39
FileCopyA 39
FileCopyA function 39
FileCopyW 40
FileCopyW function 40
FileCreateRewriteA 40
FileCreateRewriteA function 40
FileCreateRewriteW 40
FileCreateRewriteW function 40
FileDeleteA 41
FileDeleteA function 41
FileDeleteW 41
FileDeleteW function 41
FileExistsA 42
FileExistsA function 42
FileExistsW 42
FileExistsW function 42
FileExtensionIsA 42
FileExtensionIsA function 42
FileExtensionIsW 43
FileExtensionIsW function 43
FileGetSizeA 43
FileGetSizeA function 43
FileGetSizeW 43
FileGetSizeW function 43
FileIsExeA 44
FileIsExeA function 44
FileIsExeW 44
FileIsExeW function 44
FileIsIconA 45
FileIsIconA function 45
FileIsIconW 45
FileIsIconW function 45
FileIsImageA 45
FileIsImageA function 45
FileIsImageW 46
FileIsImageW function 46
FileIsLink 46
FileIsLink function 46
FileOperations.h 181
FileOrFolderExistsA 46
FileOrFolderExistsA function 46
FileOrFolderExistsW 47
FileOrFolderExistsW function 47

FileRenameA 47
FileRenameA function 47
FileRenameW 47
FileRenameW function 47
Files 177
FileWriteA 48
FileWriteA function 48
FileWriteCharA 48
FileWriteCharA function 48
FileWriteCharW 48
FileWriteCharW function 48
FileWriteNewLineA 49
FileWriteNewLineA function 49
FileWriteNewLineW 49
FileWriteNewLineW function 49
FileWriteW 49
FileWriteW function 49
FONT_BOLD 172
FONT_BOLD macro 172
FONT_ITALIC 172
FONT_ITALIC macro 172
FONT_NORMAL 172
FONT_NORMAL macro 172
FONT_STRIKEOUT 173
FONT_STRIKEOUT macro 173
FONT_UNDERLINE 173
FONT_UNDERLINE macro 173
fpreset 50
fpreset function 50
FreeContainer 50
FreeContainer function 50
FullPathA 50
FullPathA function 50
FullPathW 51
FullPathW function 51
Functions 1

G

GenerateAuthenticationSignatureA 51
GenerateAuthenticationSignatureA function 51
GenerateAuthenticationSignatureExA 51
GenerateAuthenticationSignatureExA function 51
GenerateAuthenticationSignatureExW 52
GenerateAuthenticationSignatureExW function 52
GenerateAuthenticationSignatureW 52
GenerateAuthenticationSignatureW function 52
GenerateBMPA 53
GenerateBMPA function 53
GenerateBMPW 53
GenerateBMPW function 53
GenerateNonRepudiationSignatureA 54
GenerateNonRepudiationSignatureA function 54
GenerateNonRepudiationSignatureExA 54
GenerateNonRepudiationSignatureExA function 54
GenerateNonRepudiationSignatureExW 55
GenerateNonRepudiationSignatureExW function 55
GenerateNonRepudiationSignatureW 55
GenerateNonRepudiationSignatureW function 55
GeneratePNGA 56
GeneratePNGA function 56
GeneratePNGW 56
GeneratePNGW function 56
GenerateQRCodeA 57
GenerateQRCodeA function 57
GenerateQRCodeExA 57
GenerateQRCodeExA function 57
GenerateQRCodeExW 57
GenerateQRCodeExW function 57
GenerateQRCodeW 58
GenerateQRCodeW function 58
GetAllFiles 58
GetAllFiles function 58
GetCardBufferA 59
GetCardBufferA function 59
GetCardBufferSize 59
GetCardBufferSize function 59
GetCardBufferW 59
GetCardBufferW function 59
GetCardSerialNumber 60
GetCardSerialNumber function 60
GetCardVersion 60
GetCardVersion function 60
GetEncodedCertificateSize 61
GetEncodedCertificateSize function 61

GetEncodedPhotoSize 61
GetEncodedPhotoSize function 61
GetFileMD5A 62
GetFileMD5A function 62
GetFileMD5W 62
GetFileMD5W function 62
GetFilesCountA 62
GetFilesCountA function 62
GetFilesCountW 63
GetFilesCountW function 63
GetFileSHA1A 63
GetFileSHA1A function 63
GetFileSHA1W 64
GetFileSHA1W function 64
GetFileSHA256 173
GetFileSHA256 macro 173
GetFileSHA256A 64
GetFileSHA256A function 64
GetFileSHA256W 64
GetFileSHA256W function 64
GetHBitmapA 65
GetHBitmapA function 65
GetHBitmapW 65
GetHBitmapW function 65
GetISOCodeA 66
GetISOCodeA function 66
GetISOCodeW 66
GetISOCodeW function 66
GetMD5 67
GetMD5 function 67
GetPNGA 67
GetPNGA function 67
GetPNGW 68
GetPNGW function 68
GetReaderIndexA 68
GetReaderIndexA function 68
GetReaderIndexW 68
GetReaderIndexW function 68
GetReaderNameA 69
GetReaderNameA function 69
GetReaderNameLenA 69
GetReaderNameLenA function 69
GetReaderNameLenW 70
GetReaderNameLenW function 70
GetReaderNameW 70
GetReaderNameW function 70
GetReadersCount 70
GetReadersCount function 70
GetSelectedReaderIndex 71
GetSelectedReaderIndex function 71
GetSHA1 71
GetSHA1 function 71
GetSHA256 71
GetSHA256 function 71
GetStartupA 72
GetStartupA function 72
GetStartupW 72
GetStartupW function 72
GetSupportSIS 73
GetSupportSIS function 73
GetTextLineSize 73
GetTextLineSize function 73
GetTextSize 73
GetTextSize function 73
GetTextSizeEx 73
GetTextSizeEx function 73
Graphics.h 183

H

HibernateWindows 74
HibernateWindows function 74

I

IgnoreHardwareEvents 74
IgnoreHardwareEvents function 74
IgnoreServiceEvents 74
IgnoreServiceEvents function 74
IID_PPV_ARG 173
IID_PPV_ARG macro 173
InitializeContainer 75
InitializeContainer function 75
IsAnimatedGIFA 75
IsAnimatedGIFA function 75
IsAnimatedGIFW 75

IsAnimatedGIFW function 75
IsCardActivated 76
IsCardActivated function 76
IsCardActivatedEx 76
IsCardActivatedEx function 76
IsCardPresent 76
IsCardPresent function 76
IsCardPresentEx 77
IsCardPresentEx function 77
IsCardStillInserted 77
IsCardStillInserted function 77
IsCardStillInsertedEx 77
IsCardStillInsertedEx function 77
IsCitrixSession 78
IsCitrixSession function 78
IsConnectedToInternet 78
IsConnectedToInternet function 78
IsDirectoryA 78
IsDirectoryA function 78
IsDirectoryW 78
IsDirectoryW function 78
IsEIDCard 79
IsEIDCard function 79
IsEIDCardEx 79
IsEIDCardEx function 79
IsEngineActive 79
IsEngineActive function 79
IsFemaleA 80
IsFemaleA function 80
IsFemaleW 80
IsFemaleW function 80
IsMaleA 80
IsMaleA function 80
IsMaleW 81
IsMaleW function 81
IsMediaCenter 81
IsMediaCenter function 81
IsMetroActive 81
IsMetroActive function 81
IsMultiTouchReady 82
IsMultiTouchReady function 82
IsNativeWin64 82
IsNativeWin64 function 82
IsRemoteSession 82
IsRemoteSession function 82
IsSISCard 82
IsSISCard function 82
IsSISCardEx 83
IsSISCardEx function 83
IsTabletPC 83
IsTabletPC function 83
IsUnicodeFileA 83
IsUnicodeFileA function 83
IsUnicodeFileW 84
IsUnicodeFileW function 84
IsValidFileNameA 84
IsValidFileNameA function 84
IsValidFileNameW 84
IsValidFileNameW function 84
IsValidPathNameA 85
IsValidPathNameA function 85
IsValidPathNameW 85
IsValidPathNameW function 85
IsWindows10 86
IsWindows10 function 86
IsWindows7 86
IsWindows7 function 86
IsWindows8 86
IsWindows8 function 86
IsWindowsVista 86
IsWindowsVista function 86
IsWindowsXP 87
IsWindowsXP function 87
IsWindowsXPSP2 87
IsWindowsXPSP2 function 87
IsWow64 87
IsWow64 function 87

L

LayeredWndProcA 87
LayeredWndProcA function 87
LayeredWndProcW 87
LayeredWndProcW function 87
LoadBitmapJPG 88

LoadBitmapJPG function 88
LoadBitmapPNG 88
LoadBitmapPNG function 88
LoadCertificateA 88
LoadCertificateA function 88
LoadCertificateW 89
LoadCertificateW function 89
LoadIdentityA 89
LoadIdentityA function 89
LoadIdentityW 89
LoadIdentityW function 89
LoadPhotoA 90
LoadPhotoA function 90
LoadPhotoW 90
LoadPhotoW function 90
LoadPNGResource 90
LoadPNGResource function 90

M

Macros 160
MakeCompatibleBitmap 91
MakeCompatibleBitmap function 91
MakeSoundFromFileA 91
MakeSoundFromFileA function 91
MakeSoundFromFileW 91
MakeSoundFromFileW function 91
MakeSoundFromResourceA 92
MakeSoundFromResourceA function 92
MakeSoundFromResourceW 92
MakeSoundFromResourceW function 92

N

NationalityConverter.h 184

P

PEidAddressA 150
PEidAddressA structure 150
PEidAddressW 150
PEidAddressW structure 150
PEidCertificate 150
PEidCertificate structure 150
PEidIdentityA 151
PEidIdentityA structure 151
PEidIdentityExA 152
PEidIdentityExA structure 152
PEidIdentityExW 154
PEidIdentityExW structure 154
PEidIdentityW 155
PEidIdentityW structure 155
PeidPicture 157
PeidPicture structure 157
PErrorInformation 157
PErrorInformation structure 157
PointsToPixels 92
PointsToPixels function 92
PortAvailable 93
PortAvailable function 93
PSISRecordA 157
PSISRecordA structure 157
PSISRecordW 158
PSISRecordW structure 158

Q

quricol.h 184

R

ReadAddressA 93
ReadAddressA function 93
ReadAddressExA 93
ReadAddressExA function 93
ReadAddressExW 94
ReadAddressExW function 94
ReadAddressW 94
ReadAddressW function 94
ReadAuthenticationCertificate 94
ReadAuthenticationCertificate function 94
ReadAuthenticationCertificateEx 95
ReadAuthenticationCertificateEx function 95
ReadBufferFromFileA 95
ReadBufferFromFileA function 95
ReadBufferFromFileW 95
ReadBufferFromFileW function 95
ReadCaCertificate 96
ReadCaCertificate function 96

ReadCaCertificateEx 96
ReadCaCertificateEx function 96
ReadIdentityA 97
ReadIdentityA function 97
ReadIdentityExA 97
ReadIdentityExA function 97
ReadIdentityExW 97
ReadIdentityExW function 97
ReadIdentityW 98
ReadIdentityW function 98
ReadNonRepudiationCertificate 98
ReadNonRepudiationCertificate function 98
ReadNonRepudiationCertificateEx 99
ReadNonRepudiationCertificateEx function 99
ReadPhoto 99
ReadPhoto function 99
ReadPhotoAsBitmap 99
ReadPhotoAsBitmap function 99
ReadPhotoAsBitmapEx 100
ReadPhotoAsBitmapEx function 100
ReadPhotoEx 100
ReadPhotoEx function 100
ReadRootCaCertificate 101
ReadRootCaCertificate function 101
ReadRootCaCertificateEx 101
ReadRootCaCertificateEx function 101
ReadRrnCertificate 101
ReadRrnCertificate function 101
ReadRrnCertificateEx 102
ReadRrnCertificateEx function 102
ReadSISCardA 102
ReadSISCardA function 102
ReadSISCardExA 103
ReadSISCardExA function 103
ReadSISCardExW 103
ReadSISCardExW function 103
ReadSISCardW 103
ReadSISCardW function 103
RecycleBinEmpty 104
RecycleBinEmpty function 104
ReloadReadersList 104
ReloadReadersList function 104
RemoveCallback 104
RemoveCallback function 104
RemoveStartupA 105
RemoveStartupA function 105
RemoveStartupW 105
RemoveStartupW function 105
RestoreWindowSubclassA 105
RestoreWindowSubclassA function 105
RestoreWindowSubclassW 106
RestoreWindowSubclassW function 106

S

SaveAuthenticationCertificateA 106
SaveAuthenticationCertificateA function 106
SaveAuthenticationCertificateExW 106
SaveAuthenticationCertificateExW function 106
SaveAuthenticationCertificateW 107
SaveAuthenticationCertificateW function 107
SaveCaCertificateA 107
SaveCaCertificateA function 107
SaveCaCertificateExW 107
SaveCaCertificateExW function 107
SaveCaCertificateW 108
SaveCaCertificateW function 108
SaveCardToToXMLStreamEx 174
SaveCardToToXMLStreamEx macro 174
SaveCardToToXMLStreamExA 108
SaveCardToToXMLStreamExA function 108
SaveCardToToXMLStreamExW 108
SaveCardToToXMLStreamExW function 108
SaveCardToXmlA 109
SaveCardToXmlA function 109
SaveCardToXmlExA 109
SaveCardToXmlExA function 109
SaveCardToXmlExW 110
SaveCardToXmlExW function 110
SaveCardToXmlW 110
SaveCardToXmlW function 110
SaveContainer 111
SaveContainer function 111
SaveIdentityA 111
SaveIdentityA function 111

SaveIdentityW 111
SaveIdentityW function 111
SaveNonRepudiationCertificateA 112
SaveNonRepudiationCertificateA function 112
SaveNonRepudiationCertificateExW 112
SaveNonRepudiationCertificateExW function 112
SaveNonRepudiationCertificateW 112
SaveNonRepudiationCertificateW function 112
SavePersonCsvToStream 174
SavePersonCsvToStream macro 174
SavePersonCsvToStreamA 113
SavePersonCsvToStreamA function 113
SavePersonCsvToStreamW 113
SavePersonCsvToStreamW function 113
SavePersonToCsvA 114
SavePersonToCsvA function 114
SavePersonToCsvExA 114
SavePersonToCsvExA function 114
SavePersonToCsvExW 114
SavePersonToCsvExW function 114
SavePersonToCsvW 115
SavePersonToCsvW function 115
SavePhotoA 115
SavePhotoA function 115
SavePhotoAsBitmapA 116
SavePhotoAsBitmapA function 116
SavePhotoAsBitmapExA 116
SavePhotoAsBitmapExA function 116
SavePhotoAsBitmapExW 116
SavePhotoAsBitmapExW function 116
SavePhotoAsBitmapW 117
SavePhotoAsBitmapW function 117
SavePhotoAsJpegA 117
SavePhotoAsJpegA function 117
SavePhotoAsJpegExA 117
SavePhotoAsJpegExA function 117
SavePhotoAsJpegExW 118
SavePhotoAsJpegExW function 118
SavePhotoAsJpegW 118
SavePhotoAsJpegW function 118
SavePhotoW 119
SavePhotoW function 119
SaveRootCaCertificateA 119
SaveRootCaCertificateA function 119
SaveRootCaCertificateExW 119
SaveRootCaCertificateExW function 119
SaveRootCaCertificateW 120
SaveRootCaCertificateW function 120
SaveRrnCertificateA 120
SaveRrnCertificateA function 120
SaveRrnCertificateExW 120
SaveRrnCertificateExW function 120
SaveRrnCertificateW 121
SaveRrnCertificateW function 121
SelectReader 121
SelectReader function 121
SelectReaderByNameA 121
SelectReaderByNameA function 121
SelectReaderByNameW 122
SelectReaderByNameW function 122
SendAPDU 122
SendAPDU function 122
SetCallback 122
SetCallback function 122
SetMWCompatibility 123
SetMWCompatibility function 123
SetStartupA 123
SetStartupA function 123
SetStartupW 123
SetStartupW function 123
SetSupportSIS 124
SetSupportSIS function 124
ShellCopyFileA 124
ShellCopyFileA function 124
ShellCopyFileW 125
ShellCopyFileW function 125
ShutdownWindows 125
ShutdownWindows function 125
SIS_FIELD_MAX_BIRTHDATE_LEN 174
SIS_FIELD_MAX_BIRTHDATE_LEN macro 174
SIS_FIELD_MAX_CAPTUREDATE_LEN 174
SIS_FIELD_MAX_CAPTUREDATE_LEN macro 174
SIS_FIELD_MAX_CARDNUMBER_LEN 175
SIS_FIELD_MAX_CARDNUMBER_LEN macro 175

SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN 175
SIS_FIELD_MAX_SOCIAL_SECURITY_NUMBER_LEN macro 175
SIS_FIELD_MAX_VALIDBEGIN_LEN 175
SIS_FIELD_MAX_VALIDBEGIN_LEN macro 175
SIS_FIELD_MAX_VALIDEND_LEN 175
SIS_FIELD_MAX_VALIDEND_LEN macro 175
SIS_MAX_CARDNAME_LEN 176
SIS_MAX_CARDNAME_LEN macro 176
SIS_MAX_FIRSTNAMES_LEN 176
SIS_MAX_FIRSTNAMES_LEN macro 176
SIS_MAX_INITIAL_LEN 176
SIS_MAX_INITIAL_LEN macro 176
SIS_MAX_NAME_LEN 176
SIS_MAX_NAME_LEN macro 176
SIS_MAX_SEX_LEN 176
SIS_MAX_SEX_LEN macro 176
SISRecordA 159
SISRecordA structure 159
SISRecordW 160
SISRecordW structure 160
StartEngine 126
StartEngine function 126
StopEngine 126
StopEngine function 126
StretchNativeBitmap 126
StretchNativeBitmap function 126
StripFileNameA 126
StripFileNameA function 126
StripFileNameW 127
StripFileNameW function 127
structErrorInformation 132
structErrorInformation structure 132
Structs, Records, Enums 131
SuspendWindows 127
SuspendWindows function 127
Swelio.h 184
System.h 189
SystemInfo.h 190

T

tagCardEventType 132
tagCardEventType enumeration 132
tagEidAddressA 133
tagEidAddressA structure 133
tagEidAddressW 133
tagEidAddressW structure 133
tagEidCertificate 133
tagEidCertificate structure 133
tagEidIdentityA 134
tagEidIdentityA structure 134
tagEidIdentityExA 135
tagEidIdentityExA structure 135
tagEidIdentityExW 137
tagEidIdentityExW structure 137
tagEidIdentityW 138
tagEidIdentityW structure 138
tagEidPicture 140
tagEidPicture structure 140
tagSISRecordA 140
tagSISRecordA structure 140
tagSISRecordW 141
tagSISRecordW structure 141
TurnMonitorOff 127
TurnMonitorOff function 127
TurnMonitorOn 128
TurnMonitorOn function 128

U

UpdateWindowPosition 128
UpdateWindowPosition function 128

V

VerifyPinA 128
VerifyPinA function 128
VerifyPinExA 128
VerifyPinExA function 128
VerifyPinExW 129
VerifyPinExW function 129
VerifyPinW 129
VerifyPinW function 129
VerifySignature 130
VerifySignature function 130

W

WIDTHBYTES 177
WIDTHBYTES macro 177
WriteBufferToFileA 130
WriteBufferToFileA function 130
WriteBufferToFileW 130
WriteBufferToFileW function 130